IB Physics HL Notes

Kenny Lin / BusyPizza

October 16, 2026

Contents

0 Experimental Premises **5**
0.1 SI Units & Prefixes . . . 5
0.2 Experimental error . . . 5
0.3 Representation of uncertanties . . . 6
0.4 Uncertainty calculation . . . 6
0.5 Uncertainty in graph . . . 7
0.6 Linearization of graph . . . 7

Unit A Space, time and motion **8**

1 Kinematics **8**
1.1 Displacement, distance, speed and velocity . . . 8
1.2 Uniformly accelerated motion: the equations of kinematics . . . 8
1.3 Graphs of motion . . . 9
1.4 Projectile motion . . . 9

2 Forces and Newton's laws **11**
2.1 Forces and their direction . . . 11
2.2 Newton's laws of motion . . . 13
2.3 Circular motion . . . 14

3 Work, energy and power **15**
3.1 Work . . . 15
3.2 Conservation of energy . . . 16
3.3 Power and efficiency . . . 16
3.4 Energy transfers . . . 16

4 Linear momentum **16**
4.1 Newton's second law in terms of momentum . . . 16
4.2 Impulse and force-time graphs . . . 17
4.3 Conservation of momentum . . . 17
4.4 Kinetic energy and momentum . . . 18
4.5 Two-dimensional collisions . . . 18

5 Rigid body mechanics **19**
5.1 Kinematics of rotational motion . . . 19
5.2 Rotational equilibrium and Newton's second law . . . 19
5.3 Angular momentum . . . 22

6 Relativity **22**
6.1 Reference frames and Lorentz transformations . . . 22
6.2 Effects of relativity . . . 23
6.3 Spacetime / Minkowski diagrams . . . 25

Unit B The particulate nature of matter **28**

7 Thermal energy transfers **28**
7.1 Particles, temperature and energy . . . 28
7.2 Specific heat capacity and change of phase . . . 29
7.3 Thermal energy transfer . . . 30

8 The greenhouse effect **32**
8.1 Radiation from real bodies . . . 32
8.2 Energy balance of the earth . . . 33

9 The gas laws **36**
9.1 Moles, molar mass and the Avogadro constant . . . 36
9.2 Ideal gases . . . 36
9.3 The Boltzmann equation . . . 38

10 Thermodynamics **39**
10.1 Internal energy . . . 39
10.2 The first law of thermodynamics . . . 40
10.3 The second law of thermodynamics . . . 41
10.4 Heat engines . . . 41

11 Current and circuits **43**
11.1 Potential difference, current and resistance . . . 43
11.2 Voltage, power and e.m.f . . . 44
11.3 Resistors in electrical circuits . . . 44
11.4 Terminal potential difference and the potential divider . . . 45

Unit C Wave behaviour **46**

12 Simple harmonic motion **46**
12.1 Simple harmonic oscillations . . . 46
12.2 Details of simple harmonic motion . . . 47
12.3 Energy in simple harmonic motion . . . 47
12.4 More about energy in SHM . . . 48

13 The wave model **48**
13.1 Mechanical pulses and waves . . . 48
13.2 Transverse and longitudinal waves . . . 48
13.3 Electromagnetic waves . . . 50
13.4 Waves extension . . . 50

14 Wave phenomena **51**
14.1 Reflection and refraction . . . 51
14.2 The principle of superposition . . . 52
14.3 Diffraction and interference . . . 52
14.4 Single-slit diffraction . . . 55
14.5 Multiple slits . . . 56

15 Standing waves and resonance **57**
15.1 Standing waves . . . 57
15.2 Standing waves on strings . . . 58
15.3 Standing waves in pipes . . . 59
15.4 Resonance and damping . . . 59

16 The Doppler effect **61**
16.1 The Doppler effect at low speeds . . . 61
16.2 The Doppler effect for sound . . . 62

Unit D Fields **63**

17 Gravitation **63**
17.1 Newton's law of gravitation . . . 63
17.2 Gravitational potential and energy . . . 65
17.3 Motion in a gravitational field . . . 66

18 Electric and magnetic fields **67**
18.1 Electric charge, force and field . . . 67
18.2 Magnetic field and force . . . 70
18.3 Electric potential and electric potential energy . . . 72

19 Motion in electric and magnetic fields **74**
19.1 Motion in an electric field . . . 74
19.2 Motion in a magnetic field . . . 75

20 Electromagnetic induction **76**
20.1 Electromagnetic induction . . . 76
20.2 Generators and alternating current . . . 79

Unit E Nuclear and quantum physics **80**

21 Atomic physics **80**
21.1 The structure of the atom . . . 80
21.2 Quantisation of angular momentum . . . 81

22 Quantum physics **82**
22.1 Photons and the photoelectric effect . . . 82
22.2 Matter waves . . . 85

23 Nuclear physics **85**
23.1 Mass defect and binding energy . . . 85
23.2 Radioactivity . . . 88
23.3 Nuclear properties and the radioactive decay law . . . 89

24 Nuclear fission **93**
24.1 Nuclear fission . . . 93

25 Nuclear fusion and stars **95**
25.1 Nuclear fusion . . . 95
25.2 Stellar properties and the Hertzsprung-Russell diagram . . . 96
25.3 Stellar evolution extension . . . 99

Appendices **101**

A Common constants **101**

B Formula list **103**
B.1 Unit A — Space, time and motion . . . 103
B.2 Unit B — The particulate nature of matter . . . 106
B.3 Unit C — Wave behaviour . . . 108
B.4 Unit D — Fields . . . 110
B.5 Unit E — Nuclear and quantum physics . . . 112

0 Experimental Premises

0.1 SI Units & Prefixes

Fundamental units:

- Length: metre (m)
- Mass: kilogram (kg)
- Time: second (s)
- Electric current: ampere (A)
- Thermodynamic temperature: kelvin (K)
- Amount of substance: mole (mol)
- Luminous intensity: candela (cd)

Derived units: can be expressed in terms of fundamental units.
Example: $F(N) = m \times a$ (kg m s^{-2});
$W = F \times d = m \cdot a \cdot d$ (kg m^2 s^{-2});
$Q = mc\Delta T$, c= (J kg^{-1} K^{-1}) = (m^2 s^{-2} K^{-1}).

Symbol	Prefix	Powers of 10	Name
E	Exa	10^{18}	quintillion
P	Peta	10^{15}	quadrillion
T	Tera	10^{12}	trillion
G	Giga	10^{9}	billion
M	Mega	10^{6}	million
k	kilo	10^{3}	thousand
h	hecto	10^{2}	hundred
da	deca	10^{1}	ten
–	–	10^{0}	one
d	deci	10^{-1}	one tenth
c	centi	10^{-2}	one hundredth
m	milli	10^{-3}	one thousandth
μ	micro	10^{-6}	one millionth
n	nano	10^{-9}	one billionth
p	pico	10^{-12}	one trillionth
f	femto	10^{-15}	one quadrillionth
a	atto	10^{-18}	one quintillionth

Homogeneity of equations: all terms in an equation must have the same dimensions (base units).
A homogenous equation might not be valid due to a missing coefficient.

0.2 Experimental error

Error is a problem which causes a measurement to be inaccurate / different from the true value.
Difference between uncertainties and errors: uncertainties are the range of values within which the true value is expected to lie, while errors are the difference between the measured value and the true value.
Random and systematic errors:
Random errors are unpredictable variations / random fluctuations in measurement results. (Readability of instruments / things changing in the surroundings)

Results in readings scattered around the true value (larger / smaller).
Systematic errors are consistent, repeatable errors that cause measurements to be consistently too high or too low. Causes all readings to be away from the true value in the same direction. (e.g. tool has zero error)

- Can be solved by recalibrating instruments.
- Can be detected using different methods of measurement.

Difference between precision 精准度 and accuracy 准确度:
Precision (repeated) refers to how close the measurements are to each other, while accuracy (correctness) refers to how close the measurements are to the true value.
Equipments:

- Vernier caliper: used to measure length with high precision
- Micrometer screw gauge: used to measure small distances with even higher precision

How to decrease errors:

1. Change a more precise instrument
2. Take more readings, repeat several times and average them

0.3 Representation of uncertanties

Verbal expression:

$$\text{Measurement} = (\text{best estimate} \pm \text{magnitudinal uncertainty})$$

Mathematical expression:

$$x = (x_{\text{avg}} \pm \Delta x)$$

where Δx is instrumental uncertainty.
This formula gives the range of measurements within which the true value is expected to lie.
Finding the value of uncertainty:

- For a single measurement, using the smallest division of the instrument
- Repeat the readings

Uncertainty summary:

- Uncertanity in a single reading = ± half of the smallest division of the instrument
- Uncertainty in physical quantities = ± smallest graduation of the instrument
- The uncertainty is like to be give to **only 1 s.f.** and the best estimate is given to the same decimal place as the uncertainty.
- Number of d.p. in measured value = number of d.p. in uncertainty

0.4 Uncertainty calculation

Uncertanty precision determination in several repeated measurements (an example of period of oscillation):

$$x_{\text{avg}} = \frac{\sum x_i}{n};\ \Delta x = \frac{x_{\max} - x_{\min}}{2}$$

(Could also incorporate the reaction time for this task specifically, usually ±0.2s)
Uncertanty ranges only add up together (e.g. $\pm 0.1 + \pm 0.1 = \pm 0.2$), **never subtract / disappear**.

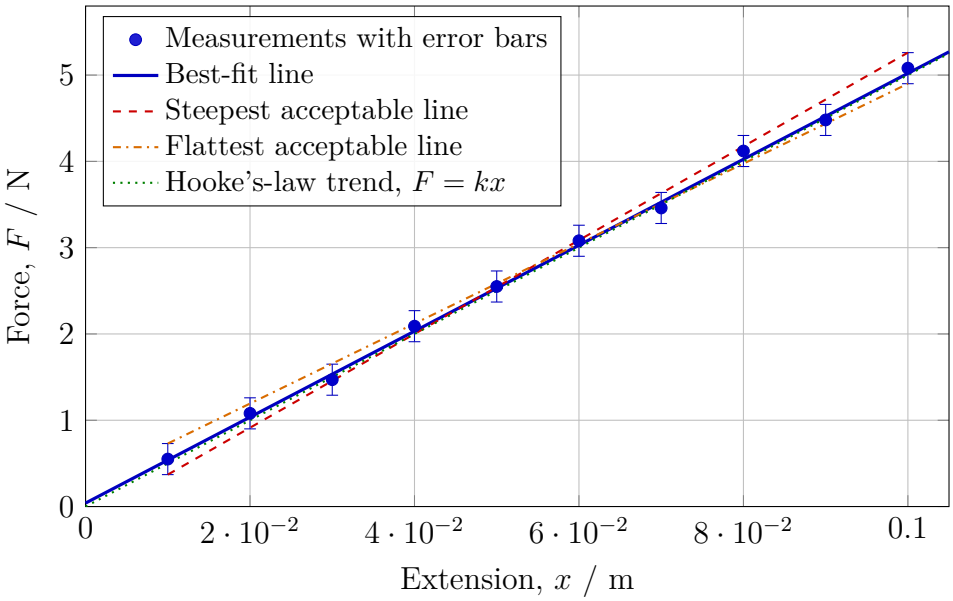
Relative / fractional (percentage) uncertainty: $\dfrac{\Delta x}{x}(\times 100\%)$

$$\frac{\Delta(x/y)}{x/y} \Big/ \frac{\Delta(xy)}{xy} = \frac{\Delta x}{x} + \frac{\Delta y}{y}$$

$$\frac{\Delta(kx^n)}{kx^n} = n \times \frac{\Delta x}{x}$$

0.5 Uncertainty in graph

Include error bars in the graph. Example of Hooke's Law applied with uncertainties:



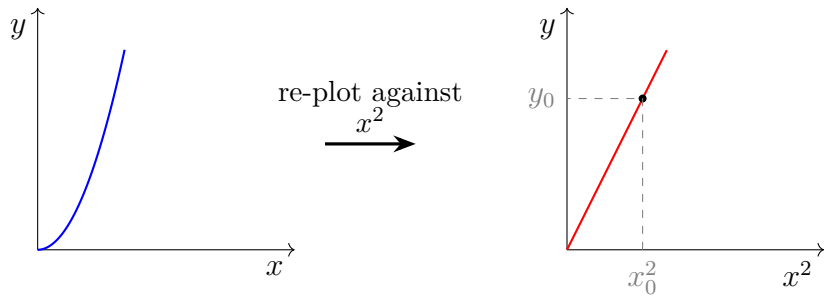
Worst acceptable line / error line (flattest: straight line passing through the highest point in first error bar and the lowest point in the last error bar; steepest: straight line passing through the lowest point in first error bar and the highest point in the last error bar)

Gradient (including uncertainties): $m \pm \dfrac{|m_1 - m_2|}{2}$

Intercept (including uncertainties): $c \pm \dfrac{|c_1 - c_2|}{2}$

0.6 Linearization of graph

- Non-logarithmic approach: transform the relationship into linear relationship via changing the x-axis being graphed against (e.g. $y \propto x^2$) This is learnt in Additional Mathematics, so it should be a revisit.
- Logarithmic approach: transform the relationship into linear ones via applying logarithmic functions (Example: $y = ax^b \Rightarrow \log y = b \log x + \log a$; Example 2: $y = ae^{bx} \Rightarrow \ln y = \ln a + bx$)



Original: $y = 2x^2$ (curve) Linearized: $y = 2x^2$, gradient $= 2$

Unit A Space, time and motion

1 Kinematics

Kinematics vs Dynamics: kinematics *describes* motion (displacement, velocity, acceleration) without regard to its cause, while dynamics *explains* motion through forces (Newton's laws).

1.1 Displacement, distance, speed and velocity

Displacement 位移: change in position, $\Delta s = s_{\text{final}} - s_{\text{start}}$.
Distance 路程: the actual length of the path followed
For uniform motion (constant speed):

- $v = \dfrac{\Delta s}{\Delta t}$, average speed is total distance divided by total time taken
- $s = s_i + vt$
- Area in a $v - t$ graph is the displacement

1.2 Uniformly accelerated motion: the equations of kinematics

For uniformly accelerated motion (constant acceleration):

- $\overline{v} = \dfrac{\Delta s}{\Delta t}$
- $v = \lim\limits_{\Delta t \to 0} \dfrac{\Delta s}{\Delta t}$
- Draw a tangent line to find the instantaneous speed.
- $\overline{a} = \dfrac{\Delta v}{\Delta t}$
- $a = \lim\limits_{\Delta t \to 0} \dfrac{\Delta v}{\Delta t}$
- $v = u + at$, where u is the initial velocity.
- $\Delta s = \left(\dfrac{u+v}{2}\right) t = ut + \dfrac{1}{2}at^2$
- $s = s_i + ut + \dfrac{1}{2}at^2$
- $v^2 = u^2 + 2a\Delta s$

Acceleration of freefall: $g = 9.8\ \text{m/s}^2$, direction is vertically downwards.
Utilize this for questions involving freefall acceleration. (When doing questions, use the negative value.)

- $s = \dfrac{1}{2}gt^2$
- $V_f = gt$
- $v^2 = 2gh$ (derived from $v^2 = u^2 + 2as$)

1.3 Graphs of motion

For any motion graph:

$$v = \frac{ds}{dt}, \qquad a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

and the area under a $v - t$ graph gives displacement, while the area under an $a - t$ graph gives change in velocity.

Take $x = 2\pi t$, where $0 \leq x \leq 2\pi$ corresponds to $0 \leq t \leq 1.0\,\text{s}$. If

$$s(t) = \sin(2\pi t),$$

then

$$v(t) = \frac{ds}{dt} = 2\pi\cos(2\pi t), \qquad a(t) = \frac{dv}{dt} = -(2\pi)^2\sin(2\pi t).$$

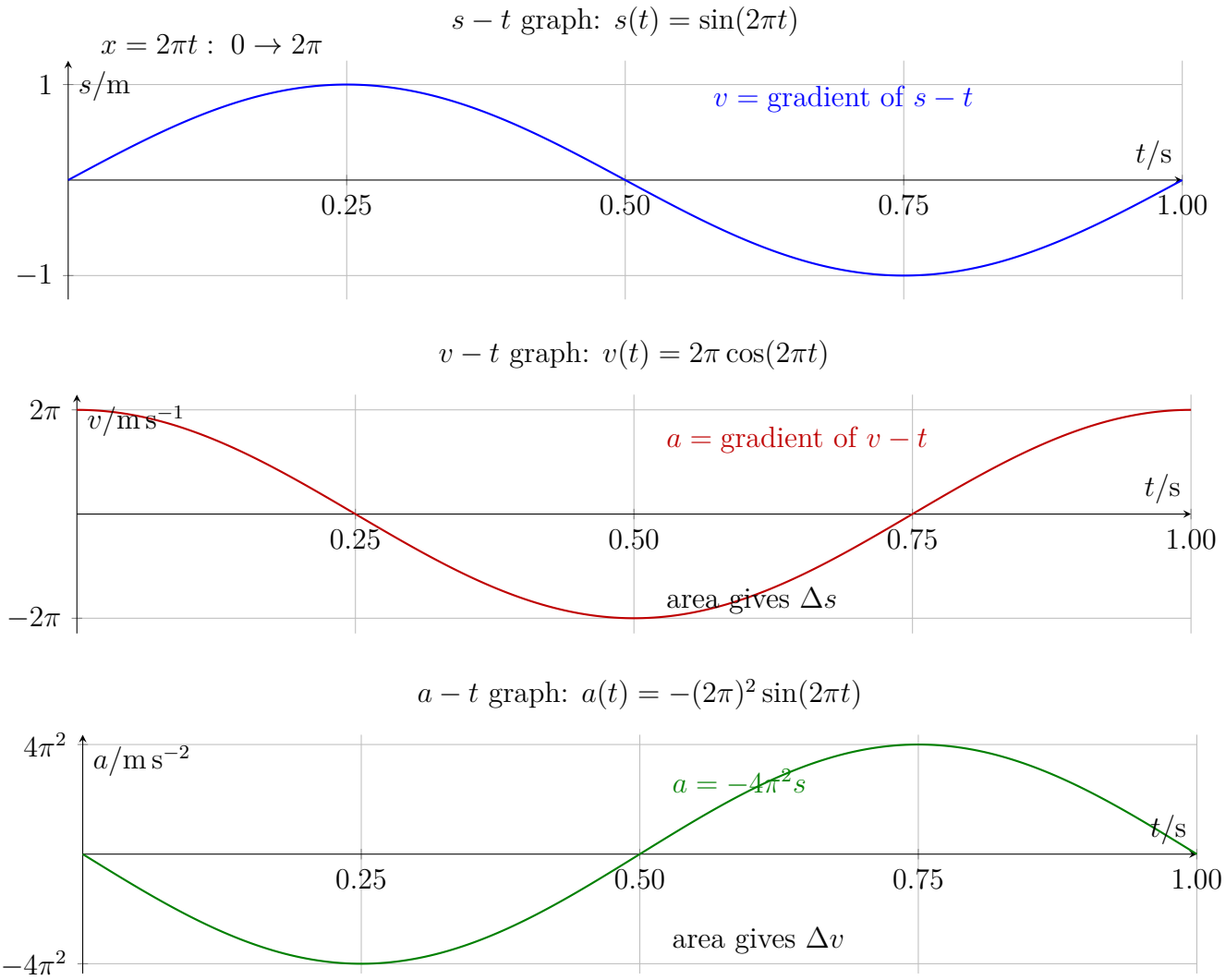


Figure 1: The $s - t$, $v - t$, and $a - t$ graphs for one cycle of sinusoidal motion.

1.4 Projectile motion

Types of projectiles:

- Up and down

- At an angle to horizontal
- Horizontal

If a projectile is launched at angle θ and initial speed u, then the direction of the velocity are:

- $V_x = u\cos\theta$
- $V_y = u\sin\theta - gt$

The position is determined by position vector $\vec{r}$ that is relative from the origin.

- $x = ut\cos\theta$
- $y = ut\sin\theta - \frac{1}{2}gt^2$
- $v_x^2 = v_y^2 - 2gy$
- $t = \frac{u\sin\theta}{g}$, this is when $v_y = 0$.
- $y_{\max} = \frac{u^2\sin^2\theta}{2g}$, the maximum height
- $t = 2\frac{u\sin\theta}{g}$, this is when the maximum displacement in the x-direction occurs.
- Maximum displacement in the x-direction (**the range**): $x = \frac{u^2\sin 2\theta}{g}$.
- We obtain the maximum range with a launch angle of $45°$.

A fluid resistance force is directed against the direction of the velocity. If there is air resistance, it is proportional to the square of the velocity. Thus, when the ball moves fast its deceleration is greater than when it moves slow. (Peak to left of the original one, pre-peak distance > post-peak distance, post-peak gradient > pre-peak gradient)

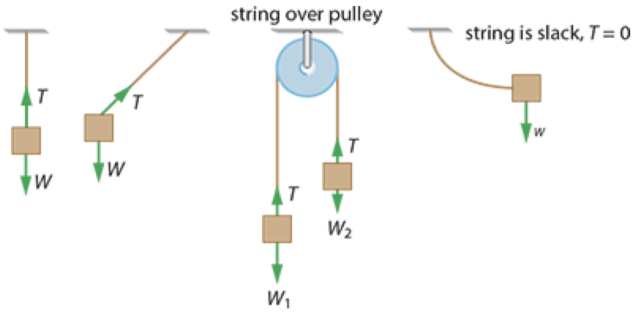
- Range decreased
- Trajectory not parabolic
- Reduced maximum height
- Vertical acceleration is no longer constant
- Horizontal deceleration
- Speeds reduced
- Determination of time of flight becomes complex and depends on the initial conditions of the flight

$F = kv$ for low speeds and $F = kv^2$ for high speeds. (For resistance)
Terminal speed: $v_T = \frac{mg}{k}$, where acceleration equals 0.

2 Forces and Newton's laws

2.1 Forces and their direction

- Weight
  - $W = mg$
  - Acts vertically downward
- String tension
  - The force that arises in any body when it is stretched or compressed, when two forces are applied in opposite directions at the ends of the string
  - Taut: under tension
  - Result of interactions between the molecules in the material, electrical in nature
  - Usually assume string is massless, tension is same on all points along the string
  - Direction: along the string

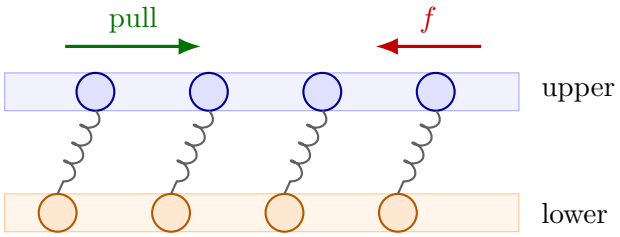


- Spring tension / Hooke's Law
  - Hooke's Law: $F = kx$, where x is the extension
  - Beyond limit of proportionality: Force no longer varies directly with extension
  - Elastic limit: The maximum stress a material can withstand and still return completely to its original shape when the load is removed.
  - Extension on Hooke's Law related content:
    - Parallel connection: $F_T = (k_A + k_B)x = k_{\text{eff}}x$
    - Series connection: $x_T = x_A + x_B = \dfrac{F}{k_A} + \dfrac{F}{k_B}$; $\dfrac{1}{k_{\text{eff}}} = \dfrac{1}{k_A} + \dfrac{1}{k_B}$
- Normal contact forces
  - When a body touches another body
  - Perpendicular to the surface of the body exerting the force, electrical nature
- Drag forces (friction in fluids)
  - Opposes the motion of a body through a fluid
  - For the case of a spherical body of radius r falling through a fluid of density ρ with speed v the drag force is given by Stoke's law:
  - $F_D = 6\pi\eta r v$, where η is the viscosity of the fluid (a measure of the forces between layers of the fluid that are moving relative to each other, unit Pa×s)
- Buoyant forces (upthrust)

- $F_B = \rho g V_{\text{imm}}$, where V_{imm} is the volume of the body that is immersed in the fluid.
- Caused by the pressure that the fluid exerts on the body.
- 王忠 Let's go: $F_{浮} = \rho_{液} g V_{排}$

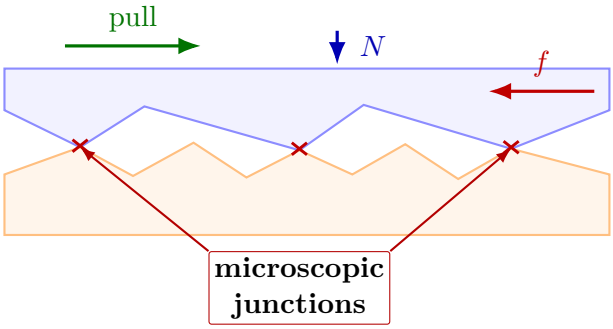
- Frictional forces
  - Generally oppose the motion of a body, electrical nature, parallel to contact surface
  - Occurs when one body slides over another / has a tendency for motion, has dynamic / static friction

Simple atoms-and-springs model



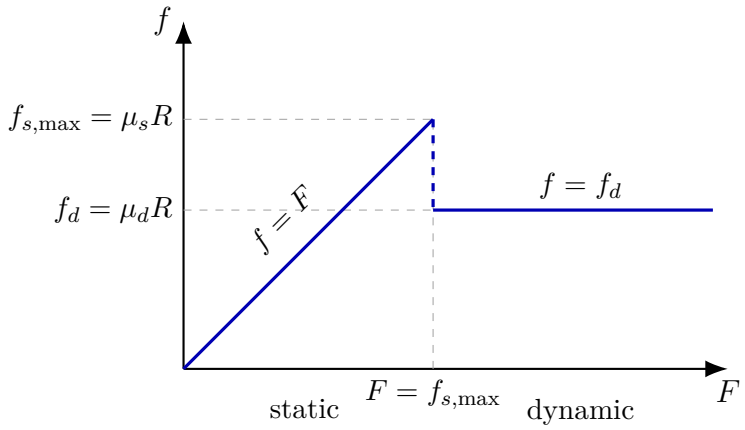
Atoms are displaced; stretched microscopic bonds act like springs and push back.

Asperities model



Real surfaces touch at asperities; junctions must shear or break before sliding occurs.

  - Friction Laws:
    1. Area of contact does not affect the frictional force.
    2. The force of dynamic friction is $f_d = \mu_d R$, where R is the normal force and μ_d is the coefficient of dynamic friction.
    3. The force of dynamic friction does not depend on the speed of sliding.
    4. The MAXimum force of static friction is $f_s = \mu_s R$, where μ_s is the coefficient of static friction and $\mu_s > \mu_d$
    5. ($\mu_s R$ does not give the force of static friction, but only the MAXimum)
    6. $0 \leq F_s \leq F_{s,\max}$; $F_d < F_{s,\max}$; $F_d = k$



- Explaining friction:
  - At the atomic level when two surfaces come into contact, small peaks on one surface cold weld with small peaks on the other surface.
  - Applying the initial sideways force, all of the cold welds oppose the motion.
  - If the force is sufficiently large, the cold welds break, and new peaks contact each other and cold weld.
  - If the surfaces remain in relative sliding motion, fewer welds have a chance to form.
  - The coefficient of friction is then defined as $\mu = \dfrac{f}{R}$

Free-body force diagrams: diagram showing the magnitude and direction of all the forces acting on a chosen body. The length of the arrow representing a given force is proportional to the magnitude of the force. Very useful when trying to solve complex questions especially involving $F = ma$ (will be covered very soon)

Equilibrium: $\sum F = 0$, the resultant / net force is zero.

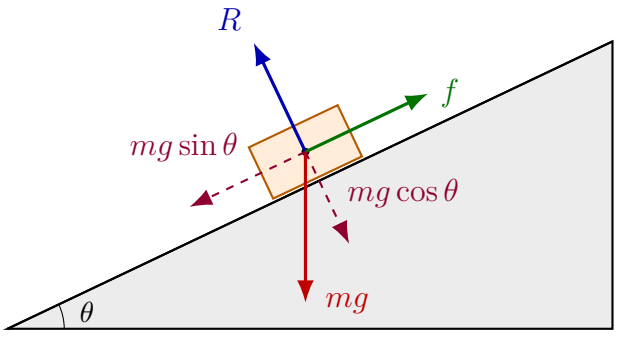
2.2 Newton's laws of motion

1. Law of inertia: When the net force on a body is zero, the body will move with constant velocity (which may be zero); in other words, it will move on a straight line with constant speed (which may be zero). An object will remain at erst or move at constant velocity unless it is acted on by a resultant force.
2. The net force on a body of constant mass is proportional to that body's acceleration and is in the same direction as the acceleration. The resultant force on one object is equal to the rate of change of momentum.
$$\vec{F} = m\vec{a}; F = \frac{\Delta p}{t} = \frac{mv - mu}{t}$$
3. If body X exerts a force on body Y, then body Y will exert an equal and opposite force on body X. When two bodies interact, they exert on each other the forces of the same type, equal in magnitude but opposite in direction simultaneously.

Solving equilibrium problems involving various forces by spliting the forces into F_x and F_y components.
How to solve an $F = ma$ problem:

1. Make a diagram and choose the direction that you will take as positive.
2. Make a free-body diagram. (**Very important!**)
3. Find the net force by looking at the free-body diagram.
4. Set the net force equal to ma.
5. Solve for what is unknown.

Questions involving inclined plane: (assumed with friction)



where $R = mg\cos\theta$, the normal force; $f = mg\sin\theta$, the frictional force. $\mu_s = \tan\theta_{\max}$, where $\theta_{\max}$ is the largest angle of the inclined plane at which we can have equilibrium. $a = g\sin\theta$ (if plane turns frictionless)

2.3 Circular motion

Circular motion requires the presence of a force directed towards the center of the circle.

$$v = \frac{2\pi r}{T} = r\omega = 2\pi r f$$

where $T = \dfrac{2\pi r}{v} = \dfrac{2\pi}{\omega}$ is the period / time taken for one complete revolution.

$$\text{Angular speed} = \omega = \frac{\text{angle swept}}{\text{time taken}} = \frac{\Delta\theta}{\Delta t}$$

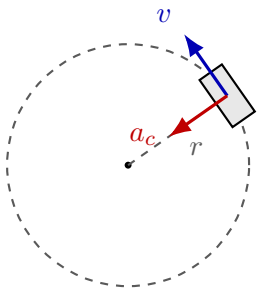
For a complete revolution, $\omega = \dfrac{2\pi}{T} = 2\pi f$.

Frequency, the number of complete revolutions in one second: $f = \dfrac{1}{T}$.

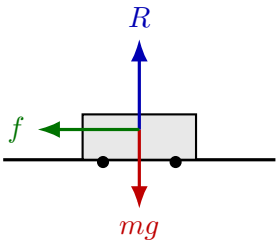
Centripetal acceleration, normal to the circle and directed toward the center of the circle:

$$a_c = v\omega = \frac{v^2}{r} = \omega^2 r = \frac{4\pi^2 r}{T^2} = 4\pi^2 r f^2$$

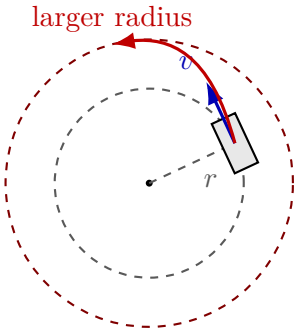
The centripetal force is directed towards the center of the circle (the direction of the net force).
A car will skid outward (i.e., will cover a circle of larger radius) if the friction force is not large enough.



top view

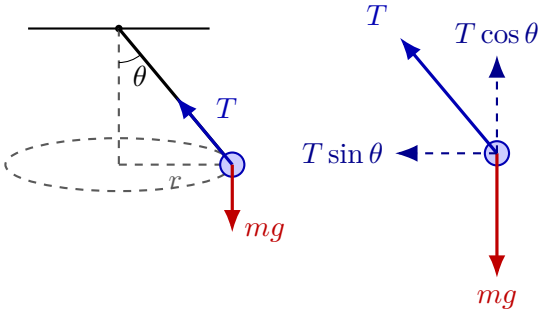


side view



speed too high

Question: A particle is tied to a string and moves with constant speed in a horizontal circle. The string is tied to the ceiling. Draw the forces on the particle.
The common mistake here is to put a horizontal force pointing toward the center and call it the centripetal force. But in fact, it already exists (the horizontal component of the tension).



The steps to solve problems on circular motion:

1. Analyze the state of motion of the studied object (if it is in a state of uniform circular motion).
2. Determine the center and the radius of the circle.
3. Analyze the forces, determine the origin of centripetal force, and calculate the resultant force in the direction to the center of the circle.
4. The resultant force provides the centripetal force of the circular motion.

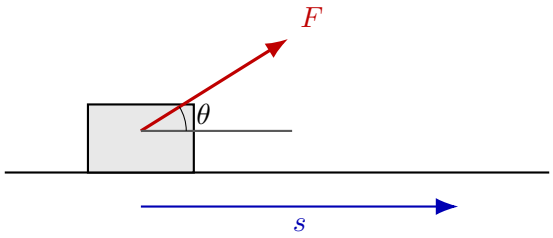
3 Work, energy and power

3.1 Work

Work done by a constant force for motion in straight line:

$$W = Fs\cos\theta = F \times d$$

The distance travelled by the body is s.
The force makes an angle θ with the displacement.
Definition: The work done by a force is the product of the force in the direction of the displacement times the distance travelled.



(向量点积：想起我了没？晚了！　怒气冲天 ing)
Value could be negative. Unit is J / N·m.

Work done by varying force: Area under a curve / integral

Work done in circular motion: work done by centripetal force is 0. (the path followed instantaneously and the direction of the force forms an angle of 90 degrees, and $\cos 90° = 0$)

$$Fs = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

where Fs is the work done by the particle.

$$E_k = \frac{1}{2}mv^2$$

Work-kinetic energy relation: $W_{\text{net}} = \Delta E_k$

The work done has resulted in a transfer of energy that has changed the kinetic energy of the particle.

Work done by gravity: 0 if the displacement is horizontal. ($W = mgs\cos 90° = 0$)
In other words: When a body is displaced such that its final position is at the same vertical height as the original position, the work done by the weight is zero.

$$\text{Work done by weight} = E_{\text{p}} = mg\Delta h$$

(Definition of gravitational potential energy: the work done by the moving force in placing a body a height h above the surface of the earth)
This implies that a body on the surface has zero potential energy.
This formula is approximate and may only be used if the height h is very small compared to the radius of the earth.
Positive if object is lowered, negative if object raised.

The work to extend a spring from its natural length by an amount x is

$$W = \text{Elastic potential energy, } E_H = \frac{1}{2}kx^2 = \frac{1}{2}k(x_2^2 - x_1^2) \text{ (if two values are present)}$$

where x is the extension and k is the spring constant.

3.2 Conservation of energy

Law of conservation of total mechanical energy, $E_T = E_k + E_p(+E_H) = \frac{1}{2}mv^2 + mg\Delta h \left(+\frac{1}{2}kx^2\right)$ only happens when friction and resistance forces are absent and no other forces act on the system from outside the system. In other words, this only happens in an isolated system.
If there are interactions between the system and its surroundings, the total mechanical energy may change. These interactions mainly involve work done W_{ext} by the surroundings and/or the transfer of thermal energy (heat), Q, to or from the surroundings. In this case:

$$\Delta E_T = W_{\text{ext}} + Q$$

3.3 Power and efficiency

Power is the rate at which work is being performed or the rate at which energy is being transferred.

$$P = \frac{\Delta W}{\Delta t} = Fv$$

where F is a constant force acting on the body and v is the instantaneous speed of the body. This is the power produced in making the body move at speed v.

$$\text{Average power developed} = \overline{F} \times \frac{v+u}{2}$$

Efficiency:

$$\eta = \frac{\text{Useful energy / power out}}{\text{Actual power / energy in}}$$

3.4 Energy transfers

The law of conservation of energy states that energy cannot be created or destroyed: it can only be transferred from one form to another.

4 Linear momentum

4.1 Newton's second law in terms of momentum

Linear momentum is defined to be the product of the mass of a body and its velocity: $\vec{p} = m\vec{v}$.
Therefore, Newton's second law can be expressed as:

$$\begin{aligned}
\vec{F_{\text{net}}} &= \frac{\Delta\vec{p}}{\Delta t} \\
&= \frac{\vec{p_F} - \vec{p_I}}{\Delta t} \\
&= \frac{m\vec{v_F} - m\vec{v_I}}{\Delta t} \\
&= m\frac{\Delta\vec{v}}{\Delta t} \\
&= m\vec{a}
\end{aligned}$$

4.2 Impulse and force-time graphs

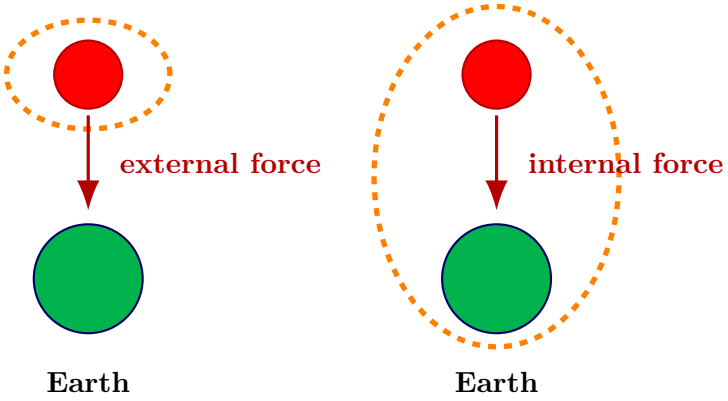
$$\vec{J} = \Delta\vec{p} = \vec{F_{\text{net}}}\Delta t$$

In other words, impulse is the area under the curve in a graph of force against time.

4.3 Conservation of momentum

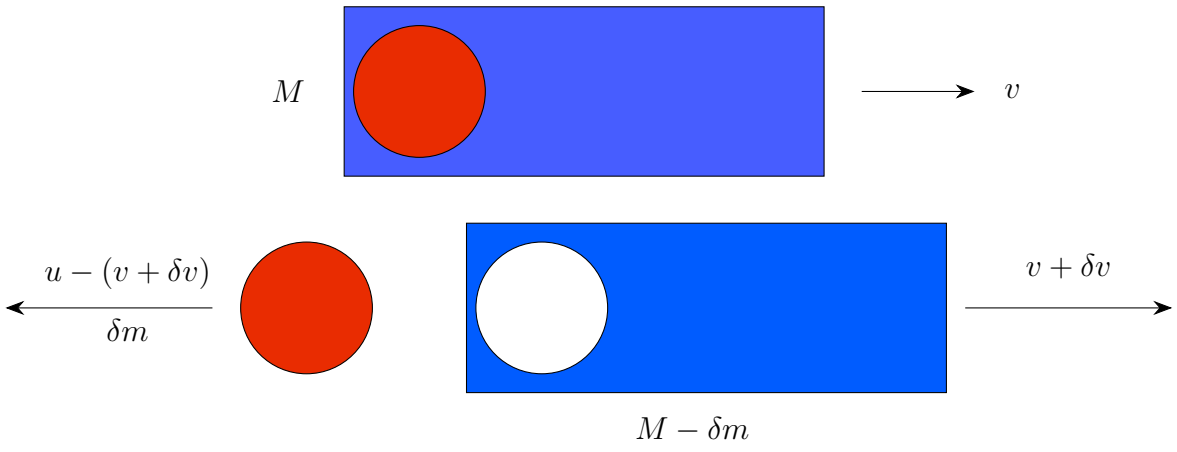
Conservation of momentum: when the net force on a system is zero the momentum does not change and it is conserved. The total momentum of a closed system of colliding objects remains constant in any direction iff no external forces act on the system.

Consider a ball that you drop from rest from a certain height. As the ball falls, its speed and momentum increase so the momentum of the ball does not stay the same. This is to be expected—there is an external force on the ball, namely its weight. So the momentum of the system that consists of just the falling ball is not conserved. If we include the earth as part of the system, then there are no external forces anymore and the total momentum will be conserved.



Extension: rocket equation 火箭方程

The rocket moves with speed v. The engine is turned on, and gases leave the rocket with speed u relative to the rocket. The initial mass of the rocket including the fuel is M. After a short time δt the rocket has ejected fuel of mass δm. The mass of the rocket is, therefore, reduced to $M - \delta m$ and its speed increased to $v + \delta v$.



Applying the law of conservation of momentum gives:

$$
\begin{aligned}
Mv &= (M-\delta m)(v+\delta v) - \delta m(\underbrace{u-v-\delta v}_{\text{speed relative to the ground}}) \\
Mv &= Mv + M\delta v - v\delta m - \delta m\delta v - u\delta m + v\delta m + \delta m\delta v \\
M\delta v &= u\delta m \\
\delta v &= \frac{\delta m}{M}u \\
\lim_{\delta t\to 0}\frac{\delta v}{\delta t} &= \frac{\delta m}{\delta t}\frac{u}{M} \Rightarrow \boxed{M\frac{dv}{dt} = u\frac{dm}{dt}}
\end{aligned}
$$

4.4 Kinetic energy and momentum

We observe that whereas momentum is conserved in all cases, kinetic energy is not.
When kinetic energy is conserved, the collision is said to be an elastic collision. Relative speed of approach is equal to the relative speed of separation in elastic collision.

$$
\begin{cases}
\frac{1}{2}m_a u_a^2 + \frac{1}{2}m_b u_b^2 = \frac{1}{2}m_a v_a^2 + \frac{1}{2}m_b v_b^2; \\
u_a - u_b = v_b - v_a;
\end{cases}
$$

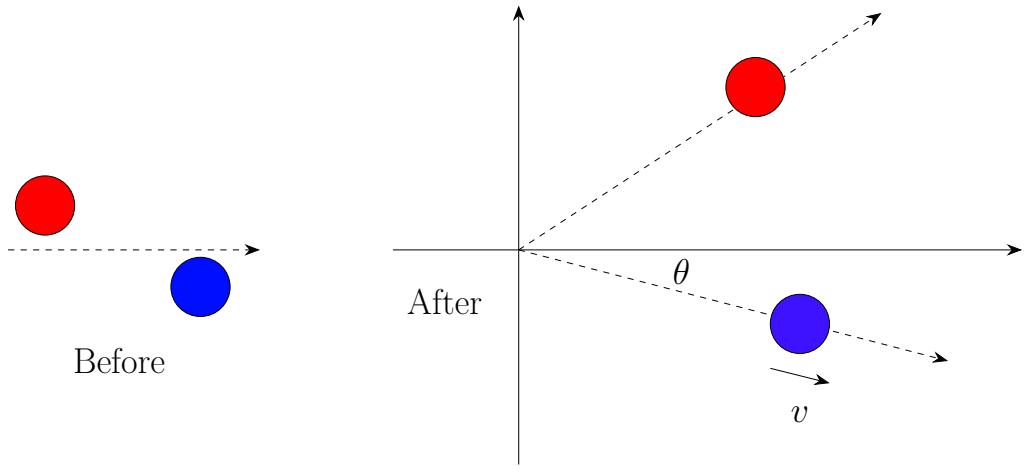
When kinetic energy is not conserved / lost (transformed into e.g., thermal / deformation / sound energy), the collision is said to be an inelastic collision.

$$
\frac{1}{2}m_a u_a^2 + \frac{1}{2}m_b u_b^2 > \frac{1}{2}m_a v_a^2 + \frac{1}{2}m_b v_b^2;
$$

(When the bodies stick together after a collision, the collision is said to be totally inelastic and the maximum possible E_k is lost.)
Kinetic energy could be calculated as:

$$
E_k = \frac{1}{2}mv^2 = \frac{m^2v^2}{2m} = \frac{p^2}{2m}
$$

4.5 Two-dimensional collisions



Key idea:

$$
\begin{aligned}
\sum p_{x_{\text{before}}} &= \sum p_{x_{\text{after}}} \\
\sum p_{y_{\text{before}}} &= \sum p_{y_{\text{after}}}
\end{aligned}
$$

5 Rigid body mechanics

5.1 Kinematics of rotational motion

Use radians at all times!
Consider a particle that rotates around a circle about an axis through the center. The angular position θ of the particle is determined by an angle (defined relative to some arbitrary reference line).
The angular displacement is the change in angular position by sweeping an angle of $\Delta\theta$.
The average angular velocity is defined as

$$\overline{\omega} = \frac{\Delta\theta}{\Delta t}$$

The instantaneous angular velocity is defined as

$$\omega = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t}$$

The average angular acceleration is defined as

$$\overline{\alpha} = \frac{\Delta\omega}{\Delta t}$$

The instantaneous angular acceleration is defined as

$$\alpha = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t}$$

Correspondence between linear and angular quantities:

Linear quantity	Angular quantity
Position s	Angle θ
Linear velocity v	Angular velocity ω
Acceleration a	Angular acceleration α
$\Delta s = ut + \frac{1}{2}at^2$	$\Delta\theta = \omega_1 t + \frac{1}{2}\alpha t^2$
$\Delta s = \frac{u+v}{2}t$	$\Delta\theta = \frac{\omega_1+\omega_2}{2}t$
$v = u + at$	$\omega = \omega_1 + \alpha t$
$v^2 = u^2 + 2a\Delta s$	$\omega^2 = \omega_1^2 + 2\alpha\Delta\theta$

Equations relating linear and angular quantities:

$$v = \omega r;\ a = \alpha r$$

Note that this linear acceleration is the rate of change of speed and is not the centripetal acceleration of circular motion.

5.2 Rotational equilibrium and Newton's second law

Torque has to do with the ability of a force to produce rotation.
Defined as the product of the force times the perpendicular distance between the line of action of the force and the axis:

$$\tau = Fd = Fr\sin\theta$$

where θ is angle between the force and the vector from the axis to the point where the force is applied. Notice that if the force is directed through the axis, the torque is zero. The force cannot produce a rotation in this case.

Translational equilibrium: net force of the body is zero. The center of mass of the body remains at rest or moves in a straight line at constant speed.
Rotational equilibrium: net torque of the body is zero.
Solve equilibrium problems:

1. Satisfying translational equilibrium: forces acting downward must equal to the normal force.
2. Torque for counterclockwise and clockwise direction are equal to each other.
3. You may take torques about any axis, not just the actual axis of rotation.

Moment of inertia:

$$I = \sum m_i r_i^2$$

where m_i are the mass of the "bits" that constitutes the body, and r_i are the distance between the bits and the axis (free to rotate about for this rigid body)
The moment of inertia depends on the axis we choose. The same body will have different moments of inertia for different axes of rotation.

The moment of inertia in rotational motion $\Leftrightarrow$ Mass in linear motion

Torque in rotational motion $\Leftrightarrow$ Force in linear motion

Some special cases:

Shape	Formula of moment of inertia
Ring	$I = MR^2$
Disk / Cylinder	$I = \frac{1}{2}MR^2$
Sphere	$I = \frac{2}{5}MR^2$
Rod	$I = \frac{1}{12}ML^2$

* : $M = \sum m_i$, the total mass; R, radius; L, the length (of the rod)

The kinetic energy of rotational motion:

$$E_k = \sum \frac{1}{2} m_i v_i^2 = \frac{1}{2}\left(\sum m_i r_i^2\right)\omega^2 = \frac{1}{2} I \omega^2$$

Rolling (without slipping)

- Rotates about its axis: All points on the wheel have the same angular velocity and linear speed.
- Sliding with velocity u along a horizontal floor: every point on the wheel has a velocity u.
- Rotates as it moves forward: Every point on the circumference of the wheel now has two motions. The top point has linear velocity of $v = u + \omega R$ and the bottom point has linear velocity of $v = u - \omega R$ (point of contact has zero velocity).

In the case of rolling without slipping, $u = \frac{2\pi R}{T} = \omega R$, where R is the distance from the center of mass to the point of contact with the ground.
If the force is applied

- through the center of mass, the ball will slide but will not rotate (there is no torque to cause rotation)
- anywhere else, it will cause sliding as well as rotation.

Kinetic energy of a body that rolls without slipping:

$$E_k = \underbrace{\left(E_{k_1} = \frac{1}{2}Mv^2\right)}_{\text{translational motion of the center of mass}} + \underbrace{\left(E_{k_2} = \frac{1}{2}I\omega^2\right)}_{\text{rotational motion}}$$

where v is the speed of the center of mass.
Total mechanical energy:

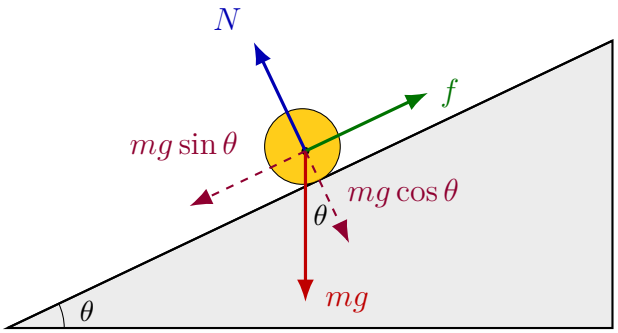
$$E_T = \underbrace{\frac{1}{2}I\omega^2}_{E_k \text{ of rotational motion}} + \underbrace{\frac{1}{2}Mv^2}_{E_k \text{ of translational motion}} + \underbrace{Mgh}_{E_p \text{ of the body}}$$

In the absence of resistance forces the total energy is conserved.
In the case of rolling without slipping, $v = \omega R$, $a = \alpha R$.

Newton's second law applied to the center of the body:

$$F_{\text{net}} = Ma \Leftrightarrow \tau_{\text{net}} = I\alpha$$

Apply these two formulas to the question and find a solution accordingly.
Linear acceleration of an object rolling without slipping down an inclined plane:



For sphere, $a = \frac{5}{7}g\sin\theta$. For other objects, follow the formula $a = \dfrac{g\sin\theta}{1 + \dfrac{I}{MR^2}}$.

For sphere, $v^2 = \frac{10}{7}gh$. For other objects, $v^2 = 2as = \dfrac{2gh}{1 + \dfrac{I}{MR^2}}$.

Work and power

$$W = F\Delta s = FR\Delta\theta = \tau\Delta\theta$$

(A constant force F is applied on a rigid body for Δt and moved the body by a distance of Δs and an angle of $\Delta\theta$)

$$P = \frac{W}{\Delta t} = \frac{\tau\Delta\theta}{\Delta t} = \tau\omega$$

5.3 Angular momentum

(A vector) It is defined as

$$L = I\omega \Rightarrow \tau_{\text{net}} = \frac{\Delta L}{\Delta t}$$

Conservation of angular momentum: When the net torque on a system is zero, the angular momentum is conserved; that is, it stays constant.

$$J = \Delta L = \tau_{\text{net}} \Delta t$$

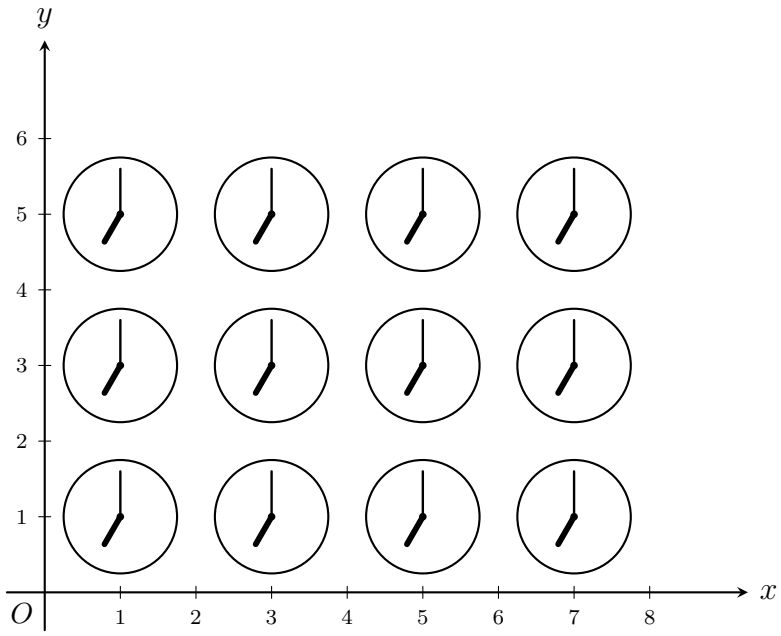
$$E_k = \frac{L^2}{2I}$$

6 Relativity

可以这么说：一切都是相对的，没有一个东西是绝对地“在动”，也没有一个东西是绝对的静止。

6.1 Reference frames and Lorentz transformations

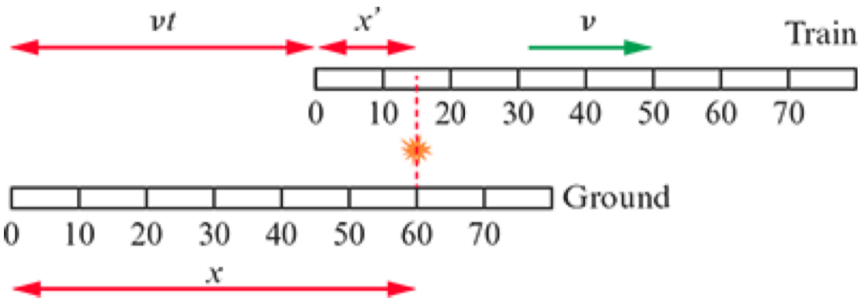
A reference frame is a set of coordinate axes and a set of clocks at every point in space. If this set is not accelerating, the frame is called an inertial reference frame.

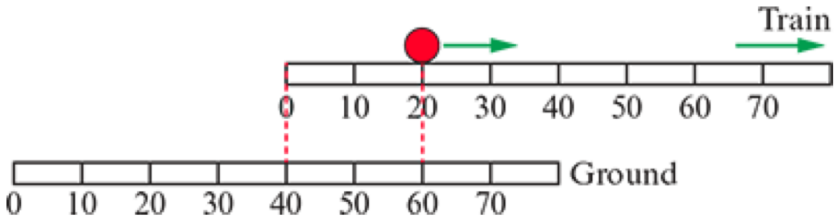


Galilean transformation equations:

$$x' = x - vt;\ t' = t;\ u' = u - v$$

where x and x' denotes the position that the event occurred to two observers.





Two observers always agree on what the time co-ordinates are; in other words, time is common to both observers.

The postulates of special relativity

- All the laws of physics are the same in all inertial frames.
- The speed of light in vacuum is the same for all inertial observers.

Lorentz transformations: (By Einstein, really)

$$x' = \gamma(x - vt);\ x = \gamma(x' + v't)$$

$$v' = \gamma\left(t - \frac{v}{c^2}x\right);\ v = \gamma\left(t' + \frac{v}{c^2}x'\right)$$

where c is the speed of light and γ is the gamma factor, $\gamma = \dfrac{1}{\sqrt{1 - \dfrac{v^2}{c^2}}}$

If v is very small compared to the speed of light, then $\gamma \approx 1$, the Lorentz equations reduce to the Galilean equations. The effects of relativity become important only at speeds close to that of light.

$$\Delta x' = \gamma(\Delta x - v\Delta t);\ \Delta v' = \gamma(\Delta t - \frac{v}{c^2}\Delta x)$$

$$\Delta x = \gamma(\Delta x' + v\Delta t');\ \Delta v = \gamma(\Delta t' + \frac{v}{c^2}\Delta x')$$

Spacetime interval

A quantity that both observers will agree on. Mathematically speaking,

$$(\Delta s)^2 = (\Delta s')^2 \Rightarrow (c\Delta t)^2 - (\Delta x)^2 = (c\Delta t')^2 - (\Delta x')^2$$

6.2 Effects of relativity

Time dilation

the interval between the ticks of a clock that moves relative to the observer in S is greater than the interval measured in S' where the clock is at rest.
When $\Delta x' = 0$, $\Delta t = \gamma\Delta t'$. In a straightforward way,

$$\text{time interval} = \gamma \times \text{proper time interval}$$

where proper time interval is the time between two events that take place at the same point in space.

Length contraction

$$L = \frac{L_0}{\gamma}$$

Note that it is only lengths in the direction of motion that are contracted.
L_0 is the proper length, the length of the rod at rest in the rest frame, frame S', $\Delta x'$.
L is the length of the rod in frame S, Δx.

Addition of velocities

Consider a frame S' that moves at constant speed v in a straight line relative to another frame S. Observers in both frames measure the velocity of an object. Observers in S' measure the velocity to be u'. Observers in S measure the velocity of the same object to be u. How are u and u' related?

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}};\ u' = \frac{u - v}{1 - \frac{uv}{c^2}}$$

Simultaneity

Suppose that as usual we have frames S and S' with frame S' moving to the right with speed v relative to S. Suppose that two events take place at the same time in frame S', $\Delta t' = 0$. The time interval between the same two events when measured in S is:

$$\Delta t = \gamma \frac{v}{c^2} \Delta x'$$

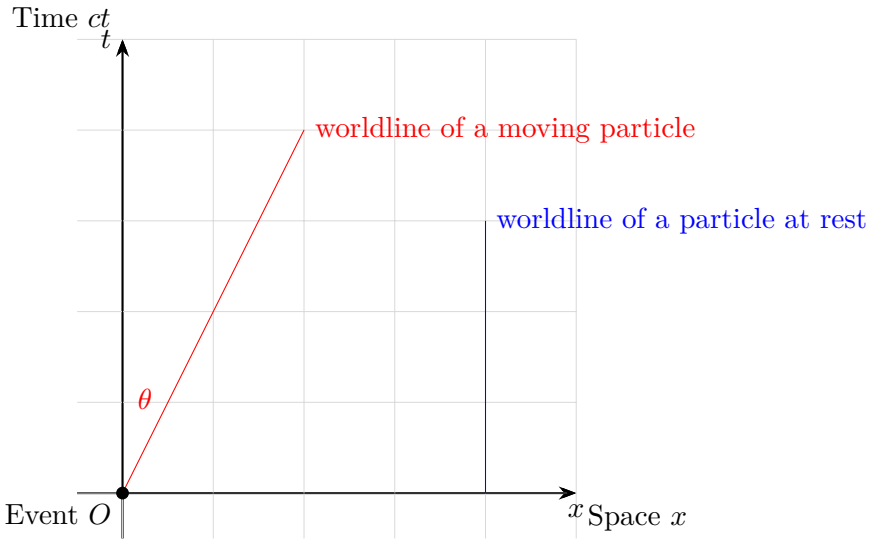
- Events that are simultaneous for one observer and which take place at different points in space, are not simultaneous for another observer in motion relative to the first. $\Delta x' \neq 0$
- If two events are simultaneous for one observer and take place at the same point in space, they are simultaneous for all other observers as well. $\Delta x' = 0$

Muon decays

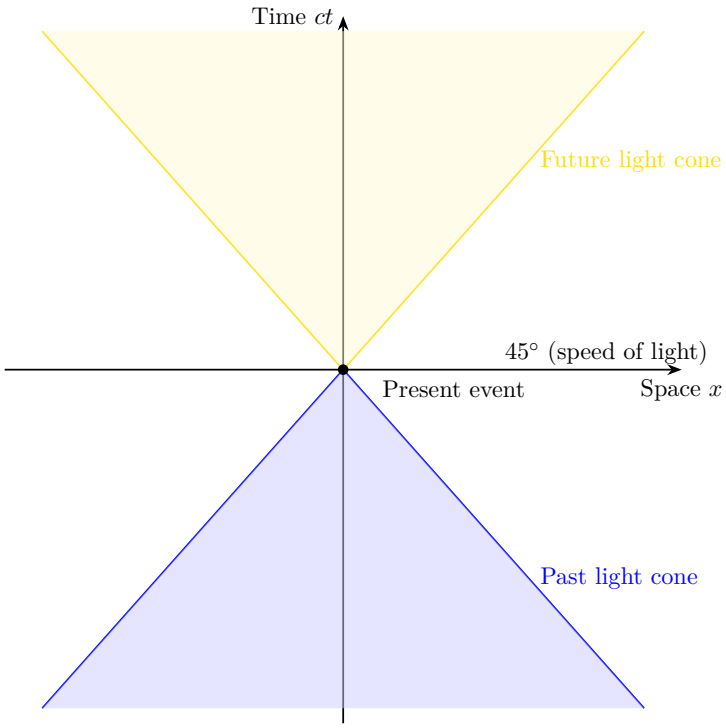
Muons are particles with properties similar to those of the electron except that they are more massive, are unstable and decay into electrons. They have an average lifetime of 2.2×10^{-6} s, and is the proper time interval between its creation and its decay.
From the point of view of an observer in the laboratory, however, the muons are moving at high speed; the lifetime is longer because of time dilation.
想象这个 μ 子从一个（3 公里高的）山上发射下来（比如说他有 $0.99c$ 的速度），地上有一个接收器。在相对论的影响下，对于观测者而言时间被减慢了（$\Delta t = \gamma \times 2.2 \times 10^{-6}$ s $= 1.56 \times 10^{-5}$ s），对于观测者而言穿越的路程增加了（4.63 km），因此 μ 子可以在被接收器接收之后再衰变。没有相对论的影响 μ 子就只能走 0.653 公里，显然不足以支撑它坚持到检测器的检测。
对于一个在 μ 子上的观测者（我们就假设真的有这样一个观测者），是地球在向上移动，因此这个山的距离对于他只有 0.42 公里，需要 1.4×10^{-6} 秒的时间遇上检测器。所以说，In this sense, muon decay experiments are direct confirmations of time dilation and indirect confirmations of length contraction.

6.3 Spacetime / Minkowski diagrams

Do not be led astray by your knowledge of Euclidean geometry!
Time is the y-axis (plotted as ct though, in meters), distance (1D) is the x-axis.



The tangent of the angle between the worldline and the ct axis gives the speed divided by the speed of light. Since nothing can exceed c, the worldline of any particle will always make an angle less than 45° with the ct axis.

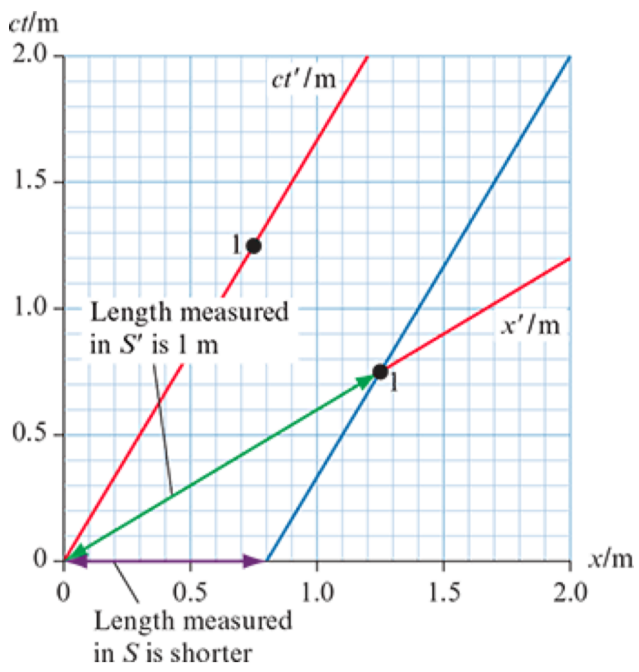


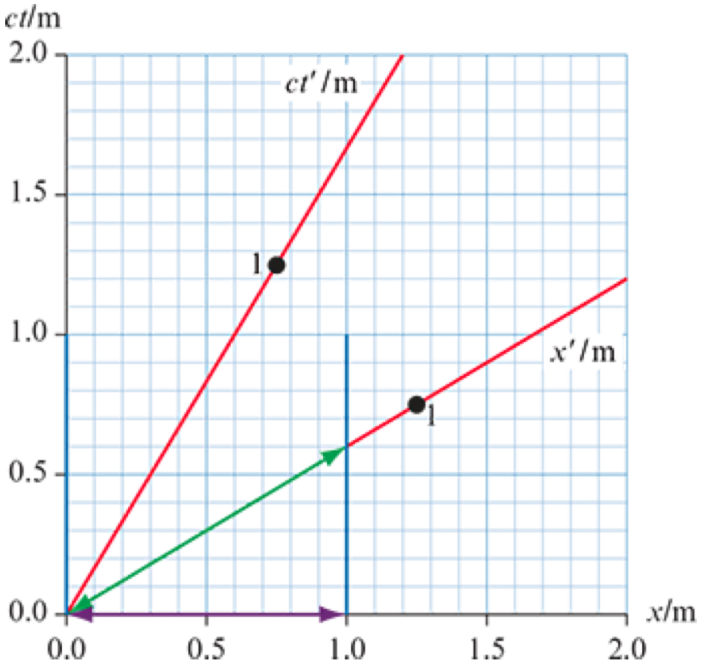
The region coloured yellow consists of all events that can be influenced by event E. Similarly, events that could have influenced E are within the region coloured blue.

重要概念：其实坐标轴可以是斜的。想象有一个 S' 以 v 的速度向右移动，且 S 和 S' 的原点重合。ct' 的坐标轴会和 ct 坐标轴形成 $\theta = \tan^{-1}\left(\frac{v}{c}\right)$ 的夹角。同样地，x' 的坐标轴会和 x 坐标轴形成 θ 的夹角。所有坐标轴均对称于光子的世界线（因为光速在所有的 Frames 里的速度都是一致的）。

Use the gamma factor and Lorentz transformation to find x and ct from x' and ct' values. ($x = \gamma(x' + vt')$, $ct = \gamma(ct' + vcx')$) 当 x 和 ct 等于伽马系数的时候，向右/向上延伸与 ct' 和 x' 相交的点就是 S' 下的 1m。

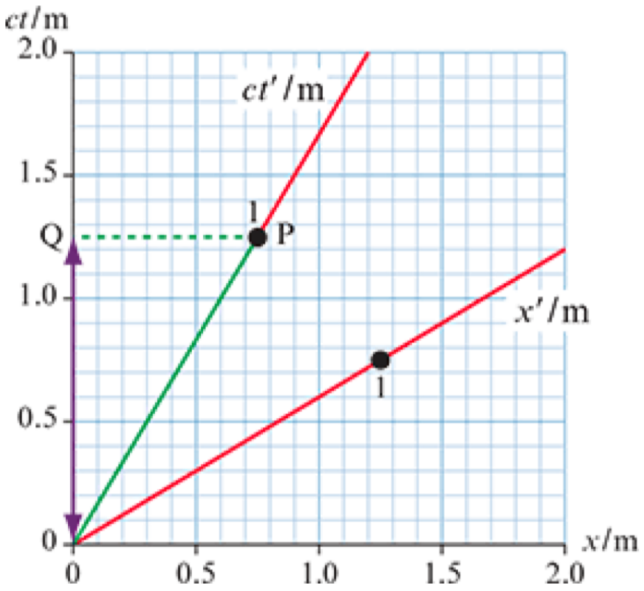
Length contraction

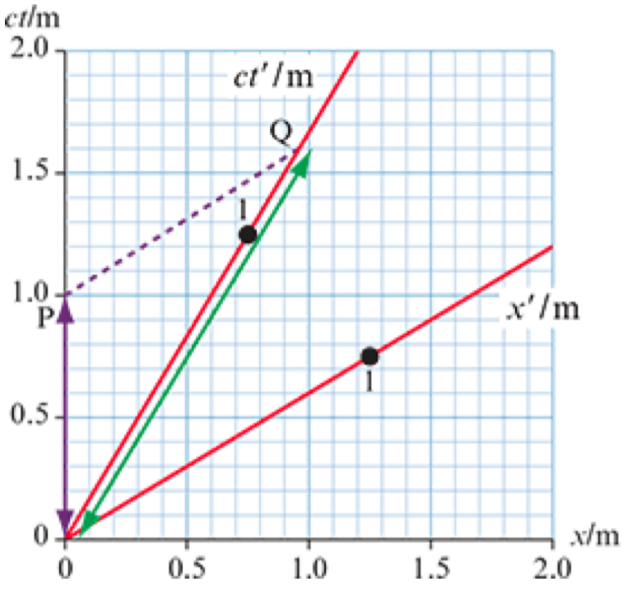




If the rod is moving its length must be measured when the position of its ends are recorded at the same time. It is only lengths in the direction of the velocity that are contracted.

Time dilation

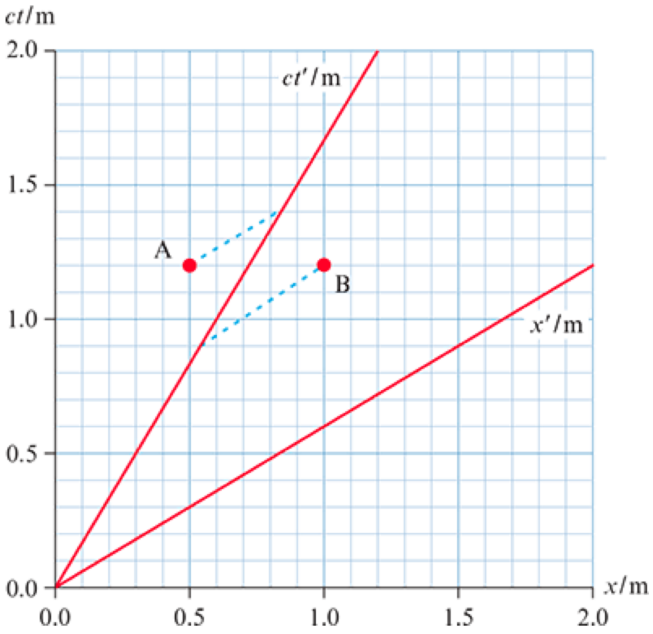




* : PQ in the second diagram is parallel to the x' axis.

Simultaneity

Events happen simultaneously in an arbitrary frame S if the ct value are the same for both events (lines on the same line parallel to the x axis).



To find out how much earlier an event (B for example) occurs use a Lorentz transformation:

$$c\Delta t' = \gamma(c\Delta t - \frac{v}{c}\Delta x)$$

This is really just a transformed form of this equation:

$$\Delta t = \gamma\frac{v}{c^2}\Delta x'$$

Unit B The particulate nature of matter

7 Thermal energy transfers

7.1 Particles, temperature and energy

States of matter: solid, liquid, gas.

Property	Solid	Liquid	Gas
Forces between particles	Strong	Weaker than solids, but still significant	Very weak (negligible except during collisions)
Particle motion	Fixed; Very difficult to break apart	Move around each other; flows, takes container shape	Free motion; forces only during collisions
Compressibility	Very nearly incompressible	Very nearly incompressible	Highly compressible
Interparticle distance	Small (reference)	Similar to solids	About 10× larger than in solids/liquids
Typical density ($\mathrm{kg\,m^{-3}}$)	7.8×10^3 (iron)	6.9×10^3 (liquid iron)	1.2 (air, depends on T and P)
Density comparison	Comparable to liquids	Comparable to solids	∼1000× smaller
Intermolecular potential energy	Large negative (deep potential well);	Negative but shallower than solids;	Approximately zero (negligible interactions)

Formula for calculating density: $\rho = \dfrac{m}{V}$.
Temperature (in kelvin) is proportional to the average random kinetic energy of the molecules.

$$\overline{E}_K = \frac{3}{2} k_B T$$

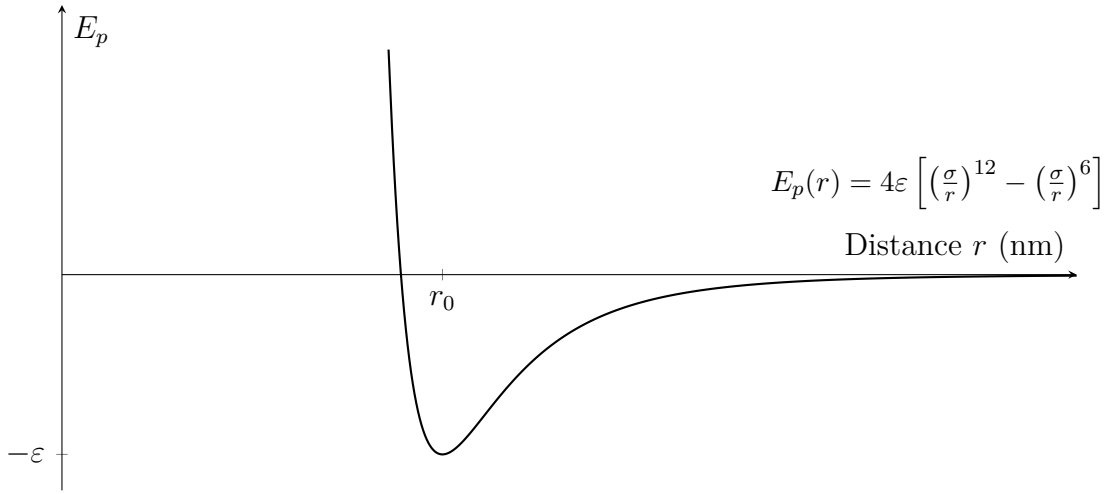
where $k_B = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$, the Boltzmann constant. This direct proportionality between temperature and the average random kinetic energy is only true for the absolute or kelvin temperature scale.

$$T\ (\text{in K}) = T\ (\text{in } ^\circ C) + 273$$

Thermal energy is the energy that gets transferred from one body to another as a result of a temperature difference. When two bodies are in contact and have different temperatures, there will be a transfer of thermal energy from the hotter to the colder body.

Internal energy of a substance is defined as the total random kinetic energy of the particles of a substance, plus the total intermolecular potential energy of the particles. The internal energy of a system can change as a result of thermal energy added or taken out but also as a result of work performed.

We need to do work to increase the separation of the particles in a solid. This work goes into intermolecular potential energy, E_p. The average separation of the molecules corresponds to the minimum of the curve, about 1.2 nm for this graph. For separations larger than 1.2 nm the forces between molecules are attractive. For smaller separations, the forces are repulsive, indicating that the molecules cannot get that close to each other. For gas, the potential energy is 0.



7.2 Specific heat capacity and change of phase

Specific heat capacity: the energy required to increase the temperature of a unit mass of the body by one kelvin.

$$Q = mc\Delta t$$

where Q is the amount of heat needed, m is the mass, Δt is the change in temperature.

Changes of phases:

- melting: when a solid changes to a liquid (thermal energy is transferred to the solid)
- freezing: when a liquid changes into a solid (thermal energy is transferred away from the liquid)
- vaporisation (or boiling): when a liquid changes into vapour (thermal energy is transferred to the liquid)
- condensation: when a vapour changes into a liquid (thermal energy is transferred away from the vapour).
- evaporation: vaporisation taking place at the surface of a liquid at any temperature (rate increases with higher surface area and higher temperature)

Latent heat: The energy supplied or removed during a phase change (does not increase / decrease the temperature of the substance)
Specific latent heat: the energy required to change the phase of 1 kg of a substance at constant temperature and pressure.

$$Q = mL$$

where m is the mass, Q is the energy applied, L_F is the **specific latent heat of fusion** (for melting / freezing), L_V is the **specific latent heat of vaporization** (for vaporization / condensation)
Generally speaking, $L_V > L_F$.

What happens during change in phase:

- The thermal energy supplied increases the intermolecular potential energy, internal energy increases
- This work increases the separation of the molecules
- The average distance between the molecules increases
- Change in phase occurred

The transfer of thermal energy between (melted) water and ice ensures that the overall equilibrium temperature stays at 0°C.

Two methods for measuring specific heat capacity:

- Electrical method
  1. Record initial temperature of the substance
  2. Apply heat from electrical heater (with known power / voltage + current) for a certain amount of time
  3. Measure the final temperature, calculate specific heat capacity
- Method of mixtures
  1. A solid is put in a container of hot water and allowed time to reach a constant temperature
  2. The temperature of the solid is recorded (equal to temperature of water)
  3. The solid is then transferred into an insulated calorimeter 热量计 of known specific heat capacity and initial temperature which contains a liquid
  4. Final temperature of water is then recorded (when thermal equilibrium reached)
  5. $$\underbrace{Q}_{\text{substance with unknown } c_{sp}} = \underbrace{Q}_{\text{calorimeter}} + \underbrace{Q}_{\text{liquid (water)}}$$

     L_F/L_V could also be calculated this way:

     $$\underbrace{Q}_{\text{latent heat}} + \underbrace{Q}_{\text{heat for change in temperature}} = \underbrace{Q}_{\text{calorimeter}} + \underbrace{Q}_{\text{liquid (water)}}$$
  6. (It is likely that some thermal energy (heat) was dissipated to the surrounding air, therefore the real c_{sp} might be larger than the calculated value)

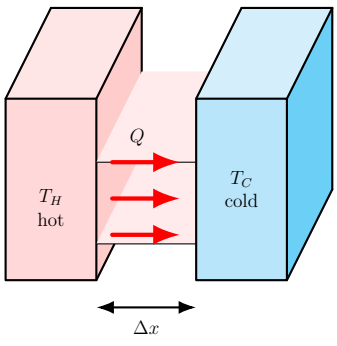
The rate of cooling is proportional to the difference in temperature between the liquid and the surroundings:

$$\frac{\Delta T}{\Delta t} = k(T - T_s)$$

where the constant k is proportional to the area of the surface of the liquid.
(This implies an exponential decrease of temperature)

7.3 Thermal energy transfer

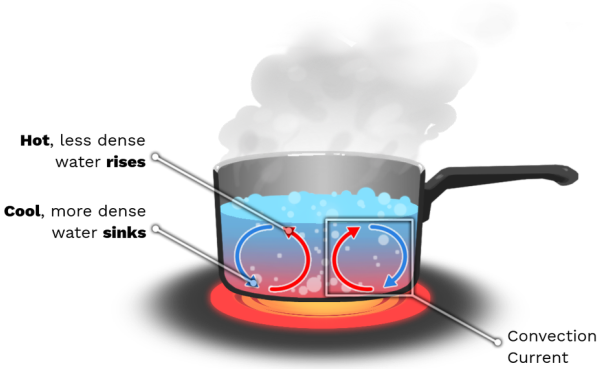
- Conduction
  - The moving electrons collide with neighbouring molecules, transferring energy to them and so increasing their average kinetic energy.



  - Molecules on the hot side of the solid vibrate about their equilibrium positions, stretching the bonds with neighbouring molecules. This stretching forces the neighbours to also begin to vibrate, and so the average kinetic energy of the neighbours increases. Energy is again transferred.
  - $$\frac{\Delta Q}{\Delta t} = kA\frac{\Delta T}{\Delta x}$$

    where k is the conductivity, A is the cross-sectional area, Δx is the width, ΔT is the temperature difference, $\Delta T = T_H - T_C$.

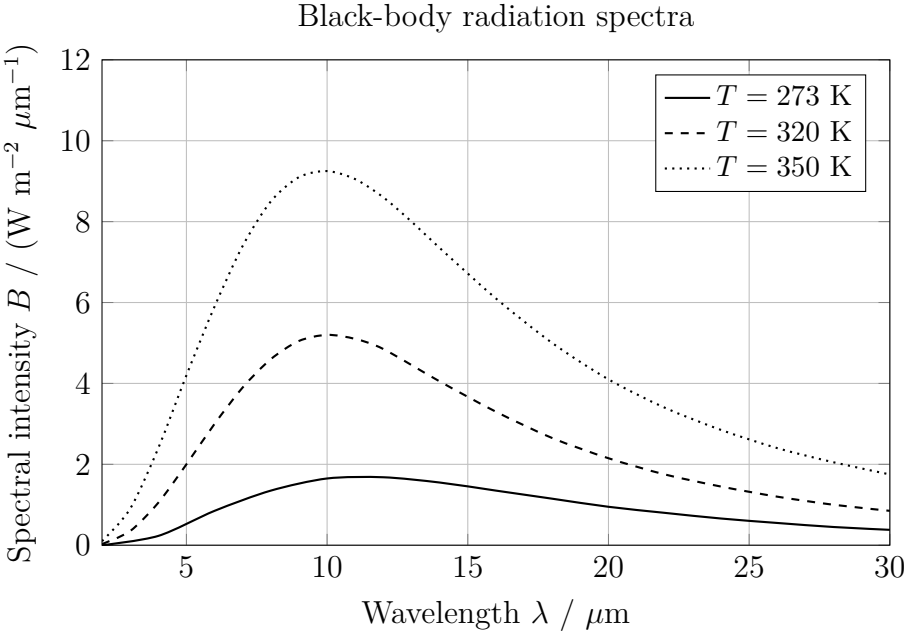
- Convection
  - Applies to fluids (liquids and gases)
  - Hotter liquids are less dense / hotter gases would expand and rise, colder fluids are more dense and sink
  - This process preforms continuously to form the convection currents



- Radiation
  - Does not require a medium for the transfer of thermal energy – can occur in vacuum
  - Radiate energy in forms of EM waves, radiation created by oscillating electric charges in the atoms of the body.
  - Stefan-Boltzmann Law: The power radiated by a black body is given by

    $$P = \sigma A T^4$$

    where A is the surface area, T is the absolute temperature of the surface, σ is the Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$
  - The radiated intensity (power radiated per unit area of the body) is thus $\frac{P}{A} = \sigma T^4$
  - Strictly, this applies to a theoretical body called a black body. This is a perfect radiator as well as a perfect absorber. A black body at low temperature radiates very little, and since it absorbs all the radiation falling on it, it looks black. At high temperature it radiates a lot and looks very bright.



* : Spectral intensity: power emitted per unit area per unit wavelength

The area under the graph is the radiated intensity. With increasing temperature, the peak of the curve occurs at lower wavelengths and the height of the peak increases. The energy radiated is electromagnetic radiation and is distributed over an infinite range of wavelengths. The peak of the spectrum corresponds to a specific wavelength $\lambda_{\max}$, called the peak wavelength.

Wien's Law:

$$\lambda_{\max} T = 2.90 \times 10^{-3}\ \mathrm{K\,m}$$

This shows that the peak wavelength λ_0 depends on temperature.

Apparent brightness (the power received per unit area of the detector):

$$B = \frac{L}{4\pi d^2} = \frac{\sigma A T^4}{4\pi d^2}$$

where L is the luminosity of the star, d is the distance from the star to the observer.
Luminosity of a star is given by

$$L = \sigma A T^4$$

where A is the surface area of the star, T is the temperature of the star.

8 The greenhouse effect

8.1 Radiation from real bodies

Real bodies radiate according to the modified Stefan-Boltzmann law:

$$P = e\sigma A T^4$$

where e is the emissivity of the body, a number between 0 and 1, the ratio of the power per unit area radiated by a body to the power per unit area radiated by a black body of the same temperature. A perfect black body has an emissivity of 1, while a perfect reflector has an emissivity of 0.
The power radiated per unit area is called (radiated) intensity, and is equal to $I = e\sigma T^4$.

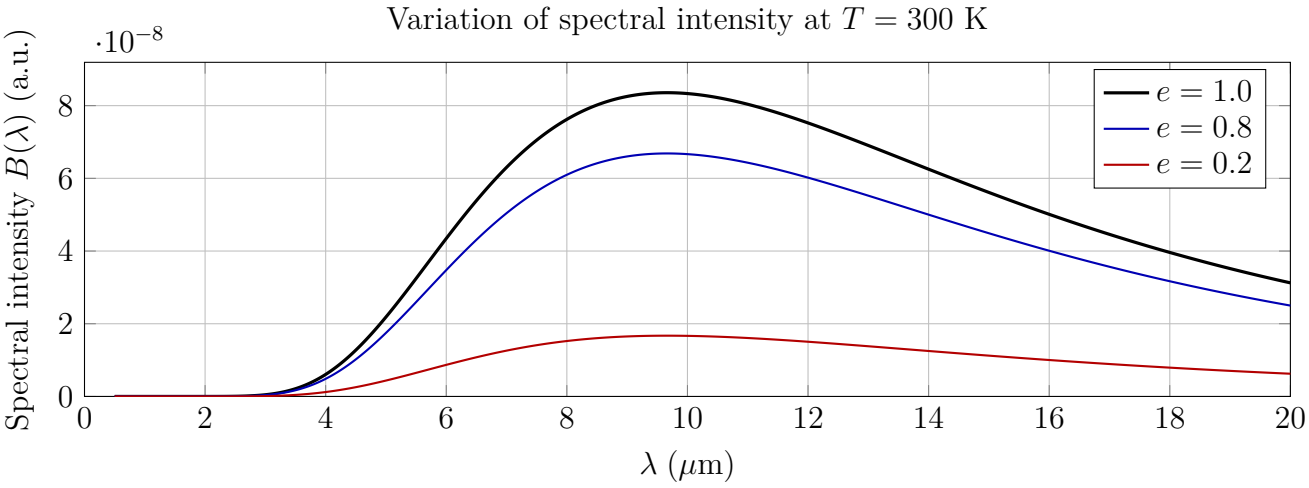
Albedo of the body is the ratio of the reflected intensity (or power) to the incident intensity (or power).

$$\alpha = \frac{\text{total reflected power}}{\text{total incident power}}$$

For a body that can only absorb or reflect, $e + \alpha = 1$.
If considering the temperature of the surrondings, T_S:

$$P_{\text{net}} = e\sigma A(T^4 - T_S^4)$$

$$I_{\text{net}} = e\sigma(T^4 - T_S^4)$$



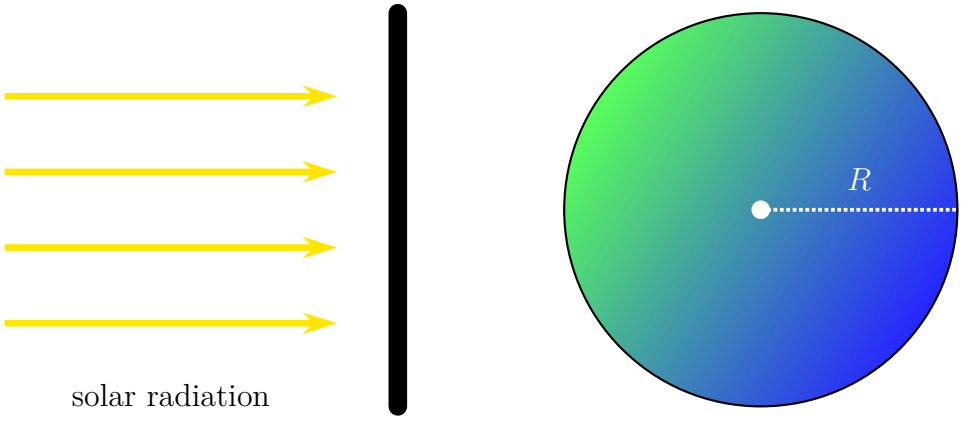
The solar constant:

$$S \approx 1400 \text{ W m}^{-2}$$

(calculated using $I = \frac{P}{4\pi d^2}$, where I is intensity, power of radiation received or emitted per unit area; $P = 3.9 \times 10^{26}$ W (total power the sun emitted), $d = 1.5 \times 10^{11}$ m $= 1$ au, average earth-sun distance)
It is the intensity of the solar radiation at the top of the earth's atmosphere.
The intensity emitted by the sun is

$$I = \frac{3.9 \times 10^{26} \text{ W}}{4\pi(7.0 \times 10^8 \text{ m})^2} = 6.3 \times 10^7 \text{ W m}^{-2}$$

8.2 Energy balance of the earth



disc of area $4\pi R^2$ surface area of the sphere $4\pi R^2$

The radiation that reaches the earth has to go through the area of a disc of radius R (the Earth's radius). The power going through the disc is given by

$$P = S\pi R^2 (= SA)$$

where S is the solar constant, A is the area of the disc.
The average intensity reaching the Earth's surface is

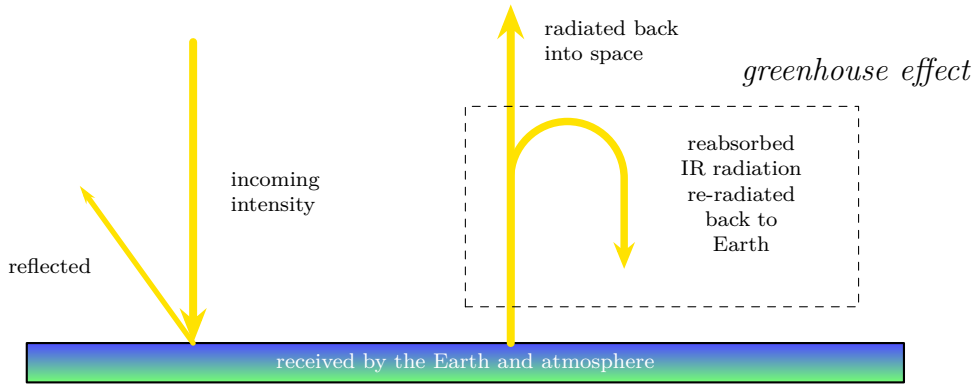
$$I_{\text{av}} = (1-\alpha)\frac{S}{4} = 245 \text{ W m}^{-2}$$

where $\alpha \approx 0.30$ (albedo of the earth varies and depends on (for example) the latitude and whether covered by clouds or not, here 0.3 is an average), and $\frac{S}{4}$ is found by diving the the power through the disc by the surface area of the earth.

Energy balance equation: (very simple, not considering greenhouse effect)

$$(1-\alpha)\frac{S}{4} = \sigma T^4$$

Greenhouse gases



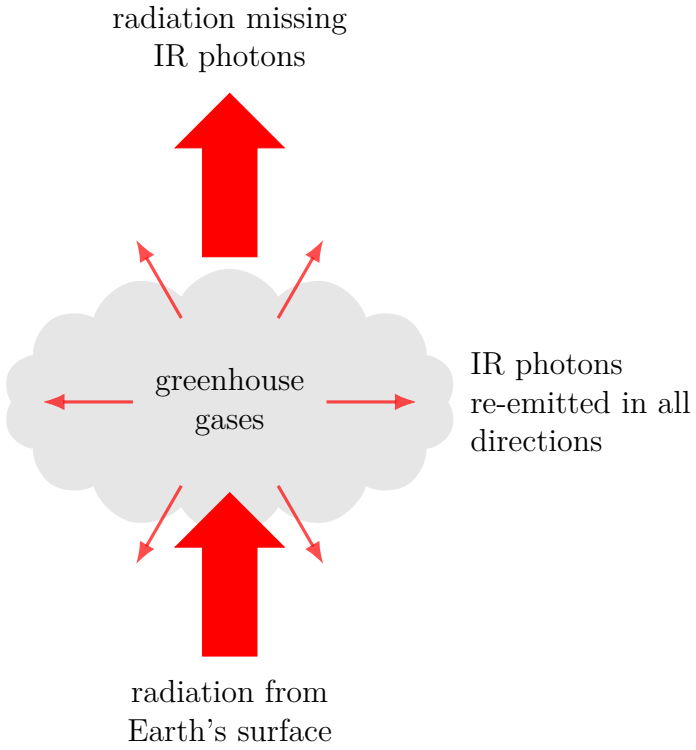
Earth's surface

A thorough process of how did the greenhouse effect occur:

1. Infrared radiation emitted from the Earth's surface is strongly absorbed by greenhouse gases in the atmosphere
2. This radiation is in turn re-radiated by these gases in all directions.
3. This means that some of this radiation is received by the earth's surface again, causing additional warming.

Enhanced greenhouse effect: additional warming due to increased quantities of the greenhouse gases in the atmosphere. The increases in the gas concentrations are due to human activity and are closely related to the burning of fossil fuels.

Greenhouse gas	Natural sources	Anthropogenic sources
H_2O	evaporation of water from oceans, rivers and lakes	irrigation
CO_2	forest fires, volcanic eruptions, evaporation of water from oceans	burning of fossil fuels, deforestation
CH_4	wetlands, ocean, lakes, rivers, termites	agriculture (rice cultivation), livestock farming, landfill sites
N_2O	oceans, forests, soils, tropical forests	agriculture (fertilizer use), burning of fossil fuels, munufacture of cement, deforestation (reduction of nitrogen fixation)



The energy of the molecules are **discrete**. The difference in energy between molecular energy levels is the energy of an infrared photon.

- Infrared photons travelling through greenhouse gases will be absorbed.
- The gas molecules that have absorbed the photons will now be excited to higher energy levels.
- But the molecules prefer to be in low-energy states, and so they will immediately make a transition to a lower-energy state by emitting the photons they absorbed.
- The emitted photons have the same energy as the absorbed photons, but they are emitted in random directions.
- Some are emitted back towards the earth, thereby warming the earth's surface.

OR

- **Resonance:** All systems have their own natural frequency of oscillation, and resonance happens when the frequency of the incoming radiation matches the natural frequency of the system.

- In the case of the greenhouse gases, the natural frequency is the frequency of the photons the greenhouse gases can absorb.
- Resonance is achieved when the photons emitted by the earth's surface have the same frequency as the natural frequency. (i.e., the energy of the photons emitted by the earth's surface is the same as the energy difference between the molecular energy levels of the greenhouse gases)
- Photons with the right energy (frequency) will be absorbed by the greenhouse gases, and then re-emitted in random directions, some of which will be back towards the earth's surface, causing additional warming.

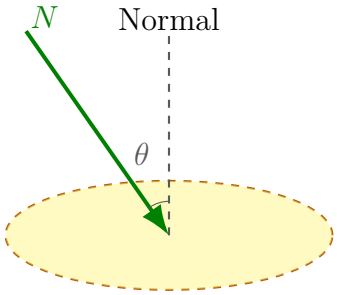
9 The gas laws

9.1 Moles, molar mass and the Avogadro constant

I assume we have some IG chemistry memory coming back?
1 mole of any substance contains as many particles (atoms / molecules, depends) as atoms in 12 g of carbon-12.
1 mol $= 6.02 \times 10^{23}$ particles
This is also known as the Avogadro constant, N_A.
$n = \dfrac{N}{N_A}$ if n is the number of moles, N is the number of particles.
The molar mass of a substance is the mass (in grams) of 1 mole of that substance.
${}^{A}_{Z}X$ has approximately a molar mass of A g/mol.
$n = \dfrac{m}{\mu}$, where μ is the molar mass, m is the mass of the substance, n is the number of moles.

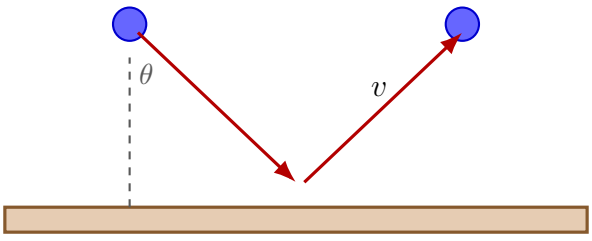
9.2 Ideal gases

Pressure: the normal force applied per unit area. Commonly known as $P = \dfrac{F}{A}$, but as shown in the diagram below, the formula should be adjusted as $P = \dfrac{F\cos\theta}{A}$.



1 atm $\approx 1.01 \times 10^5$ Pa

Lydia 丝滑连招解释压强（哎呀 hào sí huà yā）：



- $|\Delta\vec{p}| = 2mv\cos\theta$
- This change of momentum means that there is a force acting on the wall in a short amount of time
- By Newton's third law, therefore, the molecule exerted on the wall an equal and opposite force.
- Taking into account the forces due to all the molecules colliding with the walls results in a force.

Ideal gas

- The molecules are point particles, each with negligible volume.
- The molecules obey the laws of mechanics.
- There are no forces between the molecules except when the molecules collide.
- The duration of a collision is negligible compared to the time between collisions.
- The collisions of the molecules with each other and with the container walls are elastic.
- Molecules have a range of speeds and move randomly.
- They can't be liquefied / solidified.

A real gas may be approximated by an ideal gas at low density which is equivalent to low pressure and moderate temperature.
The state of a gas is linked by P, V, T, and n (number of moles).

How the ideal gas law is derived: (三巨头吓哭了)

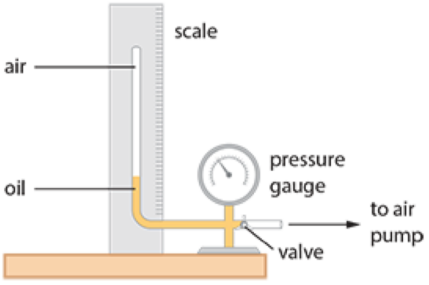
- Boyle's law: $PV =$ constant (at constant T and n)
- Charles's law: $\dfrac{V}{T} =$ constant (at constant P and n)
- Gay-Lussac's law: $\dfrac{P}{T} =$ constant (at constant V and n)

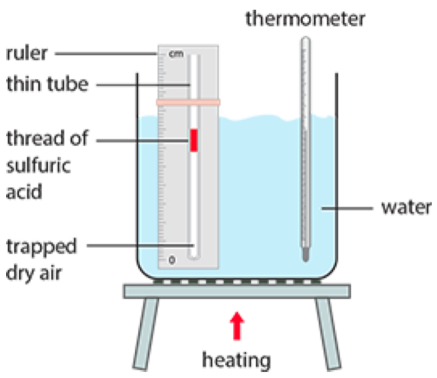
Combining these three laws gives us the ideal gas law:

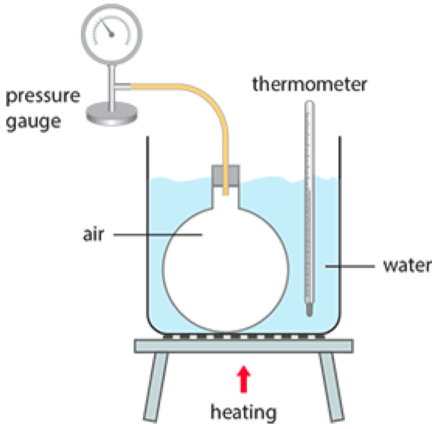
$$PV = nRT = k_B NT$$

* : Note that n is the number of moles, N is the number of particles, R is the ideal gas constant, $R = 8.31\ \mathrm{J\,mol^{-1}\,K^{-1}}$, and k_B is the Boltzmann constant, $k_B = \dfrac{R}{N_A} = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$.

In exam questions, use the equality $\dfrac{P_1V_1}{n_1T_1} = \dfrac{P_2V_2}{n_2T_2}$.

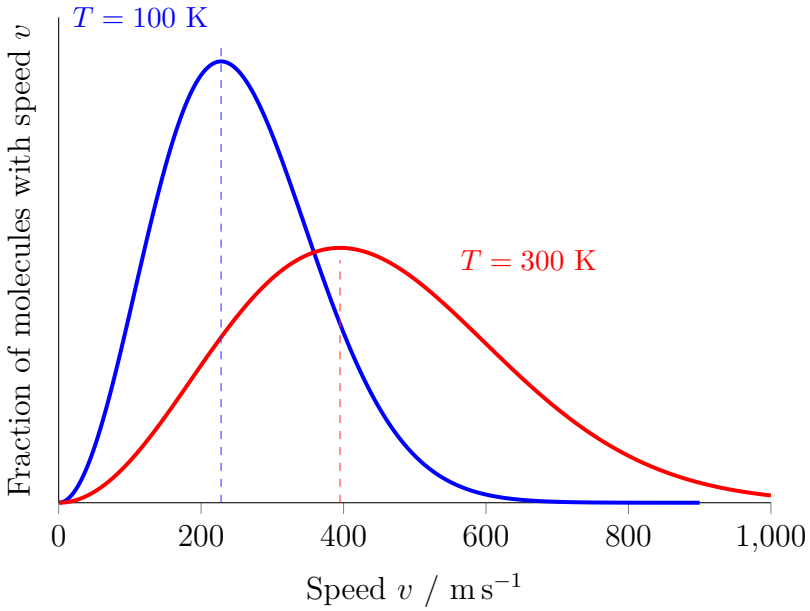






9.3 The Boltzmann equation

The molecules of a gas move about randomly with a range of speeds. The speed at the peak represents the most probable speed that would be found if you picked a molecule at random.



The root mean square speed of the molecules is given by

$$c = \sqrt{\frac{\sum_{i=1}^{n} v_i^2}{N}}$$

The average kinetic energy of the molecules is given by

$$\overline{E}_k = \frac{1}{2}mc^2$$

where m is the mass of a molecule, c is the root mean square speed of the molecules.
Note that we will use c to represent the root mean square speed, most probable speed, and the average speed.
The equation for the pressure of the gas is given by

$$P = \frac{1}{3}\rho c^2$$

下面是推导过程（考试不需要）：
$\vec{J} = 2mv_x$ (a molecule with mass m and velocity v_x collides with the wall and rebounds)
$T = 2\dfrac{L}{v}$ (time between collisions on the right side of the wall, assume the length of the container is L)
$\dfrac{J}{T} = \dfrac{2mv_x}{2\frac{L}{v}} = \dfrac{mv_x^2}{L}$ (average force exerted by the molecule on the wall over large amount of time)
$P_{\text{molecule}} = \dfrac{mv_x^2}{\frac{L^3}{N}} = \dfrac{mv_x^2}{V}$ (pressure exerted by the molecule on the wall, container is a cube)
$P_{\text{total}} = \dfrac{m}{V}\sum_{i=1}^{N} v_{x,i}^2$ (pressure exerted by all the molecules on the wall)
$\overline{v_x^2} = \dfrac{\sum_{i=1}^{N} v_{x,i}^2}{N}$ (average of v_x^2) $\Rightarrow P = \dfrac{mN}{V}\overline{v_x^2}$
$\because \overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2}$ and $\overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2} = c^2 \therefore \overline{v_x^2} = \dfrac{c^2}{3} \Rightarrow P = \dfrac{1}{3}\dfrac{mN}{V}c^2 = \dfrac{1}{3}\rho c^2$

The average kinetic energy of the molecules is also given by

$$\overline{E}_k = \frac{3}{2}k_B T$$

where k_B is the Boltzmann constant, T is the absolute temperature of the gas.
This exceptional equation relates a microscopic quantity, the average kinetic energy of molecules, to a macroscopic quantity, temperature. It also says that at $T = 0$, all molecular motion stops.

The total internal energy of an ideal gas (i.e. total random kinetic energy) is given by

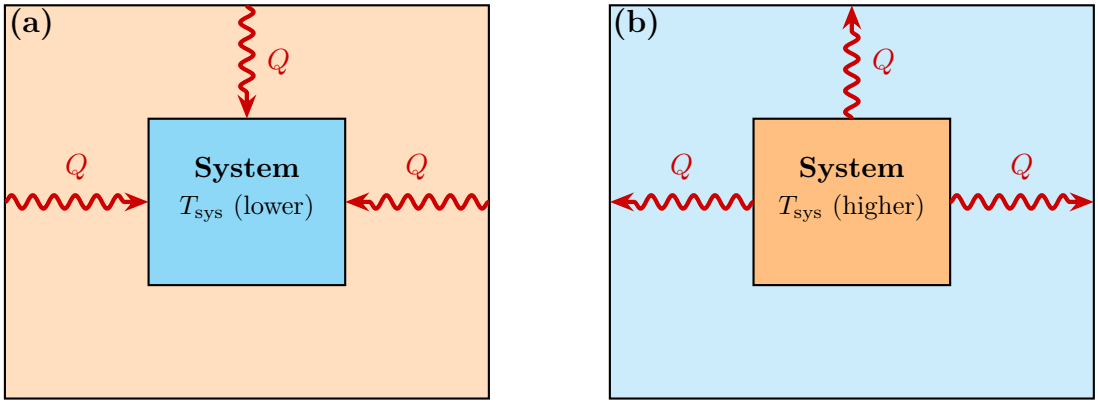
$$U = \frac{3}{2}Nk_B T = \frac{3}{2}nRT = \frac{3}{2}PV$$

10 Thermodynamics

10.1 Internal energy

The change in internal energy: (ideal gas)

$$\Delta U = \frac{3}{2}Nk_B \Delta T = \frac{3}{2}nR\Delta T = \frac{3}{2}P\Delta V = \frac{3}{2}V\Delta P$$



$T_{sys} < T_{surr}$
Heat *enters* the system

$T_{sys} > T_{surr}$
Heat *leaves* the system

Concepts regarding *systems*:

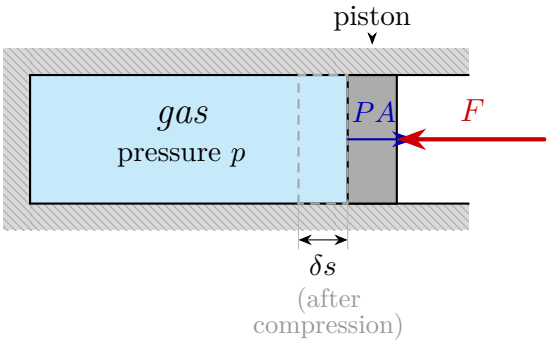
- Systems: the complete set of objects under consideration
- Closed system: a system that can exchange energy but not matter with its surroundings, mass can not enter or leave the system.
- Isolated system: a system that can exchange neither energy nor matter with its surroundings
- If all the parameters defining the system are given, we speak of the system being in a particular state
- Any process that changes the state of a system is called a thermodynamic process

Internal energy is a property of the particular state of the system under consideration and therefore is a state function.Heat and work are related to changes in the state of the system, not to the state itself.

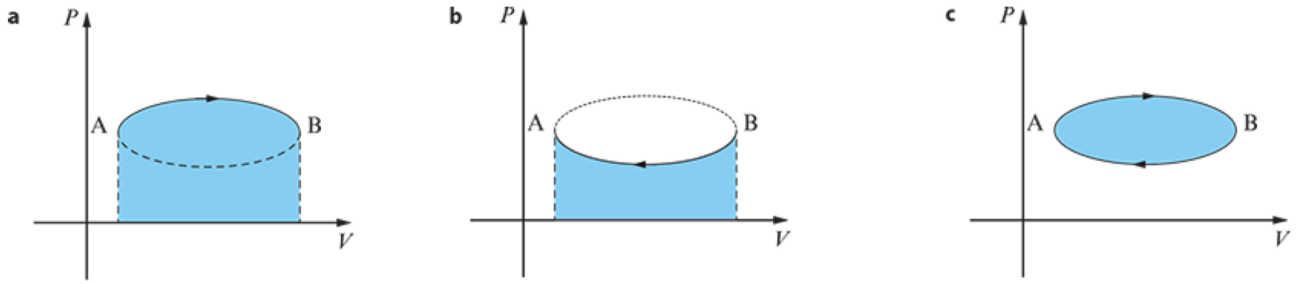
Process	Description	Equation
Isothermal	Temperature is constant	$P_1V_1 = P_2V_2$
Isobaric	Pressure is constant	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
Isovolumetric	Volume is constant	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
Adiabatic	No heat exchange	$P_1V_1^{\gamma} = P_2V_2^{\gamma}$, $\gamma = \frac{5}{3}$ $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$, $\gamma - 1 = \frac{2}{3}$

Work done on / by the gas:

$$W = F\Delta s = PA\Delta s = P\Delta V$$



It could also be calculated by finding the area under the curve in a P-V graph. (when pressure is not constant)



For a closed loop in a pressure-volume diagram, the work done is the area of the loop.
(Work done expanding is shown in a, work done compressing is shown in b)

10.2 The first law of thermodynamics

The first law is just the law of conservation of energy applied to thermodynamics.

$$\underbrace{Q}_{\text{heat}} = \underbrace{\Delta U}_{\text{change in internal energy}} + \underbrace{W}_{\text{work done}}$$

- $Q > 0$: heat is supplied to the system; $Q < 0$: heat is removed from the system
- $\Delta U > 0$: internal energy / temperature increases; $\Delta U < 0$: internal energy / temperature decreases
- $W > 0$: work is done by the system (gas expands); $W < 0$: work is done on the system (gas compressed)

10.3 The second law of thermodynamics

克劳德·香农（Claude Shannon）说过：”熵是信息的度量。”永远的现代信息学之父，致敬。
It is quite extraordinary that this formula of thermodynamics has featured extensively in Claude Shannon's theory of information, a crucial part of modern telecommunications theory.

Entropy: a measure of the disorder of a system, or the number of ways the system can be arranged.
The change in entropy is given by

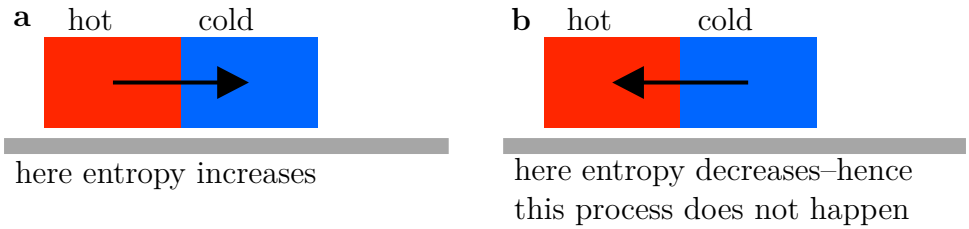
$$\Delta S = \frac{Q}{T}$$

where Q is the quantity of heat given to or removed from the system at a temperature T (in K).

- When heat is supplied to the system, $Q > 0$, $\Delta S > 0$, the entropy of the system increases.
- When heat is removed from the system, $Q < 0$, $\Delta S < 0$, the entropy of the system decreases.

For a reversible process that returns the system to its original state, the total change in entropy is zero (an example is the isothermal expansion / compression of gas).
Extension: $\Delta S = \int \frac{dQ}{T}$ when the temperature is not constant.

The entropy of an isolated system never decreases. Entropy increases in realistic irreversible processes and stays the same in theoretical, idealized, reversible processes.



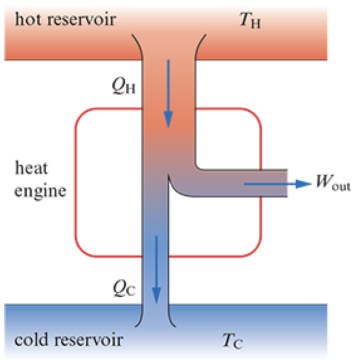
We denote $\Omega(H)$ the number of microstates that correspond to a macrostate H. (for example, the possible arrangements of N coins consisting of H heads)
Entropy is defined as:

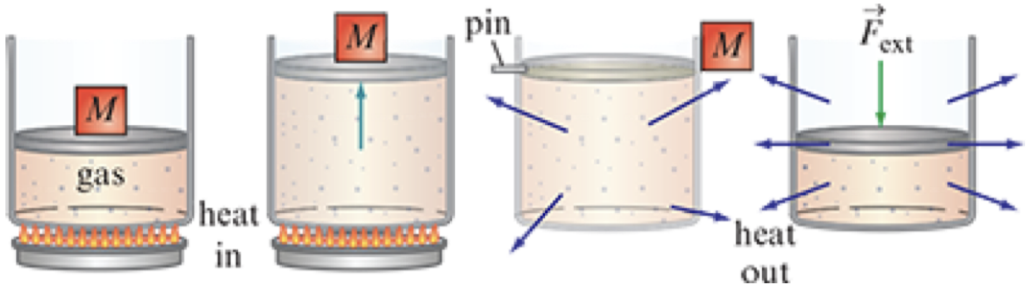
$$S = k_B \ln \Omega$$

where Ω is the number of microstates that correspond to the macrostate (e.g. total energy of the gas) of the system.
We see that the states which are likely to occur have large entropy.

10.4 Heat engines

Definition: a device that transfers thermal energy to mechanical or electrical energy, which can then be used to do mechanical work.





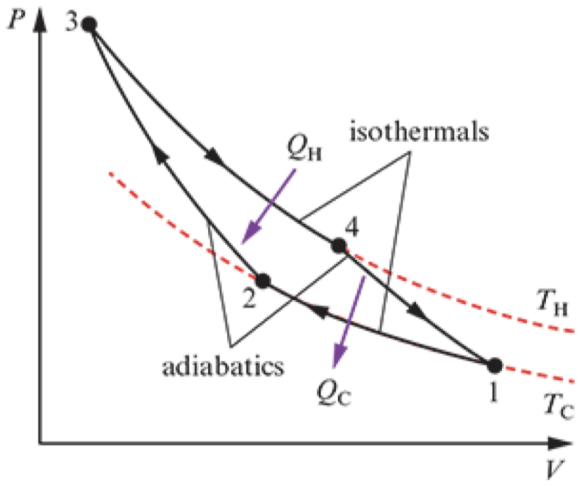
- A hot reservoir at temperature T_H transfers heat Q_H to the engine.
- The gas is expanded by the heat, doing work W on the surroundings (in this case raising a load).
- Gas is allowed to cool down to a lower temperature T_C isovolumetrically by transferring heat Q_C to the cold reservoir (the surroundings).
- An external force compresses the gas isothermally back to the original state.

The efficiency of a heat engine:

$$\eta = \frac{\text{useful energy output}}{\text{energy input}} = \frac{W}{Q_h} = 1 - \frac{Q_c}{Q_h}$$

where Q_h is the heat absorbed from the hot reservoir, Q_c is the heat rejected to the cold reservoir, and W is the work done by the engine. This shows that it is impossible to have a heat engine with 100% efficiency.

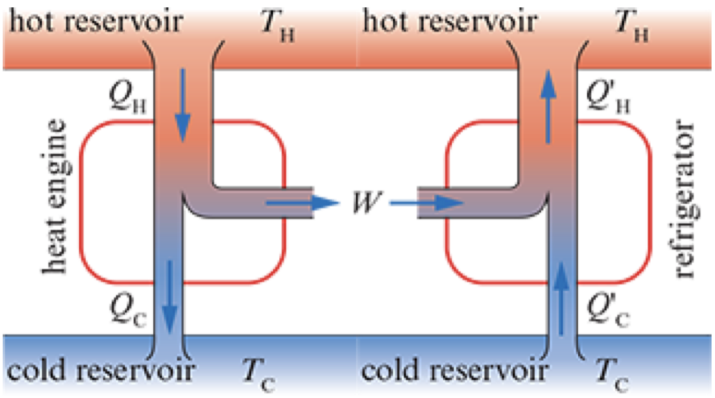
Carnot cycle: limit on the efficiency of heat engines



- The engine starts its cycle at 1.
- The gas is compressed isothermally until 2. During this stage heat Q_C leaves the gas and work is done on the gas.
- From 2 to 3 it is compressed adiabatically; work is again done on the gas.
- From 3 to 4 the gas expands isothermally receiving heat Q_H from a hot reservoir; work is done by the gas.
- From 4 to 1 the gas expands adiabatically; work is done by the gas.

The net work done by the engine is $W_{\text{net}} = Q_H - Q_C$
The change in entropy is $\Delta S = \Delta S_{1\to2} + \Delta S_{2\to3} + \Delta S_{3\to4} + \Delta S_{4\to1}$
$\Delta S_{3\to4} = \frac{Q_H}{T_H}$, $\Delta S_{1\to2} = \frac{Q_C}{T_C}$, $\Delta S_{2\to3} = \Delta S_{4\to1} = 0$ (adiabatic, no heat exchange)
$\because \Delta S = 0 \therefore \frac{Q_H}{T_H} = \frac{Q_C}{T_C}$
$\because \frac{Q_C}{Q_H} = \frac{T_C}{T_H} \therefore \eta = 1 - \frac{T_C}{T_H}$
This shows that the efficiency of a Carnot engine can never be 100% because $T_C \neq 0$ and $T_H \neq \infty$.

Extension: proof of the Carnot efficiency being the maximum efficiency by contradiction



If the Carnot refrigerator ran as a heat engine, the efficiency would be $\frac{W}{Q'_H}$.

If we assume the engine on the left has higher efficiency than the Carnot efficiency, then $\frac{W}{Q_H} > \frac{W}{Q'_H}$.

$\Rightarrow Q'_H > Q_H \Rightarrow Q'_H - Q_H > 0$

$\because W = Q'_H - Q'_C = Q_H - Q_C \therefore Q'_C - Q_C = Q'_H - Q_H = Q > 0$

The combination has the effect of transferring heat from the cold to the hot reservoir without performance of work, which is against the second law.

So the assumption of an engine with better efficiency than the Carnot efficiency must be false.

11 Current and circuits

11.1 Potential difference, current and resistance

$$V = \frac{W}{q}$$

where W is the work done to move a charge q through the component.

The kinetic energy gained when the charge is released (electric force will push it away) is qV.

Conductors: materials that allow electric charge to flow through them easily, many free electrons per unit volume. (for metals, 10^{28} free electron per cubic meter at v of 10^5 m/s)

Insulators: materials that do not allow electric charge to flow through them easily, few or no free electrons per unit volume.

DC: direct current, the current flows / motion of electrons in one direction only.

AC: alternating current, the current / motion of electrons changes direction periodically.

$$I = \frac{\Delta q}{\Delta t}$$

$$R = \frac{V}{I}$$

Ohm's law: $I \propto V$ when temperature and R are constant for ohmic conductors.

The resistance of a wire is given by

$$R = \rho \frac{L}{A}$$

where ρ is the resistivity of the material, L is the length of the wire, and A is the cross-sectional area of the wire.

LDR: light dependent resistor, the resistance decreases as the intensity of light increases.

NTC Thermistor: the resistance decreases as the temperature increases.
Interpreting resistance at a microscopic level:

- Electrons suffer collisions with the atoms of the wire
- Collisions are inelastic, the electrons lose kinetic energy to the atoms, which causes the wire to heat up ($\uparrow$ random $E_K \Rightarrow \uparrow T$)
- Atoms in the conductor vibrate more means an increased probability of collisions
- This reduces the speed of the electrons and therefore reduces the current, which is equivalent to an increase in resistance.

11.2 Voltage, power and e.m.f

$$V = IR$$

$$\text{electric power: } P = \frac{W}{\Delta t} = \frac{\Delta q V}{\Delta t} = IV = RI^2 = \frac{V^2}{R}$$

This formula could also be used to calculate the power dissipated by a wire / electrical appliance.

Note that the conventional current is defined as flowing from positive to negative, which is opposite to the actual flow of electrons.
The electromotive force $(e.m.f)$ ε is the work done per unit charge by a source of electrical energy (e.g. battery) in moving a charge around a complete circuit.

$$\varepsilon = \frac{W}{q} = \frac{P}{I}$$

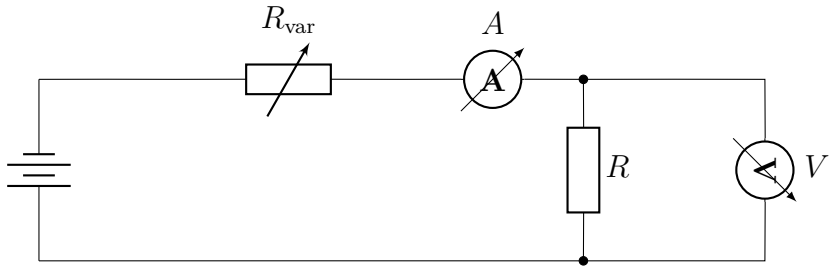
11.3 Resistors in electrical circuits

	Parallel	**Series**
Voltage	Same for all components, $V = IR_T$	Split according to the resistance of the components
Current	Same for all components, $I = \dfrac{V}{\sum R}$	Different for each branch, $\sum I = V \sum \dfrac{1}{R}$
Resistance	Total resistance $= R_T = \sum R$	Total resistance $= R_T = \dfrac{1}{\sum \frac{1}{R}}$

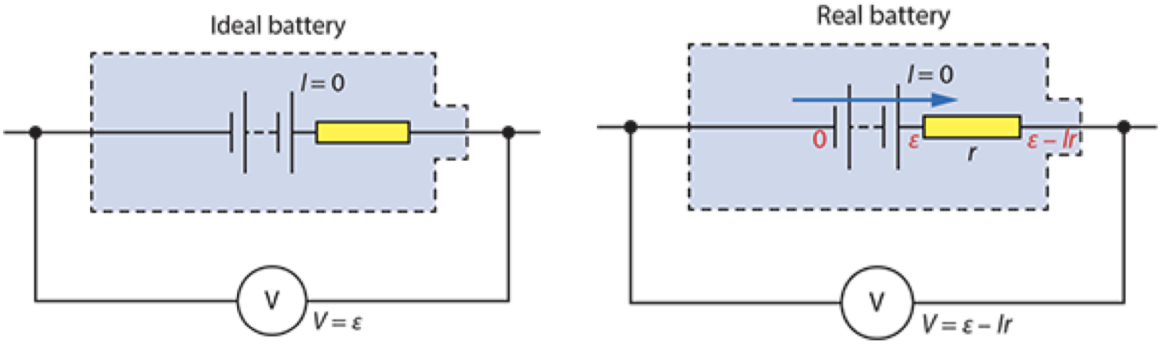
Symbol	**Description**	**Symbol**	**Description**	**Symbol**	**Description**
	Connecting wire		Junction		Galvanometer
	Cell		Crossing		Potentiometer
	Battery		Lamp		Variable resistor
	Resistor	V	Voltmeter		LDR
	DC power supply	A	Ammeter		Thermistor
	AC power supply		Switch		

11.4 Terminal potential difference and the potential divider

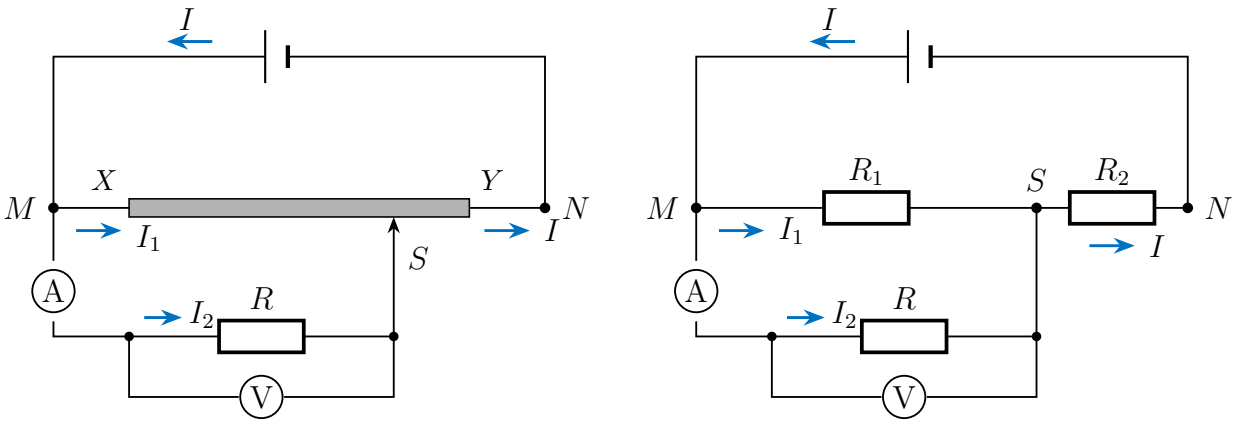
Ammeter: connected in series, after the component being measured, an ideal ammeter has 0 resistance.
Voltmeter: connected in parallel to the component being measured, an ideal voltmeter has infinite resistance.



The correct arrangement for measuring the current through and potential difference across a resistor.



We need to consider the existence of internal resistance in the real world: chemicals in cells (for example) also creates resistance in the cell.
The potential difference across the terminals of a battery is equal to the $e.m.f$ when there is no current in the circuit. It is less than the $e.m.f$ when there is current.
An $V - I$ graph can be used: ε is the y-intercept, r is the magnitude of the gradient of the line.

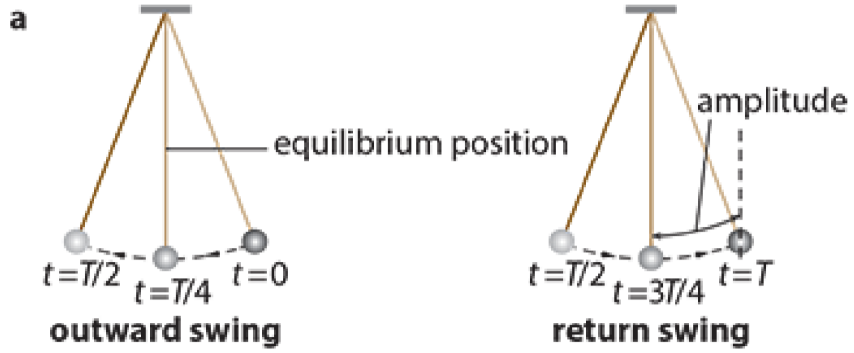


By varying where the slider S connects to XY, different potential differences and currents are obtained for the device R.
The potential difference across R now ranges from 0 to ε.

Unit C Wave behaviour

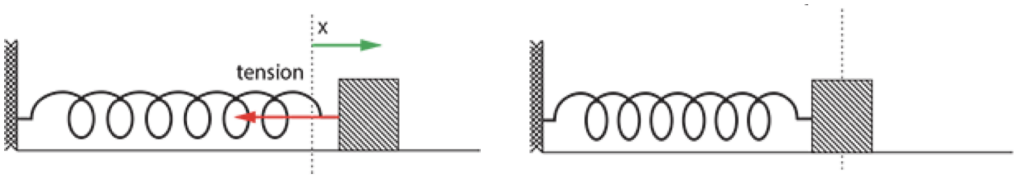
12 Simple harmonic motion

12.1 Simple harmonic oscillations



Oscillations refer to back and forth motion that repeats. The body moves back and forth around an equilibrium position.

- Period, $T = 2\pi\sqrt{\frac{L}{g}}$ (L is the length of the string), time taken to complete one full oscillation
- Amplitude, A, which is the maximum displacement away from the equilibrium position.
- Frequency, $f = \frac{1}{T}$, number of oscillations in 1s.
- Acceleration (and the restoring force) is proportional to the opposite of displacement. $a = -\omega^2 x$ (x is the displacement)
- Angular frequency $\omega = \frac{2\pi}{T} = 2\pi f$



In this case:

- $a = -\frac{k}{m}x$
- $\omega = \sqrt{\frac{k}{m}}$
- $T = 2\pi\sqrt{\frac{m}{k}}$

In SHM:

- the period does not depend on amplitude
- the amplitude stays constant as time goes on.

A graph of acceleration against displacement gives a straight line through origin with a slope of $-\omega^2$.

Phase difference between two oscillations: $\Delta\phi = \frac{\Delta t}{T} \times 2\pi$ rad, where Δt is the separation in time between two neighbouring peaks. (2+ same SHMs have different peaks, started with different displacements)

12.2 Details of simple harmonic motion

Displacement x as a function of time is expressed as

$$x = x_0 \sin(\omega t + \phi)$$

where x_0 is the amplitude and ϕ is the phase angle.

$$v = \omega x_0 \cos(\omega t + \phi);\ v_{\max} = \omega x_0$$

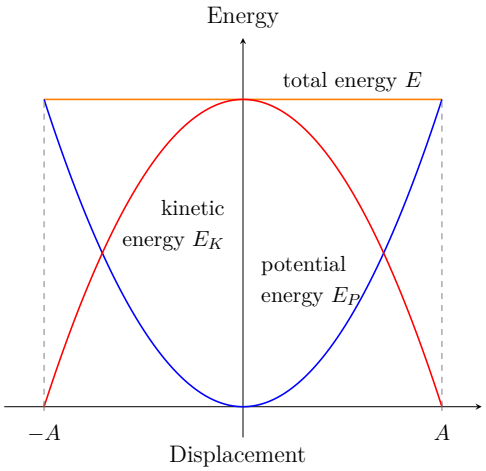
$$a = -\omega^2 x_0 \sin(\omega t + \phi);\ a_{\max} = \omega^2 x_0$$

Phase difference can be expressed as magnitude of differences between two phase angles.

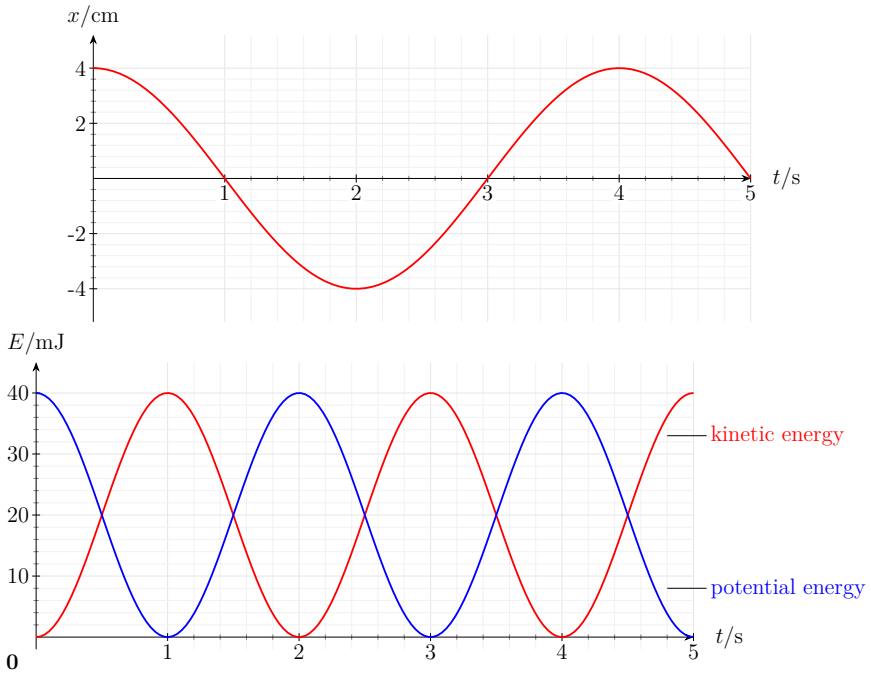
$$\Delta\phi = |\phi_1 - \phi_2|$$

12.3 Energy in simple harmonic motion

$E_T = E_P + E_K =$ constant (in the absence of other resistance forces, energy is conserved)



You will need to deduce how kinetic and potential energy varies with time from a given displacement-time graph:



12.4 More about energy in SHM

$$E_K = \frac{1}{2}m\omega^2(x_0^2 - x^2)$$

$$E_P = \frac{1}{2}m\omega^2 x^2$$

$$E_T = \frac{1}{2}m\omega^2 x_0^2$$

$$v = \pm\omega\sqrt{x_0^2 - x^2}$$

13 The wave model

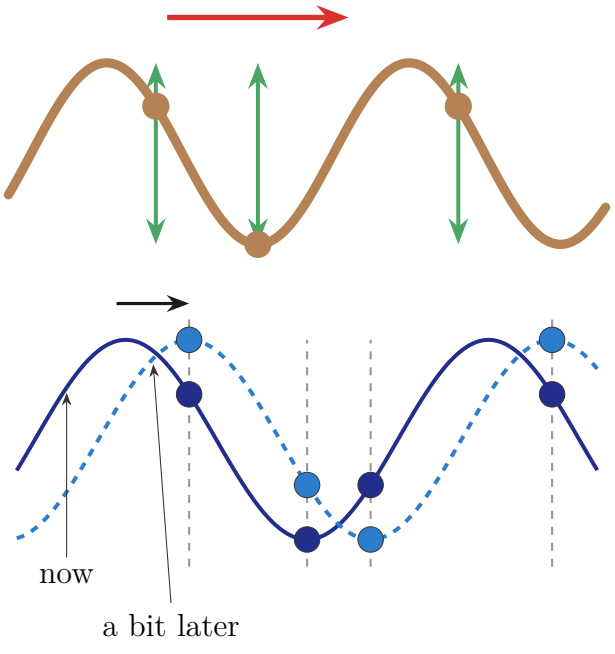
13.1 Mechanical pulses and waves

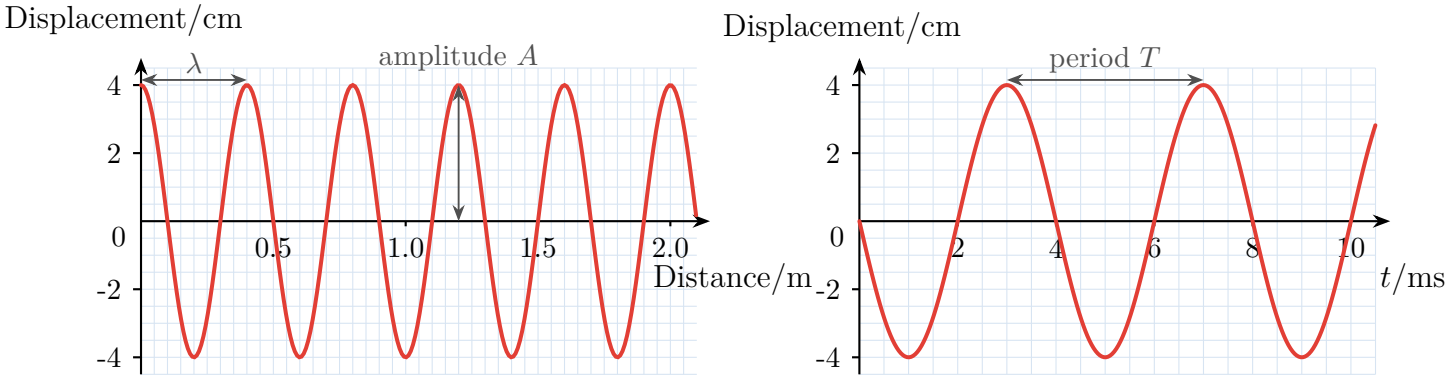
A mechanical wave is a disturbance that travels in a medium through oscillations of the particles of the medium. A wave transfers energy and momentum. The direction of propagation of the wave is the direction of energy transfer.

- Wavelength, λ, length of one full wave / distance from crest to crest or trough to trough
- Period, T, time to create one full wave
- Frequency, $f = \frac{1}{T}$, the number of full waves created in 1s
- Speed, $v = \frac{\lambda}{T} = f\lambda$

13.2 Transverse and longitudinal waves

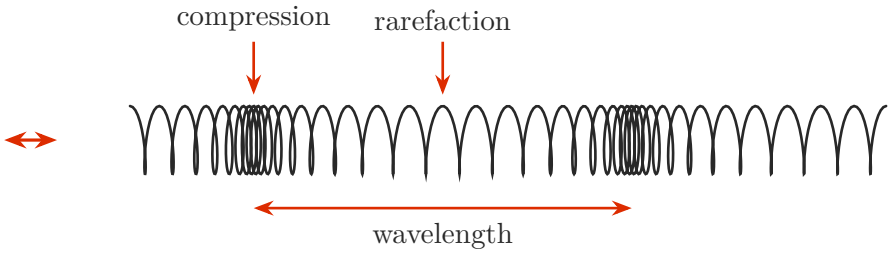
Transverse: the displacement of the particles of the medium is at right angles to the direction of energy transfer.
As a wave propagates through a medium, the particles of the medium perform SHM with a frequency that is the same as that of the wave.



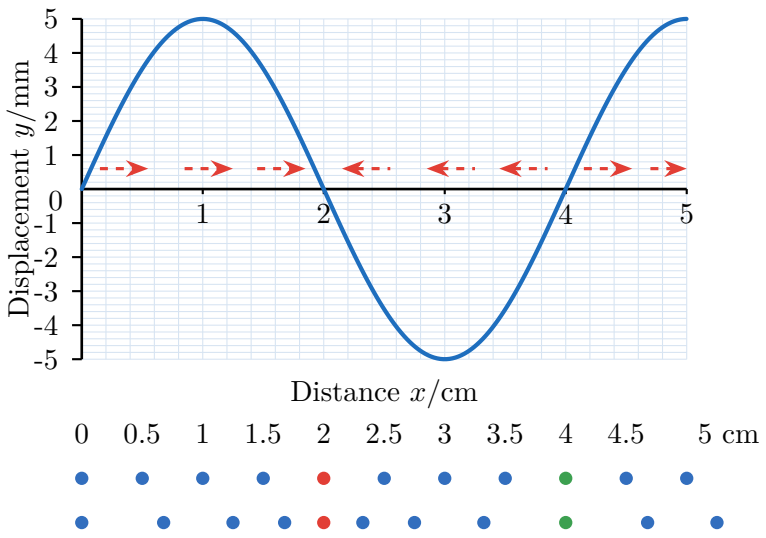


Displacement-distance and Displacement-time (displacement of one specific point)

Longitudinal waves: the displacement is parallel to the direction of energy transfer.



Compression: a region of higher than normal density
Rarefaction: a region of lower than normal density
You cannot tell whether a wave is transverse or longitudinal by looking at displacement-distance or displacement-time graphs. The graphs look the same for both.



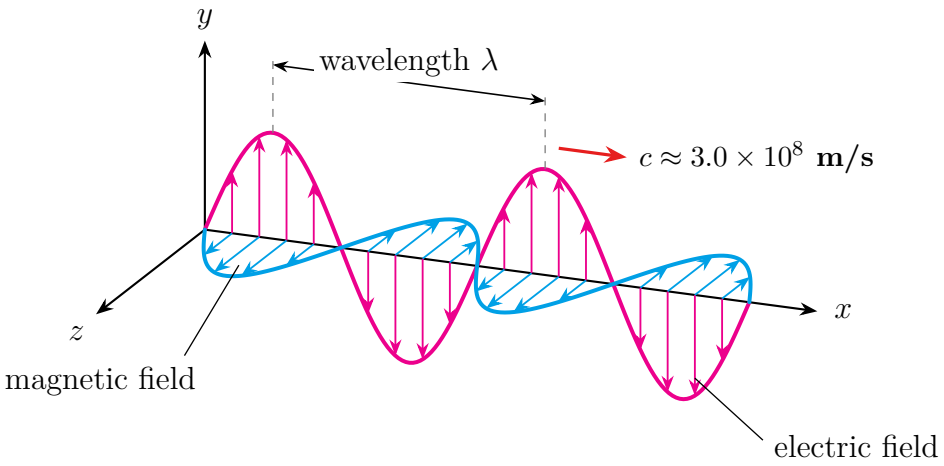
Positive displacement means moving right, negative displacement means moving left.

Phase difference is defined as

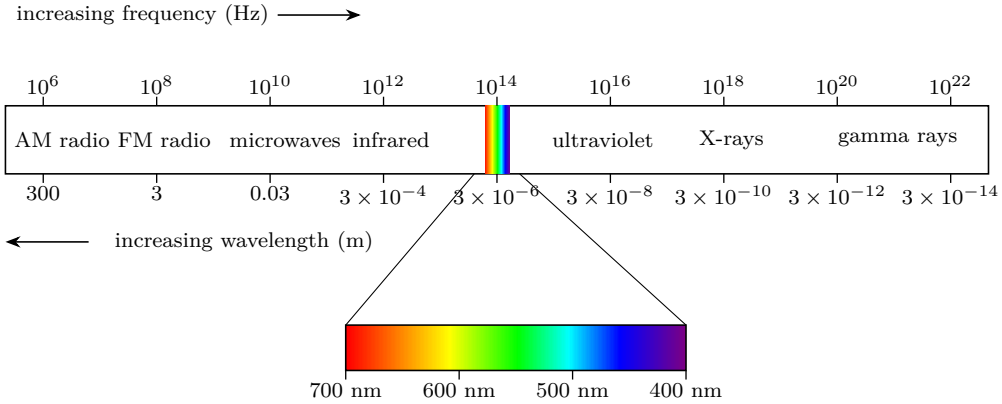
$$\Delta\phi = 2\pi\frac{\Delta x}{\lambda}$$

where Δx is the difference between consecutive peaks. The same formula can also be used to find the phase difference between two different points on the same wave.

13.3 Electromagnetic waves



An oscillating electric field produces an oscillating magnetic field such that the two are at right angles to each other and both propagate in space at the speed of light. Two fields are in phase.

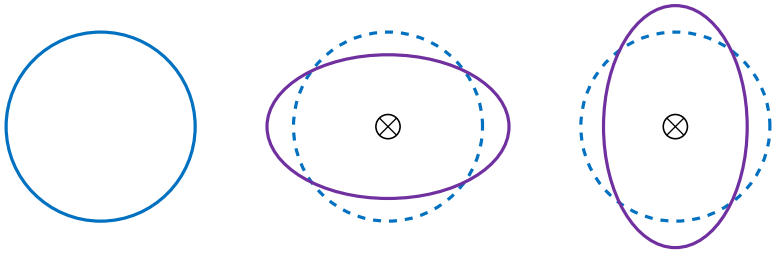


EM waves are transverse waves, and they can travel in a vacuum. They are produced when electric charges accelerate or during transitions in molecular, atomic or nuclear energy levels.

13.4 Waves extension

Gravitational waves

These waves would be generated whenever very large masses such as neutron stars and black holes collide with each other. Gravity waves are transverse and travel at the speed of light. The passage of a gravity wave through space distorts space, stretching and compressing it.



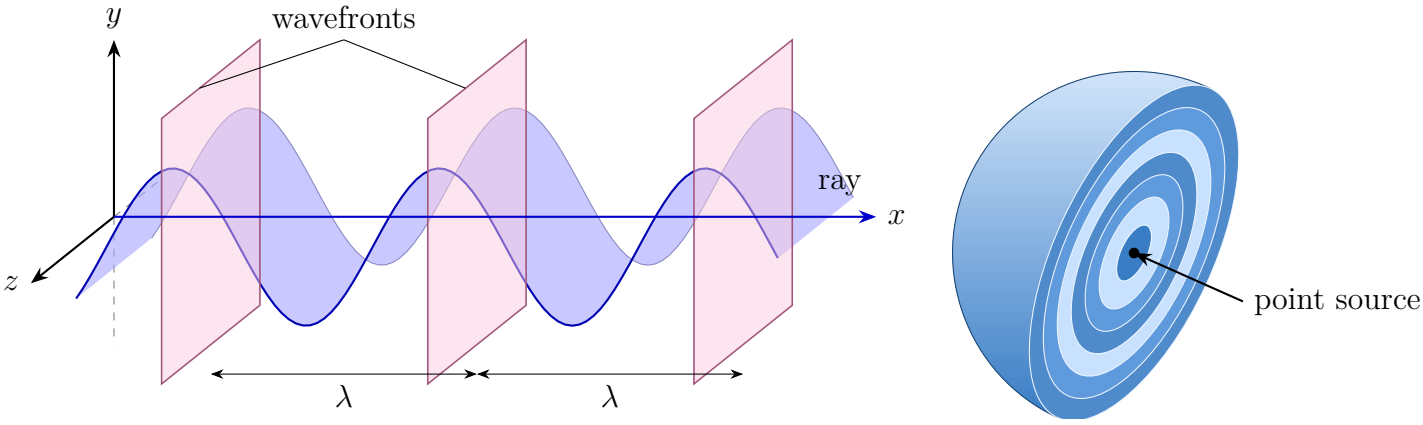
Matter waves

According to quantum theory, **all particles have wave properties**. The waves corresponding to particles are called matter waves. These are waves of probability: the displacement of the wave is related to the probability of finding a particle near a particular position.

14 Wave phenomena

14.1 Reflection and refraction

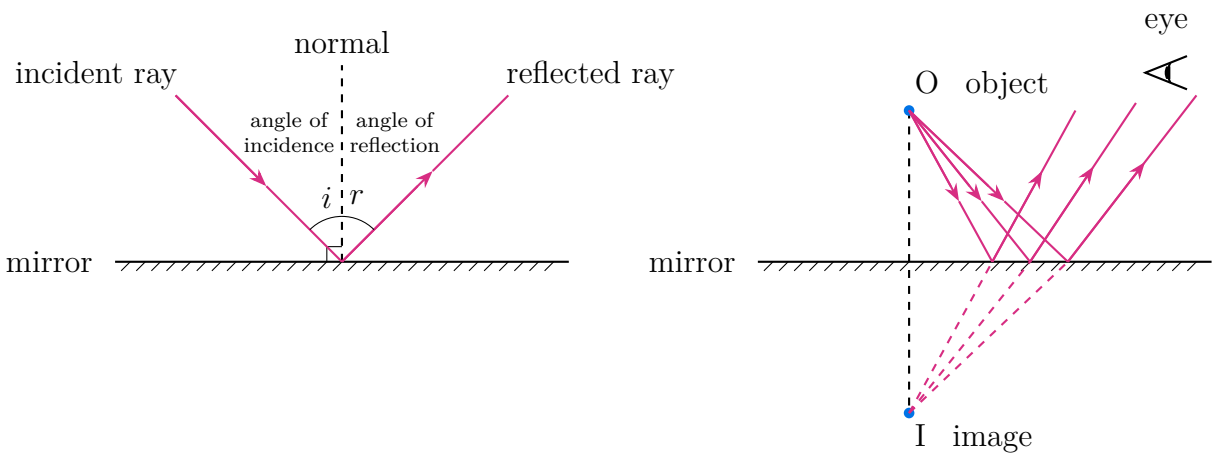
Wavefronts: planes will be normal to the direction of the wave.
Rays: lines at right angles to the wavefronts show the direction of wave propagation.



A point source emits waves in all directions. The wavefronts from a point source are spherical / cylindrical.

Reflection

Law of reflection: angle of incidence = angle of reflection, $i = r$.
Reflection takes place when the reflecting surface is sufficiently smooth. This means that the wavelength of the incident wave has to be larger than the size of any irregularities of the surface.
The wavelength of the reflected waves is the same as that of the incident wave.



Refraction

Refraction: the passage of light or any other wave from one medium into another where it has a different speed. It changes the direction of the incident ray. The wavelength changes for the wave in the medium. When a ray strikes an interface between two media, there is both reflection and refraction.

- If the speed of wave in medium is lower, the ray bends towards the normal.
- If the speed of wave in medium is higher, the ray bends away from the normal.

Snell's Law

$$\frac{\sin\theta_1}{c_1} = \frac{\sin\theta_2}{c_2} \Rightarrow n_1 \sin\theta_1 = n_2 \sin\theta_2$$

where n is the respective refractive index (must > 1, $\frac{\text{speed in vacuum}}{\text{speed in medium}}$).

Total Internal Reflection, TIR

TIR occurs at the boundary between two media when all the incident ray in medium 1 is reflected back into medium 1. For TIR to occur, the incident material must be denser than the second material. For incident angle $> c$, no ray will be refracted and TIR would occur.

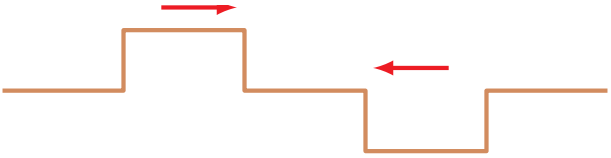
Applications of TIR include periscopes & **Optic fiber** (let's go CS)

When the angle of refraction is exactly 90° the light is refracted along the boundary. At this point, the angle of incidence is known as **the critical angle** c**.** Equation:

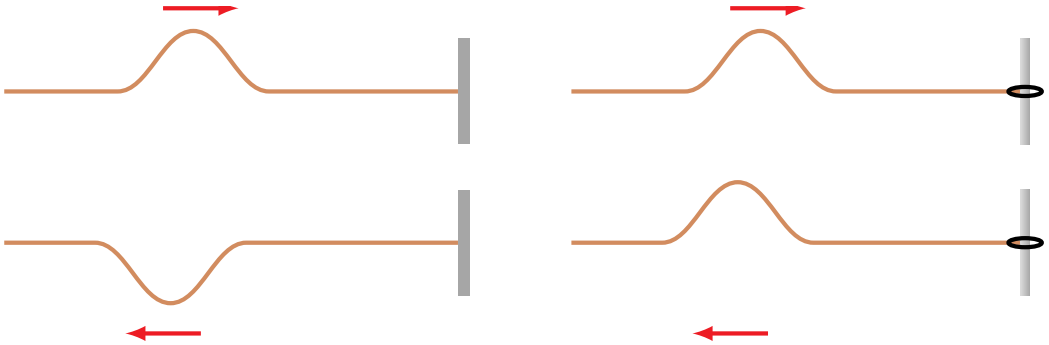
$$c = \sin^{-1}\left(\frac{1}{n}\right) \Leftrightarrow n = \frac{1}{\sin c}$$

14.2 The principle of superposition

想起了一些关于量子叠加态的内容..... $|k\rangle$
When two or more waves of the same type arrive at a given point in space at the same time, the displacement of the medium at that point is the algebraic sum of the individual displacements.



Reflection of waves

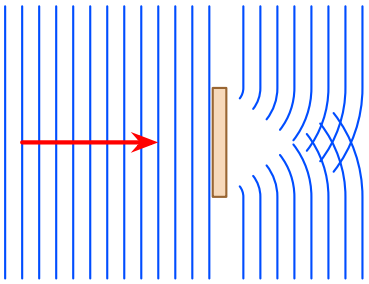


Fixed end, inverted (反作用力), phase change of 180°

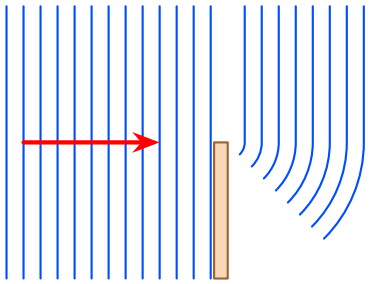
End free to move, no phase change, end falls

14.3 Diffraction and interference

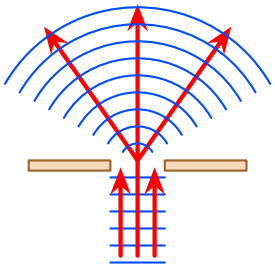
Diffraction: The spreading of a wave as it goes past an obstacle or through an aperture.
Diffraction is most prominent when the wavelength λ is larger than or equal to the gap width.



diffraction around an obstacle



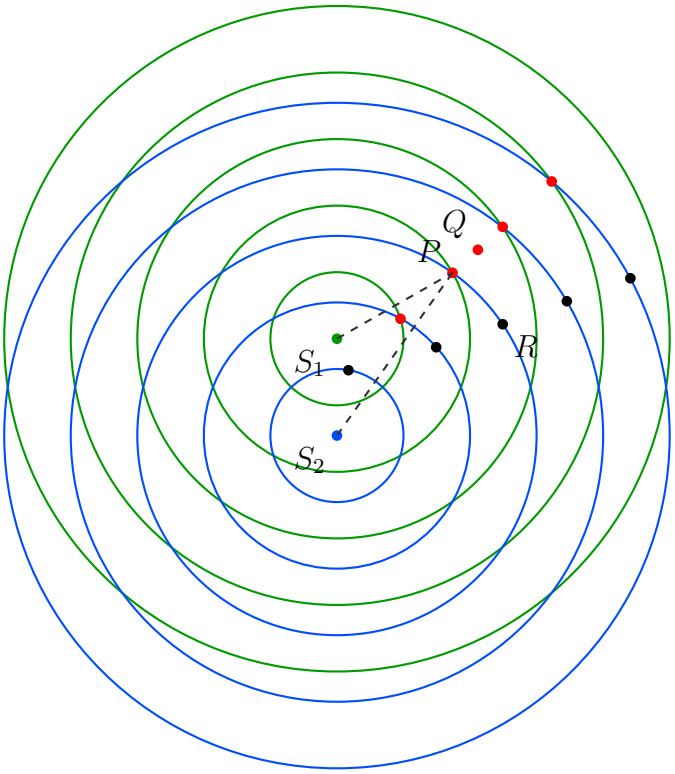
diffraction at an edge

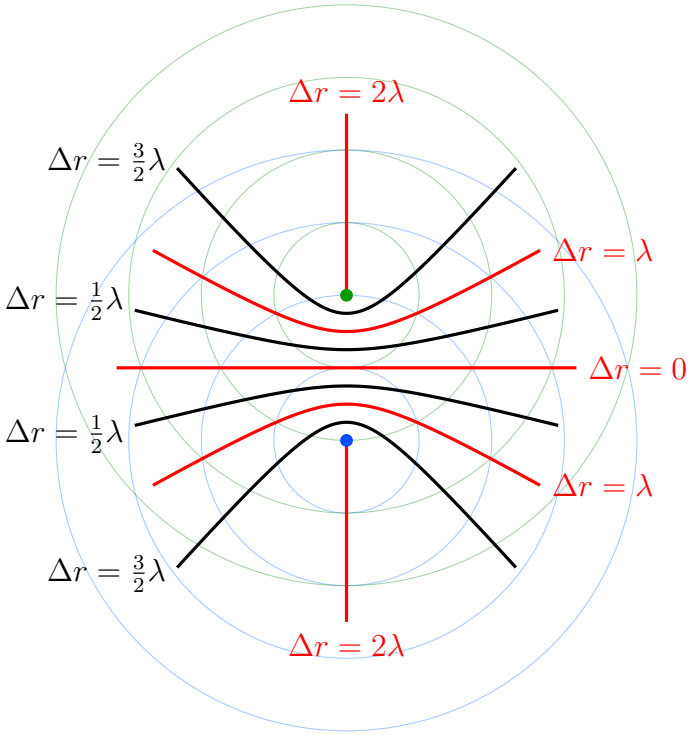


diffraction through a slit

Double-source interference

When two identical waves meet at the same point in space, the principle of superposition states that the resulting wave has a displacement that is the sum of the individual displacements. The resulting amplitude can be anything from zero to twice the amplitude of one of the waves. All waves show interference.





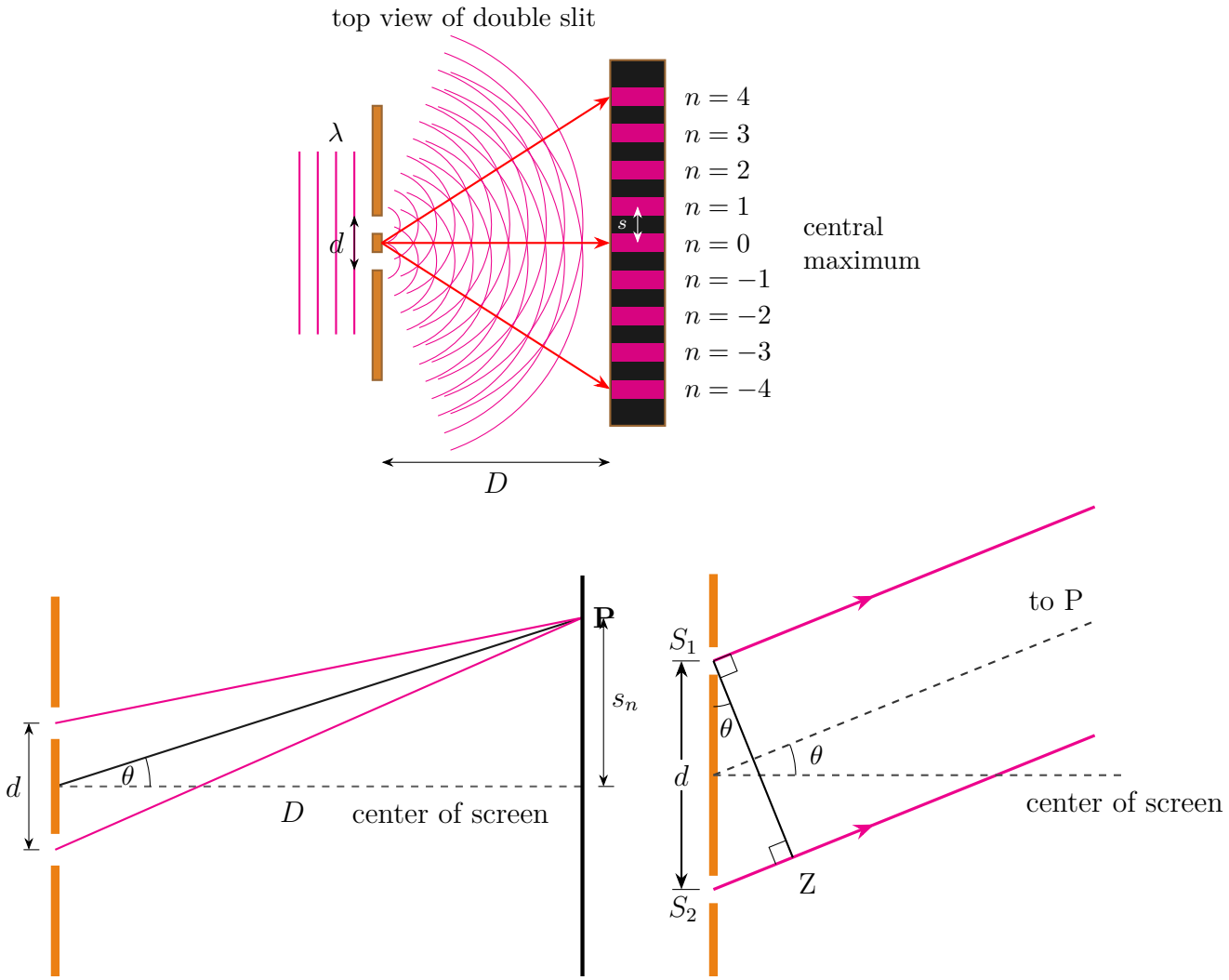
* : S_1 and S_2 are identical sources.

The path difference: the difference in distance of the point from two sources, $\Delta r = |S_1P - S_2P|$. (For P, path difference is equivalent to λ) Note that at P, waves arrive as crests; at point Q, waves arrive at troughs. There is **constructive interference** for the points in red and **destructive interference** for those in black.

- $|S_1P - S_2P| = n\lambda$ with $n \in \mathbb{N}$ for path difference in constructive interference. It happens when two waves are in phase.
- $|S_1P - S_2P| = \left(n + \frac{1}{2}\right)\lambda$ with $n \in \mathbb{N}$ for path difference in destructive interference. Happens when two waves are **exactly** out of phase.
- In other cases, the resultant amplitude will be a value between 0 to $2A$.

Complete destructive interference takes place only when the two waves have equal amplitudes.

Young's Double Slit Experiment



The value of n indicates that the path difference is $n\lambda$.

Here, the path difference is $S_2Z = d\sin\theta$.
Deduction process for the linear separation on a screen of two consecutive maxima, s:

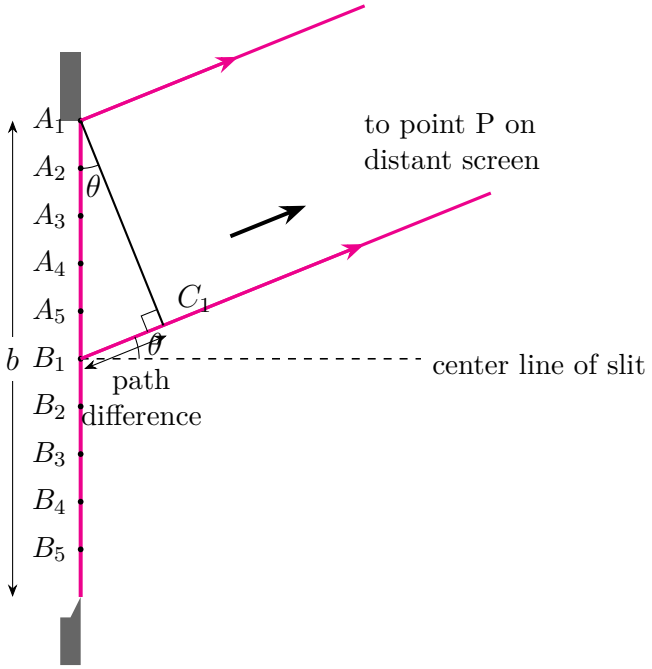
- Because d is small, $\sin\theta \approx \tan\theta \approx \theta$.
- From left diagram, $\tan\theta = \dfrac{s_n}{D} \Rightarrow s_n = D\sin\theta$
- $\sin\theta = \dfrac{n\lambda}{d} \Rightarrow s_n = \dfrac{n\lambda D}{d}$
- $s = \dfrac{\lambda D}{d}$, the maxima of the interference pattern are equally separated.

Coherent waves: phase difference between two waves (originated from the same source) does not change with time.

An intensity of one unit corresponds to the intensity from only one slit. At points of constructive interference the amplitude is double that of just one wave. Since intensity is proportional to the square of the amplitude, the intensity is four times as large.

14.4 Single-slit diffraction

Diffraction is negligible is λ is much smaller than the gap width and obvious when $\lambda \geq$ gap width.

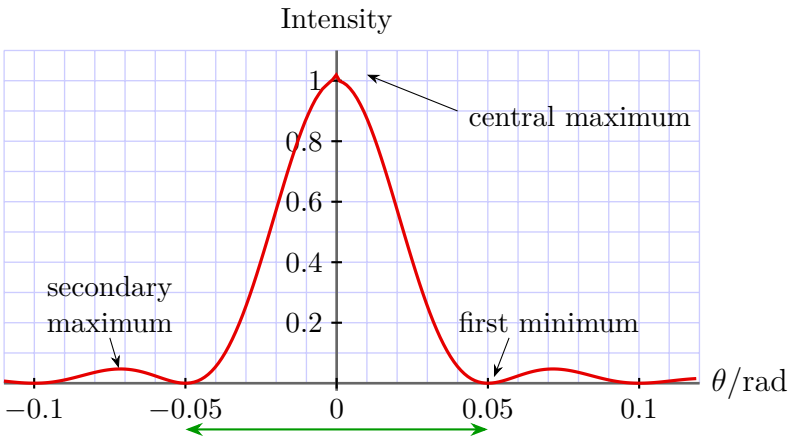


Points from A_1 to A_5 and from B_1 to B_5 are in phase (lies on the same wavefront).
The results of the interference of the waves can be seen on a distant screen.

The path difference between ray A_1 and B_1 is equal to $B_1C_1 = \frac{b}{2}\sin\theta$. This would result in complete destructive interference if the path difference is half of the wavelength. $\frac{b}{2}\sin\theta = \frac{\lambda}{2} \Rightarrow b\sin\theta = \lambda$

For every point on the upper half of the slit there is a corresponding point on the lower half, so we see that all the points on the wavefront will result in complete destructive interference if the first pair results in destructive interference.

Approximate formula for the angle of the first diffraction minimum: $\theta = \frac{\lambda}{b}$, where b is the slit width.



Intensity in single slit diffraction for red light.

The intensity of the first secondary maximum is only about 4.5% of the intensity at the central maximum.

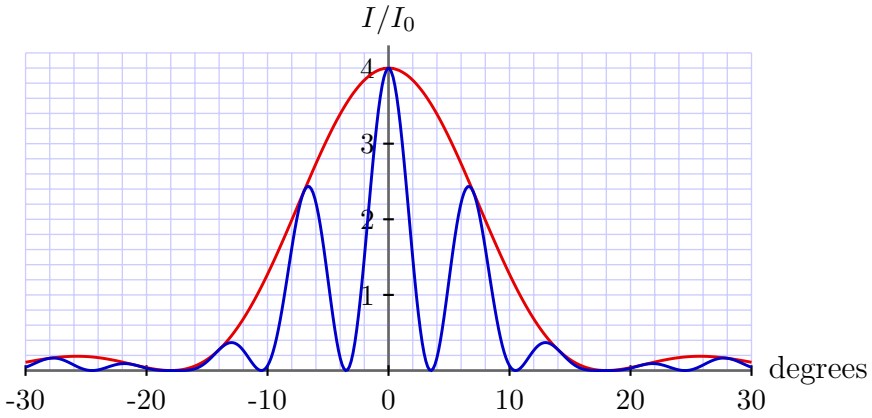
The angular width of the central maximum (measured by the green arrow) is given by $\frac{2\lambda}{b}$.

If we increase the slit width, more light goes through so the intensity will increase and the angular width will decrease.

Blue light has a shorter wavelength and so with blue light the first minimum occurs at smaller angle.

Effect of (non-negligible) slit width on two-slit intensity patterns

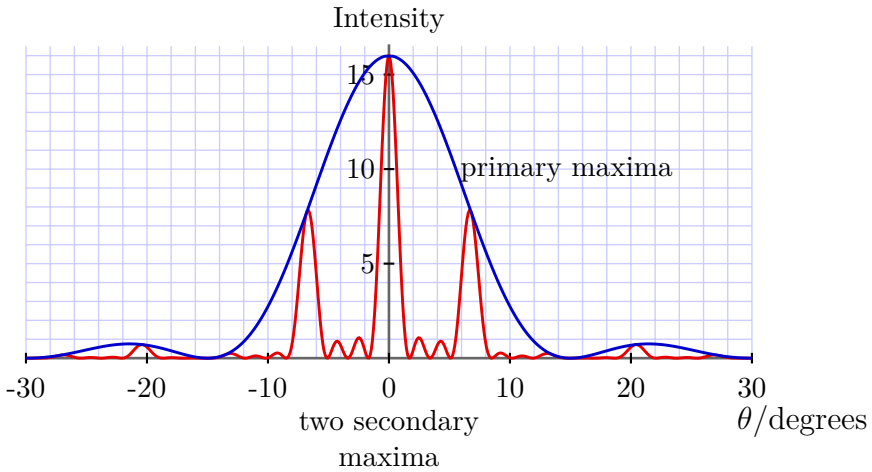
The single slit diffraction pattern modulates the two-slit pattern so that the intensity is given by the blue curve. The effect of the slit width is that the red curve becomes the envelope of the blue curve.



14.5 Multiple slits

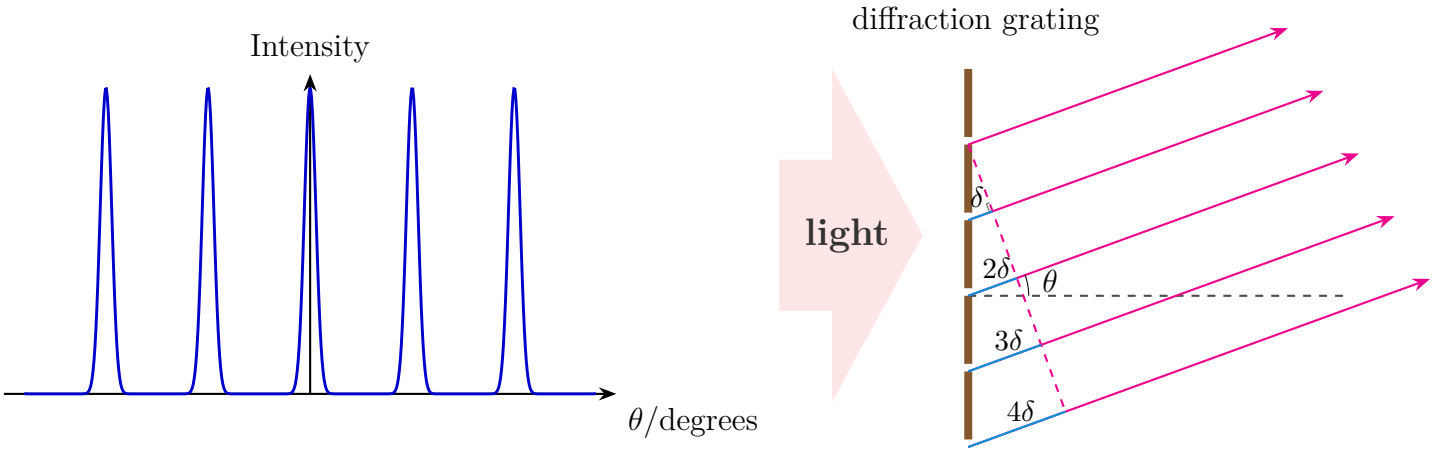
As the number of slits increases, the interference pattern increases in complexity. With N slits, there are $N-2$ secondary maxima. The intensity of the central maximum is N^2 times the intensity of just one slit by itself.
As the number of slits increases:

- The primary maxima are observed at the same angles
- The primary maxima get narrower and brighter
- The secondary maxima become unimportant



Diffraction Grating

Used to measure wavelength of light. A diffraction grating consists of a large number of parallel slits of very small width. (Instead of actual slits, modern gratings consist of a transparent slide on which rulings or grooves have been precisely cut.) The maxima in the interference pattern are sharp and bright and can easily be distinguished from their neighbours, they are called ”lines”.



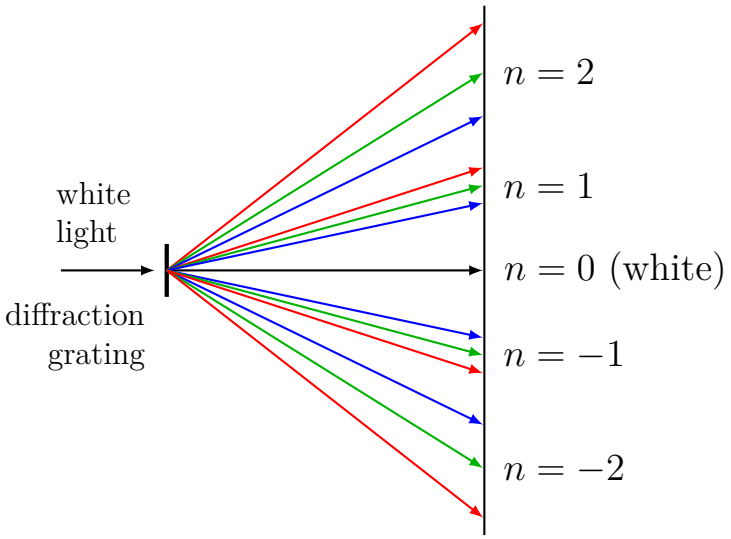
The intensity distribution of diffraction grating.
The maxima have roughly the same intensity and are very narrow.
右图中 $\delta = d\sin\theta$ 即为 path difference.

The maxima of the pattern are observed at angles that can be found by:

$$d\sin\theta = n\lambda, n \in \mathbb{Z}$$

A maximum observed at $n = k$ is called the k^{th} order maximum.
In practice, "x lines per mm" $\Leftrightarrow d = \dfrac{1}{x}$ mm.

White light incidence on diffraction grating:



The $n = 1$ and $n = 2$ orders are beginning to overlap. This is because different wavelengths in different orders can satisfy the grating equation at the same angle, so light from the second order falls on top of light from the first order.

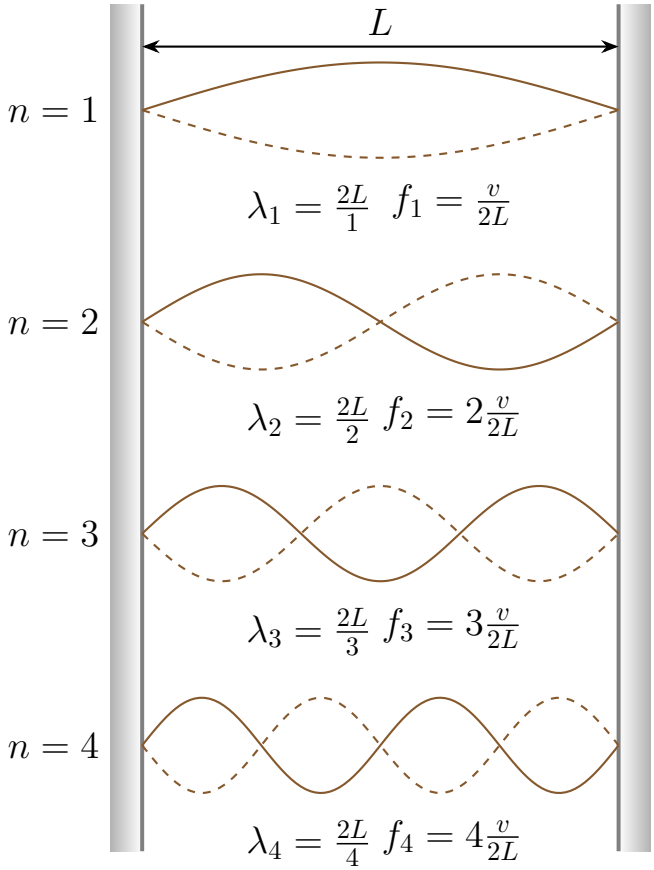
15 Standing waves and resonance

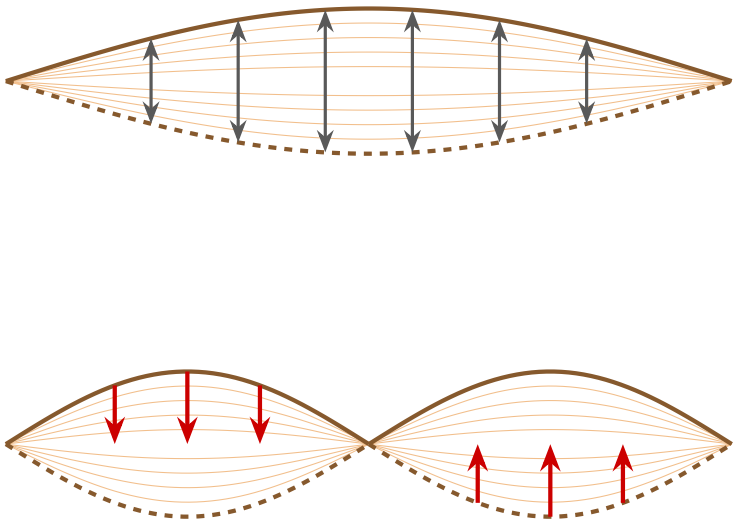
15.1 Standing waves

A standing wave is formed if two waves of the same speed, wavelength and amplitude travelling in opposite directions meet. The resulting wave has a displacement that is the sum of the displacements of the two travelling waves.
Characteristics of standing waves:

- The position of crests and troughs are always at the same position.
- Nodes: as a result of destructive interference, the displacement at that point is always zero. The distance between two consecutive nodes is half a wavelength.
- Antinodes: points halfway between two nodes, as a result of constructive interference, the displacement gets as large as possible (for an instant of time only).
- The amplitude of oscillation is different at different points on the string.
- A standing wave does not transfer energy.
- The ends of a standing wave are either nodes or antinodes. These "boundary" or "end conditions" determine the possible wavelength of the wave.
- All points on a standing wave have the same frequency.

15.2 Standing waves on strings





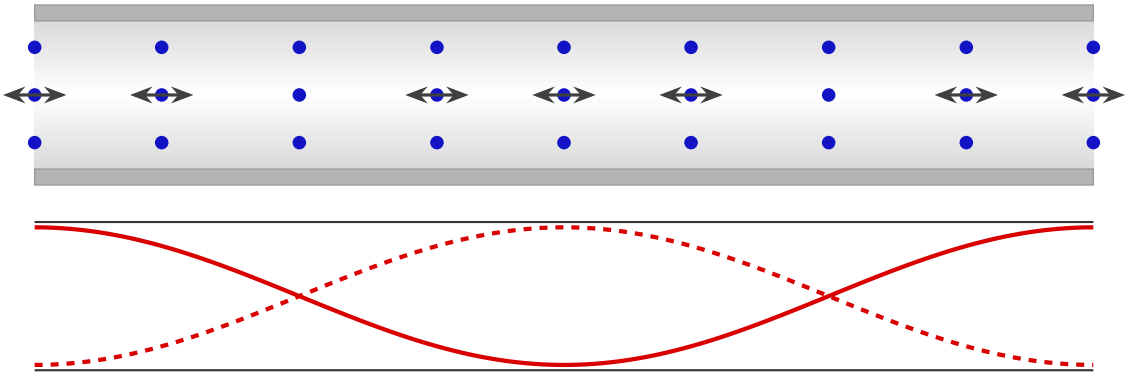
* : the λ is deduced using the fact that the distance between two consecutive nodes is half a wavelength.

Different standing wave patterns will be established on the string depending on the frequency used.
The standing wave that has the longest wavelength here is called the first harmonic.
The higher harmonics have more loops, and the wavelength decreases and the frequency increases.
Generally speaking, $\lambda_n = \dfrac{2L}{n}$. (while f is given by $\dfrac{v}{\lambda}$, where v is the speed of the travelling wave).

Different points on the standing wave oscillate with different amplitudes.
Points within the same loop oscillate in phase, while there is a 180° phase difference between points in one loop and points in the next loop.

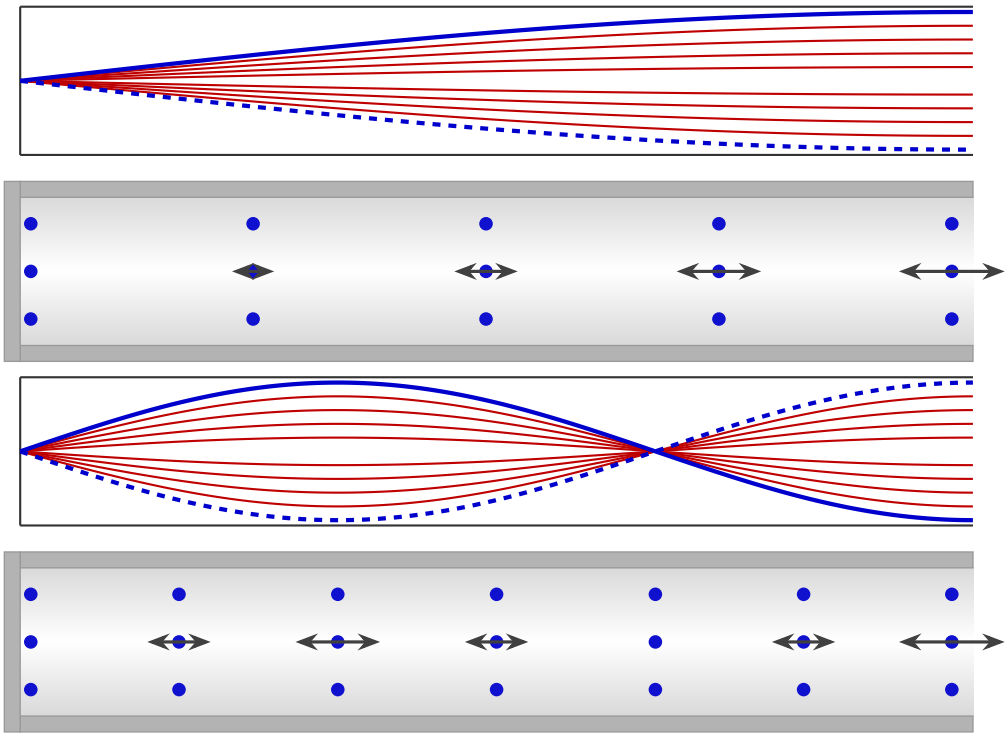
Note that when asked about sound made from the string, the f of the standing wave is the same as that of the sound.

15.3 Standing waves in pipes



The case of a pipe with both ends closed (which is not very useful) is similar to that of a string with ends fixed (two nodes at the ends). Normally, only discuss about the case of both end opening.

The dots represent molecules of air in the pipe. The double-headed arrows show how far these molecules oscillate back and forth. The molecules at the ends oscillate the most.



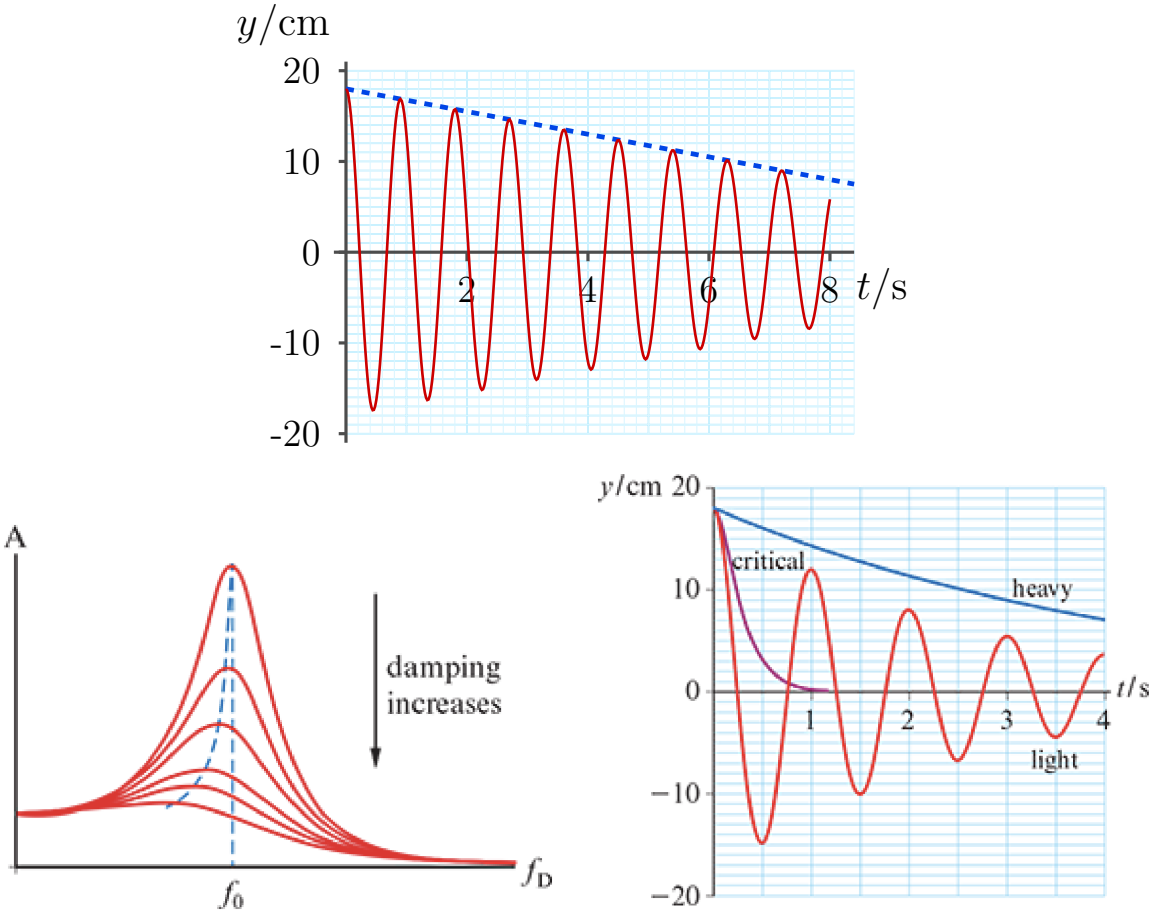
* : The case of a free end (antinode) and a closed end (node).

For this case, the allowed wavelength is given by $\lambda = \dfrac{4L}{n}$, n is an odd number.

15.4 Resonance and damping

If the externally applied forces resist the motion we will have damping, loss of energy in an oscillating system. The amplitude decreases exponentially with time.

Natural frequency f_0: the frequency that the system is free to oscillate.
The force F will be assumed to vary periodically with time with the driving frequency, f_D.
Oscillations that happen under externally applied force F is said to be forced / driven.
After the external force is first applied, the system will eventually switch to oscillations with a frequency equal to f_D. This is true both when the system is initially at rest and when it is oscillating at its own frequency before the force is applied.
The amplitude of the oscillations will depend on f_D and the amount of damping.



The degree of damping increases as we move from the top curve down.
The blue dashed curve represents the peak of the curves (maximum amplitudes).

- For very small damping, the peak of the curve occurs very close to the natural frequency of the system. When the driving frequency is equal to the natural frequency, the amplitude of oscillations is a maximum. This condition is called **Resonance**.
- As the amount of damping increases, the peak shifts to lower frequencies and becomes wider and shorter. In other words, with large damping the amplitude is a maximum when the driving frequency is slightly less than the natural frequency.
- At very low frequencies, the amplitude is essentially constant, and at high frequencies the amplitude goes to zero.

Three cases regarding damping:

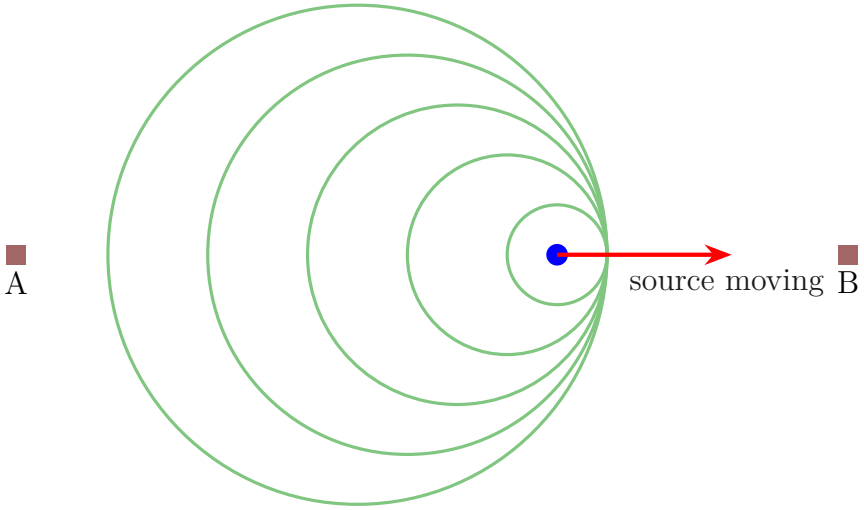
- In light damping the system returns to equilibrium after many oscillations of decreasing amplitude.
- In critical damping, the system returns to equilibrium in the shortest possible time without performing oscillations.
- In the heavy or overdamped case the system returns to equilibrium after a very long time without performing oscillations.

16 The Doppler effect

It is more pronounced if the car going by is a Formula 1 racing car! (额 V10 那会儿可能效果最明显)
Me: I found a video with F1 cars that perfectly demonstrates Doppler effect!
Lydia: You should wear a shirt that says "Doppler Effect" next Halloween...

16.1 The Doppler effect at low speeds

Definition: the change in the observed frequency of a wave when there is relative motion between the source and the observer.



The large wavefront was emitted first. The position of the source when this wavefront was emitted was at the center of this wavefront. The second wavefront was emitted later, and in this time the position of the source moved to the right. So this wavefront has its center shifted to the right, and similarly for the third wavefront.

Observer B will measure a higher frequency (number of wavefronts received per second) compared to observer A. For observer B, because of the bunching of the wavefronts, the wavelength measured will be less than the wavelength measured by the source.

Doppler effect for light

The approximate Doppler formula (when v is small compared to c):

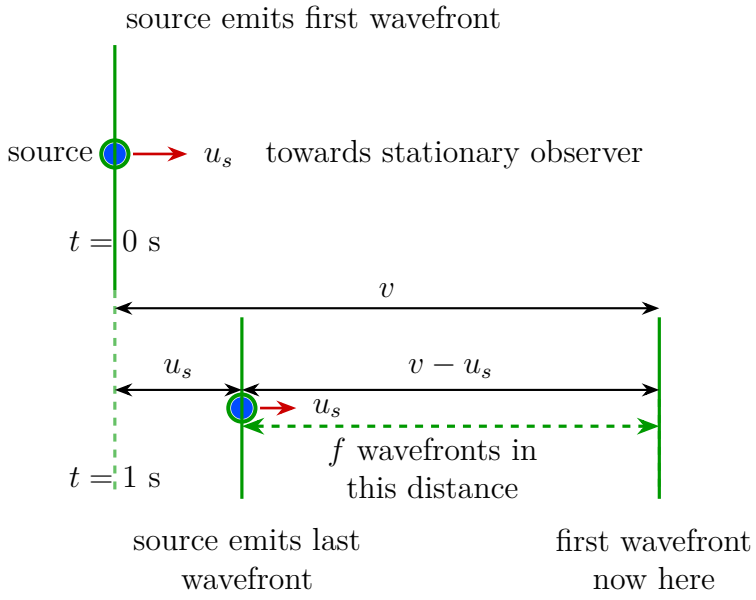
$$\frac{\Delta f}{f} = \frac{\Delta \lambda}{\lambda} \approx \frac{v}{c}$$

Δf gives the change in observed frequency, or $\Delta \lambda$ gives the change in observed wavelength.

Redshift: Increase in λ, decrease in f.
Blueshift: Decrease in λ, increase in f.

16.2 The Doppler effect for sound

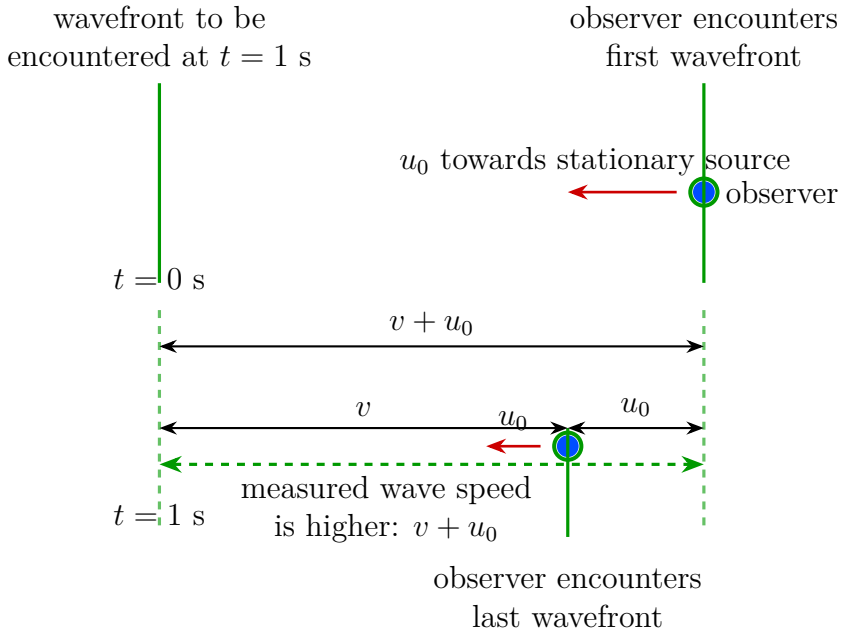
Moving source and stationary observer



The stationary observer will measure a wavelength of $\lambda' = \frac{v - u_s}{f} = \frac{v}{f'} = \lambda\left(1 \pm \frac{u_s}{v}\right)$ (where + is for source moving away, - for source approaching).

The frequency measured would then be $f' = \frac{v}{\lambda'} = f\left(\frac{v}{v \pm u_s}\right)$ (where + is for source moving away, - for source approaching).

Stationary source and moving observer



It could be seen as, from the observer's perspective, that the source is moving. Note that here, the speed of sound relative to the observer is no longer v as the medium has turned from still air to moving air.

Frequency observed would be given by $f' = f\frac{v + u_0}{v + u_0 - u_0} = f\left(\frac{v \pm u_0}{v}\right)$ (where + is for observer approaching, - for observer moving away) The wavelength would be the same.

Unit D Fields

17 Gravitation

17.1 Newton's law of gravitation

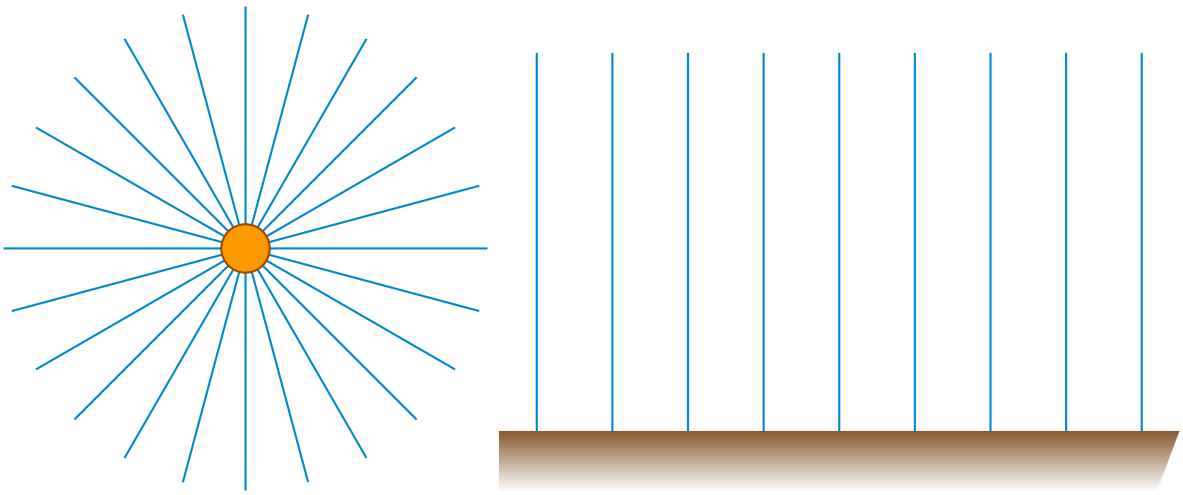
A planet that orbits around the sun experiences acceleration, and thus a force is acting on it.
The attractive force of gravitation between two point masses (idealised bodies with no physical size) are:

$$F = G\frac{M_1 M_2}{r^2}$$

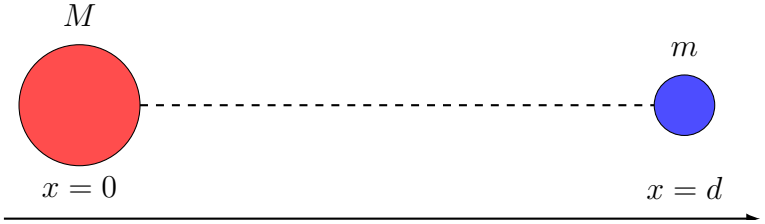
where $G = 6.67 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$, and r is the center-to-center distance between the spherical bodies.
The direction of the force is along the line joining the two masses.
The gravitational force is always attractive. The magnitude of the force on each mass is the same.
(Spherical bodies with uniform distribution of mass can be considered as point mass, as the center of mass lies within the center of the spherical body.)

Gravitational Field Strength

A mass M is said to create a gravitational field in the space around it. A formal definition: region of space where a mass experiences a (gravitational) force.
The gravitational field strength, at a certain point, is the gravitational force per unit mass experienced by a small point mass m placed at that point: $g = \frac{F}{m}$, or $\frac{GM}{r^2}$ for spherical bodies with mass M.

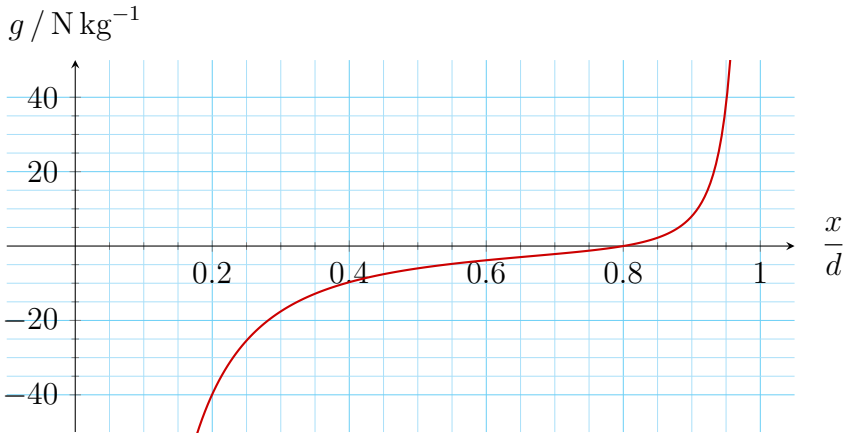


The gravitational field strength around a single point or spherical mass is radial, which means that it points towards the centre of the mass creating the field. The field is not uniform: magnitude of g decreases as we move away from M.
The other case: a uniform field with constant gravitational field strength. (e.g. constant a of freefall)



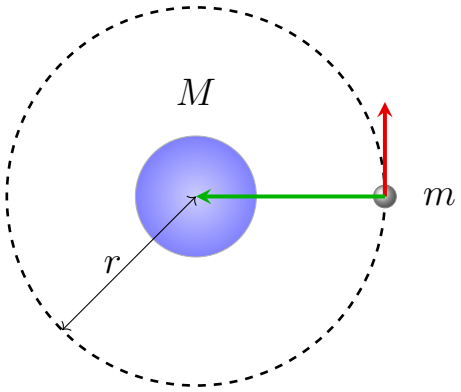
At a point that is a distance x from the centre of the larger mass the net gravitational field strength is:

$$g = -\frac{GM}{x^2} + \frac{Gm}{(d-x)^2}$$



Variation of g with distance from the larger mass.

Orbital Motion

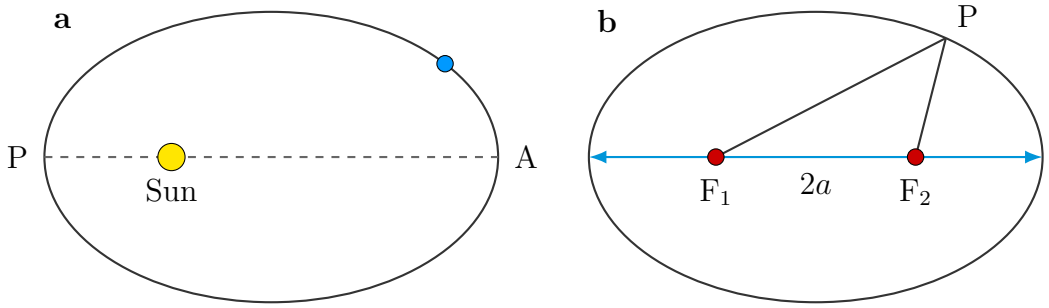


$$v = \sqrt{\frac{GM}{r}} \; ; \; T^2 = \frac{4\pi^2 r^3}{GM}$$

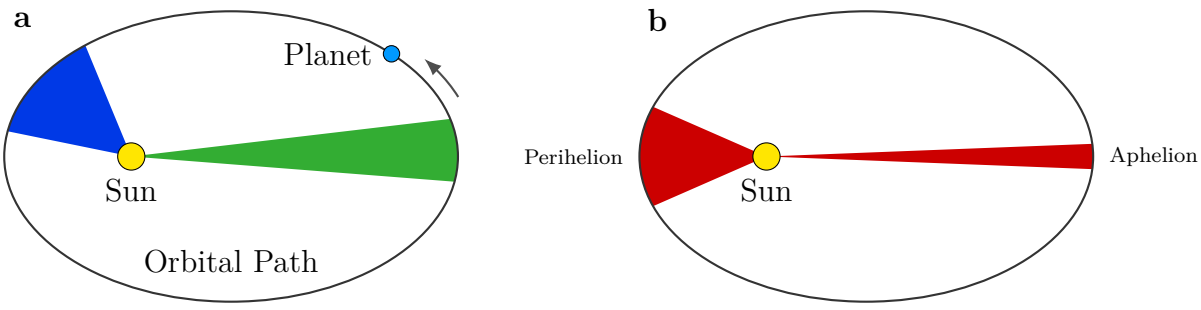
where T is the time for one complete revolution, period of revolution.

Kepler's Laws

- First Law: planets move on ellipses with the sun at one of the foci of the ellipse.



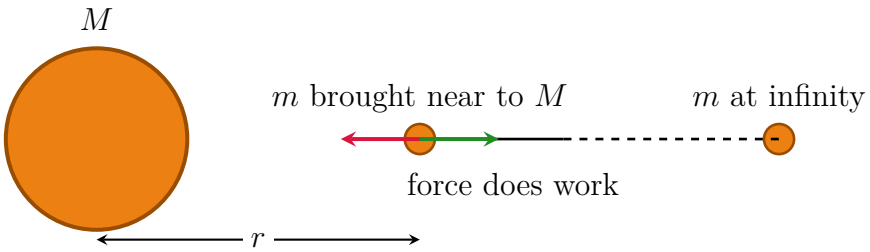
- Second Law: the line joining a planet and the sun sweeps equal areas in equal times.



- Third Law: the period of revolution of a planet around the sun is proportional to the $\frac{3}{2}$ power of the semi-major axis of the ellipse. $\left(T^2 = \frac{4\pi^2 a^3}{GM}\right)$.
  For a circular orbit with radius r, $r = a$.

17.2 Gravitational potential and energy

The gravitational potential energy of two bodies is the work that was done in bringing the bodies to their present position from when they were infinitely far apart. This energy is stored in the gravitational field.



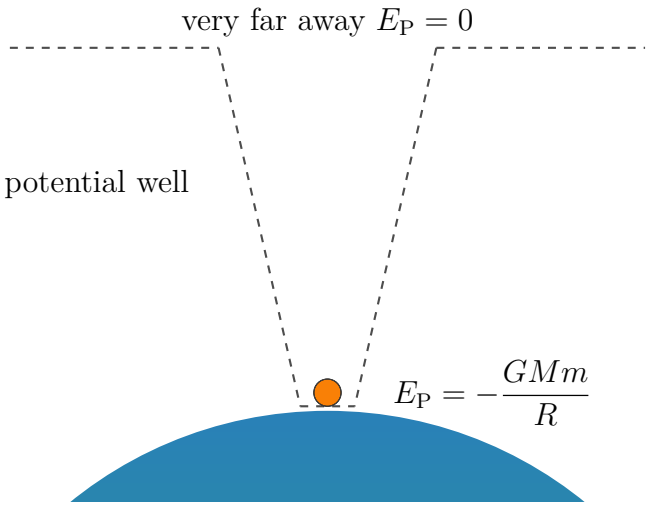
* : Assume M is fixed and that only m has moved.
This energy belongs to the pair of masses M and m and not just to one of them.
(Assume no kinetic energy is involved. Masses move at a small constant speed.)

The total work done in moving the mass m from infinity to a distance r from the centre of a spherical mass M is:

$$W = \int_{\infty}^{r} \frac{GMm}{r^2} \mathrm{d}r = -\left[\frac{GMm}{r}\right]_{\infty}^{r} = -\frac{GMm}{r} \Rightarrow E_p = -\frac{GMm}{r}$$

Why is the gravitational potential energy negative:

- Note that the force is directed to the right, while the motion is directed left.
- No potential energy initially $\Rightarrow$ Increase E_K for m and decreased potential energy overall



Gravitational Potential

The gravitational potential, V_g, at a point P in a gravitational field is the work done per unit mass in bringing a small point mass from infinity to point P.

$$V_g = \frac{W}{m} = -\frac{GM}{r};\ E_p = mV_g;\ W = m\Delta V_g$$

where W is the work done on the mass by the external agent moving the mass.

Field and potential

Gravitional field strength can be calculated using:

$$g = \frac{\Delta V_g}{\Delta r} \left(= -\frac{dV_g}{dr} \right)$$

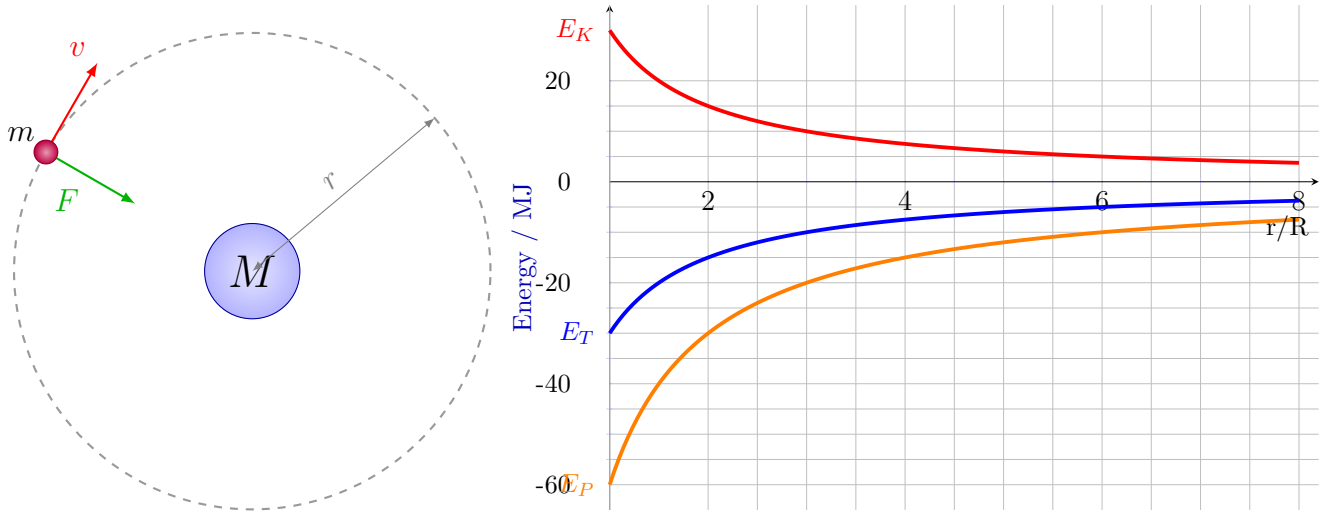
where Δr is the difference between two points close to each other and ΔV_g is the difference between gravitational potential.

Equipotential surface

An equipotential surface consists of points (that have same distance from the source) that have the same gravitational potential.

Equipotential surfaces for two equal spherical masses and two unequal masses. Field lines are normal to the equipotential surfaces. The work done to move a point mass from one point to another point on the same equipotential surface (from A to B) is zero: $\Delta V_g = 0$.

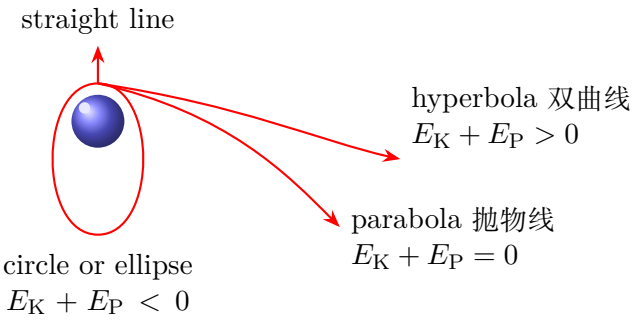
17.3 Motion in a gravitational field



$$E_T = E_K + E_p = \frac{1}{2}mv^2 - \frac{GMm}{r} = \frac{1}{2}m\left(\sqrt{\frac{GM}{r}}\right)^2 - \frac{GMm}{r} = -\frac{GMm}{2r}$$

- If the total energy is positive, the object will follow a hyperbolic path and never return.
- If the total energy is zero, the object will follow a parabolic path to infinity, where it will just about stop. It will never return.

- If the total energy is negative, the object will go into a circular or elliptical orbit (or crash into the planet if the launching speed is too low).



Escape speed

Escape speed is the minimum speed a body in a gravitational field must have in order to reach infinity.
The condition to escape from any point in a gravitational field is that the total energy at that point must be zero.

$$v_{esc} = \sqrt{\frac{2GM}{r}}$$

Reasons behind weightlessness

Essentially, the resultant force on the astronaut is zero.
In the case where W is the weight, F is the net force, N is the reaction force from the floor of the spacecraft:

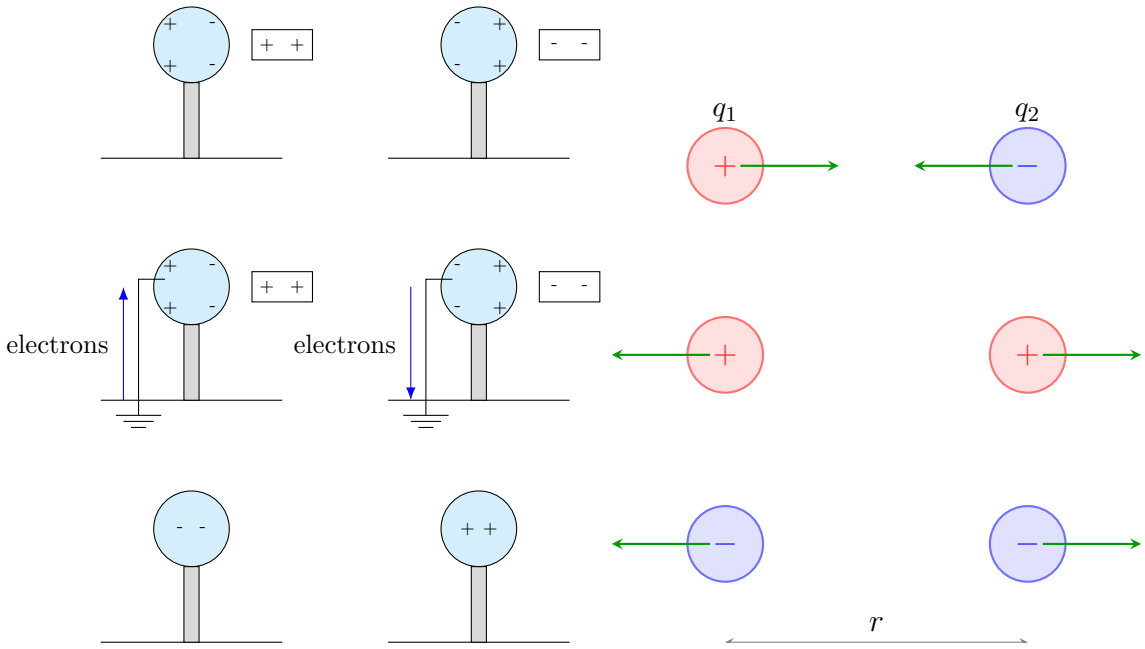
$$\begin{aligned} F &= W - N \\ &\Downarrow \\ N &= W - F \\ &= \frac{GMm}{r^2} - \frac{mv^2}{r} \\ &= \frac{m}{r}\left(\frac{GM}{r} - v^2\right) \\ \because v_{orbit}^2 &= \frac{GM}{r} \\ \therefore N &= 0 \end{aligned}$$

18 Electric and magnetic fields

18.1 Electric charge, force and field

Properties of electric charges:

1. Law of conservation applies to electric charges: they can't be created or destroyed.
2. It is a quantizied quantity: this means that the amount of electric charge on a body is always an integral multiple of a basic unit. ($1e = 1.6 \times 10^{-19}$ C)



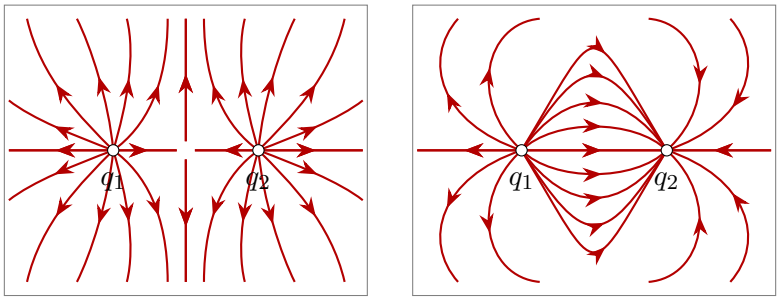
Coulomb's Law states that, the electric force between two point charges q_1 and q_2 that is separated by a distance of r is given by

$$F = k\frac{q_1 q_2}{r^2}$$

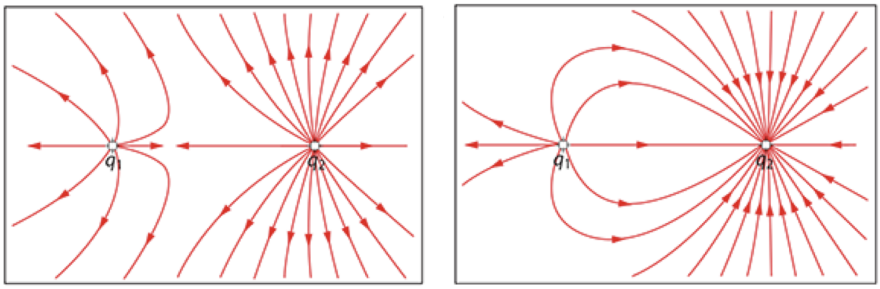
where $k = \dfrac{1}{4\pi\varepsilon}$ is the Coulomb constant (the ε is the permittivity).
($k = 8.99 \times 10^9$ N m^2 C^{-2}, $\varepsilon = 8.85 \times 10^{-12}$ C^2 N^{-1} m^{-2} (in vacuum))

Electric Field

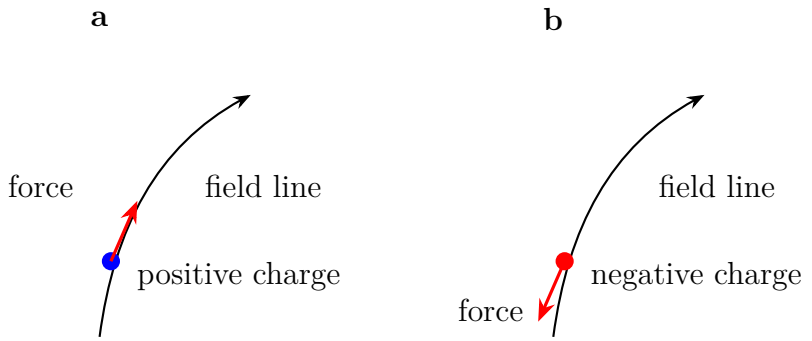
Electric Field Strength, E, is the electric force per unit charge experienced by a small, positive, point charge q: $\vec{E} = \dfrac{\vec{F}}{q}$. The direction of the electric field is the same as the direction of the force experienced by a positive charge at the given point.
The formula $E = k\dfrac{Q}{r^2}$ applies to a point charge / outside of the spherical charge.



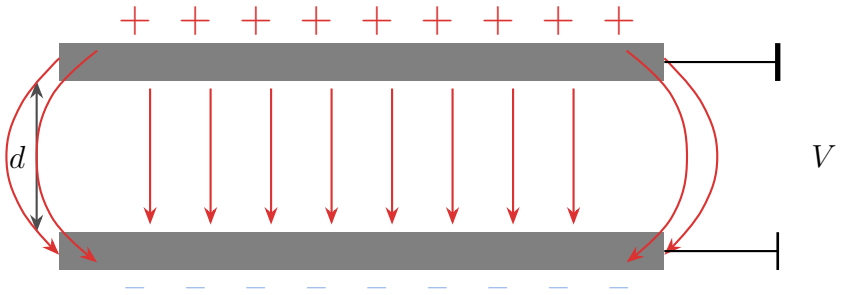
Field lines for two equal positive charges and two equal and opposite charges.



Field lines for unequal and positive charges. Field lines for unequal and opposite charges.

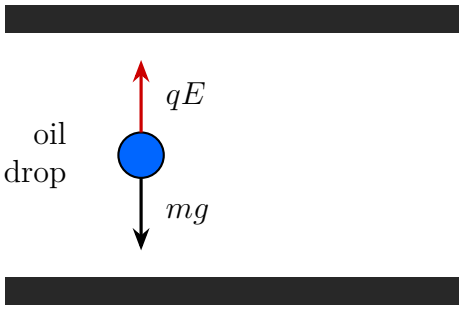


- The electric force on a positive charge is tangent to the field line in the direction of the arrow of the field line.
- The force on a negative charge is tangent to the field line in a direction opposite to the arrow.



The magnitude of the **uniform** electric field strength in between the plates is given by $E = \frac{V}{d}$.

The Millikan oil drop experiment



In this experiment, oil drops that were charged by X-rays were allowed to enter a region of uniform electric field between the parallel plates. Some oil drops were observed to be at rest in between the plates. This means that the weight of the oil drop is matched by the electric force.

$$mg = qE$$
$$mg = q\frac{V}{d}$$
$$\Downarrow$$
$$q = \frac{mgd}{V}$$

Millikan measured the mass of the oil drop by turning off the electric field and observed the oil drop that is now falling vertically down. The oil drop quickly achieves terminal speed because now the weight is matched by the drag force given by Stoke's Law:

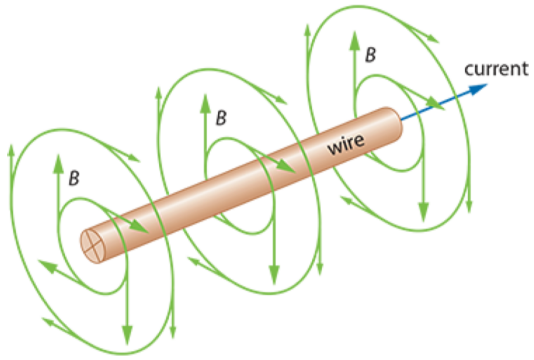
$$mg = 6\pi\eta r v$$
$$r = \frac{mg}{6\pi\eta v}$$
$$r = \frac{\rho\left(\frac{4}{3}\pi r^3\right)g}{6\pi\eta v}$$
$$r = \sqrt{\frac{9\eta v}{2\rho g}}$$

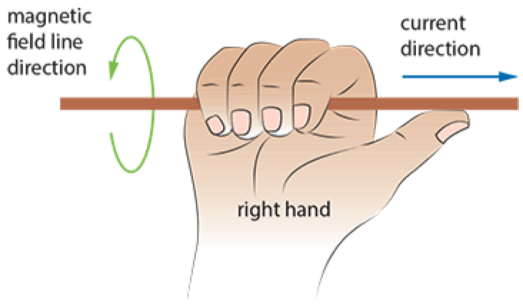
Find radius $\Rightarrow$ Find mass $\Rightarrow$ Find charge, is multiple of $e = 1.6 \times 10^{-19}$ C

18.2 Magnetic field and force

A magnetic field is produced by electric currents. An electric field is produced by electric charges. The vector representing a magnetic field is called magnetic flux density and its symbol is B.

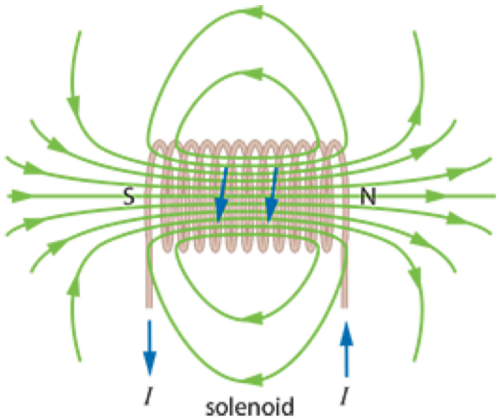
In permanent magnets, such as bar magnets and horseshoe magnets, the magnetic field is created by microscopic currents due to the motion of electrons in the atoms of the material. These microscopic currents create microscopic magnetic fields; in materials called ferromagnetic these tiny fields are aligned so that a large observable macroscopic magnetic field is produced.

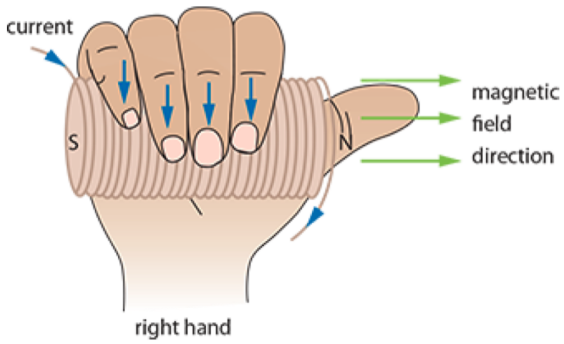




Solenoids / coils

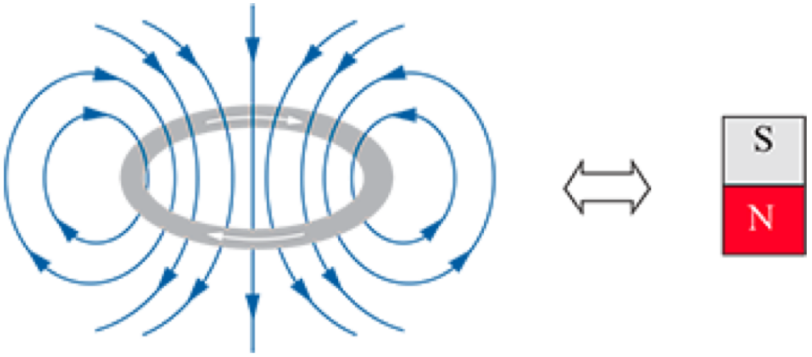
The magnetic field lines within the solenoid are fairly uniform, indicating that the field is roughly constant in both magnitude and direction. Notice the similarity between the field outside the solenoid and that around the bar magnet. Notice also that magnetic field lines always exit from a north (N) pole and enter at a south (S) pole.





The field of a single loop of current

For all practical purposes the field of a ring, just like that in the exterior of a coil, may be taken to be the same as that of a bar magnet.



Magnetic force on a moving charge

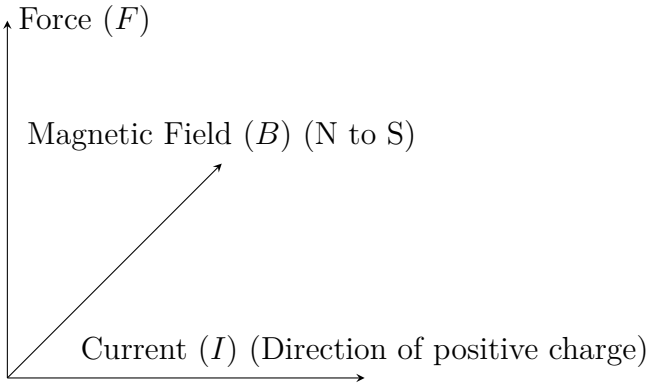
Experiments show that an electric charge moving in a region of magnetic field experiences a new type of force called a magnetic force. There is an exception to this rule: if the velocity of the charge is parallel to the direction of the field, the magnetic force is zero.

A charge q moving with speed v in a region of magnetic field of magnetic flux density B will experience a magnetic force F given by:

$$F = qvB\sin\theta$$

Leave $\theta = 90°$ for most of the time, and $F = qvB$ would be obtained this way. The force is at right angles to both the velocity vector and the magnetic field. **Use left hand rule to find the direction of the force.** Note that $\vec{F} = q\vec{v} \times \vec{B}$, the vector product of two. These rules give the direction of the magnetic force on a positive charge. To find the force on a negative charge, pretend the charge is positive and then reverse the force direction you find.

(Through this way, we can obtain the formula for magnetic flux density as $B = \dfrac{F}{qv\sin\theta}$. A magnetic flux density of 1 T produces a force of 1 N on a charge of 1 C moving at 1 m/s at right angles to the direction of the field.)

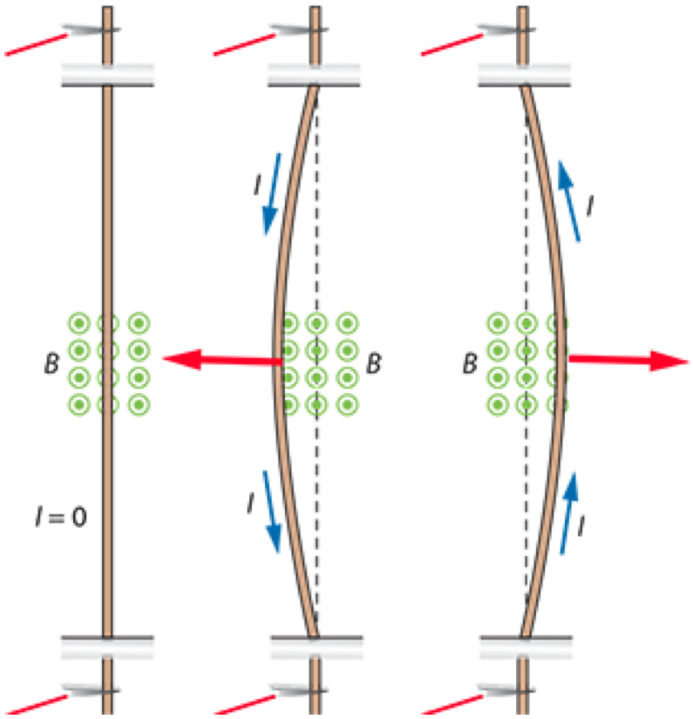


The magnetic force on a current-carrying wire

A current-carrying wire placed in a magnetic field will experience a magnetic force because there is a force on the moving charges in the wire. The formula for the magnetic force on a length L that finds itself in the region of the field is:

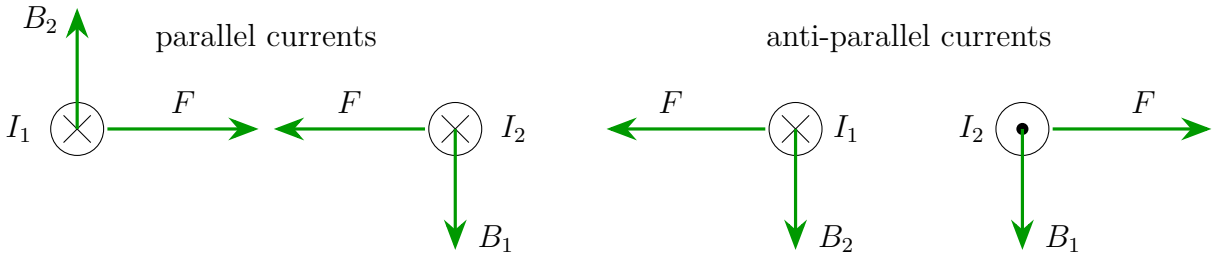
$$F = BIL\sin\theta$$

where θ is the angle between the current and the direction of the magnetic field.



The force between two current-carrying wires

The currents are different in the two wires, so the magnetic fields are different, but the two forces are equal in magnitude. (In the diagram below, $B_1 \neq B_2$.) If the currents are parallel, the forces are attractive, and if they are anti-parallel, the forces are repulsive.



Extension: Magnitude of Magnetic Flux density created

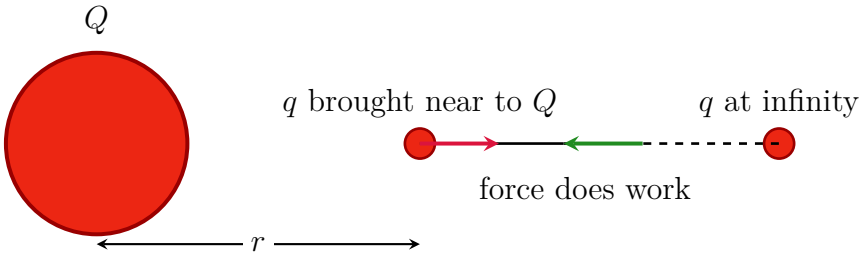
The magnitude of the magnetic flux density created at a distance r from a long straight wire carrying current I is given by the formula $B = \dfrac{\mu_0 I}{2\pi r}$, where $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1} is a constant of physics called magnetic permeability.

The magnetic flux density created by the left wire at the position of the right wire is $B_1 = \dfrac{\mu_0 I_1}{2\pi r}$. Therefore the force experienced by a length L of the right wire is $F = B_1 I_2 L = \dfrac{\mu_0 I_1 I_2 L}{2\pi r}$.

Thus, we can obtain the formula for force per unit length: $\dfrac{F}{L} = \mu_0 \dfrac{I_1 I_2}{2\pi r}$.

18.3 Electric potential and electric potential energy

An electric charge creates electric potential. Two electric charges also share electric potential energy. The formulas we derived for gravitation carry over to electricity essentially by replacing mass everywhere by charge.



The electric potential at a point P in an electric field is the amount of work done per unit charge as a small positive test charge q is moved from infinity to P: $V_e = \frac{W}{q}$.
The work done in moving a charge q from infinity to point P is the electric potential energy of the system, $E_P = qV_e \left(= \frac{kQq}{r}\right)$. (Work done is expressed as $W = \frac{kQq}{r}$ by using calculus, therefore $V_e = \frac{kQ}{r}$. The value of $k = 8.99 \times 10^9$.)
Electric potential and electric potential energy are scalar quantities, just like gravitational potential and gravitational potential energy. The work done is independent of the path followed.
The work done W in moving charge q from A to B is: $W = q\Delta V_e$, where ΔV_e is the potential difference between A and B.
Electric potential is a scalar quantity. Therefore, if we have i charges q_i with distance of r_i to the point P, then the electric potential at point P would be $V_e = k\sum \frac{q_i}{r_i}$.
For a point P outside the sphere and at a distance r from the centre of the sphere, the potential at P is $V_e = \frac{kQ}{r}$.
On the surface of the sphere the potential is $V_e = \frac{kQ}{R}$ where R is the radius of the sphere.
But at any point inside the sphere, the electric potential is constant and has the same value as the potential at the surface.

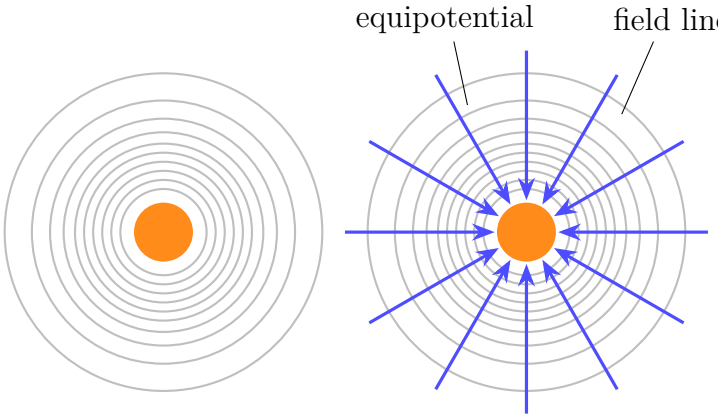
There is a deep connection between potential and field for both gravitation and electricity. To see this connection let us move a point charge q from a point P to a neighbouring point Q. There is a potential difference ΔV_e between the two points, and the points are a distance Δr apart.

$$W = qV_e = F \times d = qE\Delta r \Rightarrow E = \frac{\Delta V_e}{\Delta r}\left(-\frac{dV_e}{dr}\right)$$

In a graph showing the variation with distance of the potential, the gradient (slope) of the graph is the magnitude of the field strength. This applies to both gravitational and electric fields.

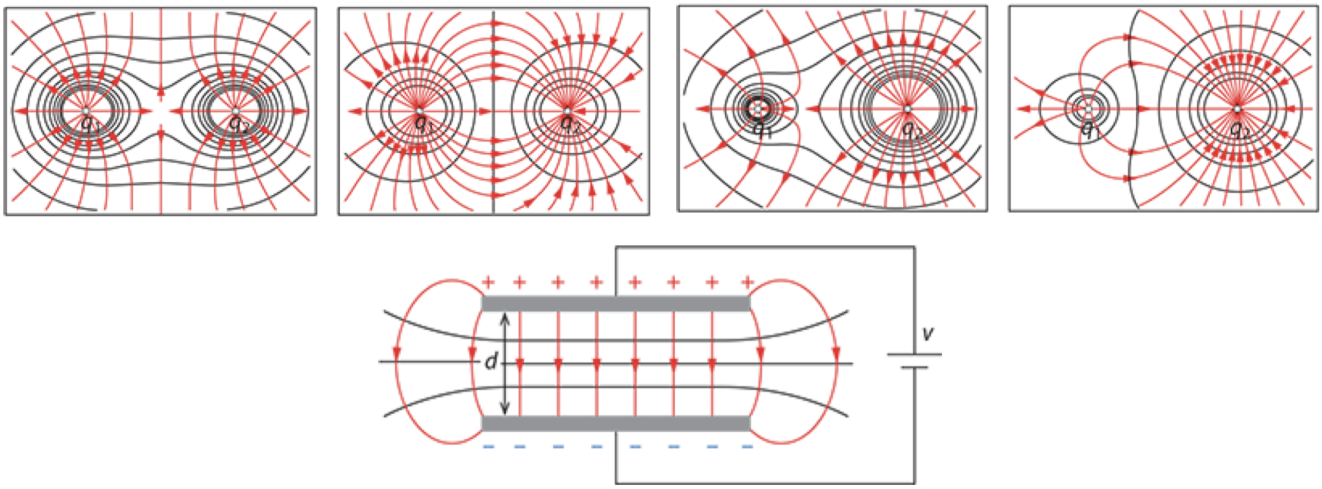
The connection between field lines and equipotential surfaces

Equipotential surfaces for a point charge.



The field lines are at right angles to the equipotential surfaces.

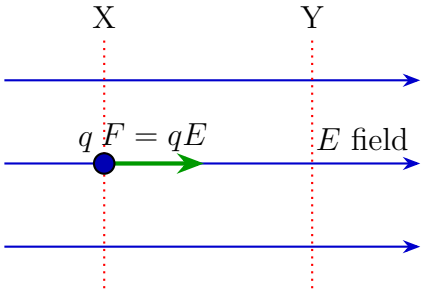
Explanation for why field lines are at right angles to the equipotential surfaces:
To move a charge from one point on an equipotential surface to another requires zero work. If the field lines were not normal to the equipotential surfaces, there would be a component of the field along the equipotential and so a force on the charge. As the charge moved this force would do work, which contradicts the fact that the work should be zero.

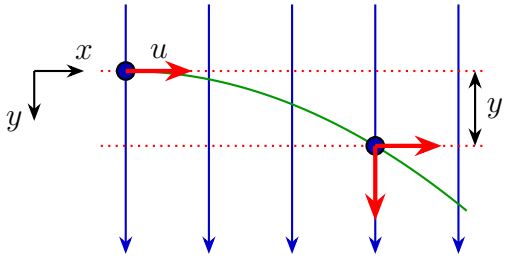


19 Motion in electric and magnetic fields

19.1 Motion in an electric field

The path of an electric charge in a uniform electric field is either a straight line or a parabolic path. We have a straight line if the charge is initially at rest or if its original velocity is parallel to the electric field. In all other cases we have a parabolic path.
A positive particle will experience a force qE in the direction of the field. It will also accelerate in the direction of the field, $a = \dfrac{qE}{m}$.
The positive charge increases its kinetic energy (= work done by eletric force, qV) as it moves to the right.





In the second figure:

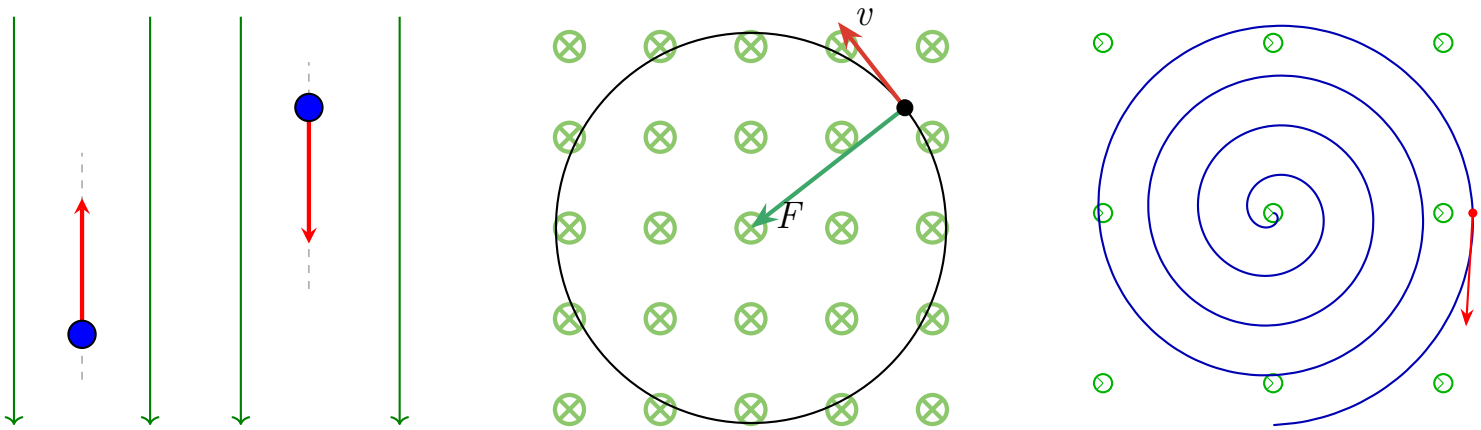
- the particle will follow a parabolic path
- the component of velocity normal to the field will remain constant: $v_x = u$
- the component of velocity parallel to the field will increase uniformly: $v_y = \dfrac{qE}{m}t$
- The kinetic energy of the particle increases as time goes on because the electric force is doing work on the particle.
- After time t, the particle will move a horizontal distance $x = ut$ and fall a vertical distance $y = \dfrac{1}{2}\dfrac{qE}{m}t^2$.

- $$\begin{aligned} E_K &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}m\left(v_x^2 + v_y^2\right) \\ &= \frac{1}{2}m\left(u^2 + \left(\frac{qE}{m}t\right)^2\right) \end{aligned}$$
- $\therefore \Delta E_K = \frac{1}{2}m\left(\frac{qEt}{m}\right)^2 = Fy$, work done by the electric force.

Equations involved in trajectory in Chapter 1 also applies here.

19.2 Motion in a magnetic field

- In the simple case where the velocity of the charged particle is parallel or anti-parallel to that of the uniform magnetic field, the magnetic force is zero and so the particle will move along a straight line.
- A positive charge in a uniform magnetic field that is directed into the plane of the paper. The velocity is at right angles to the magnetic field. The magnetic force is at right angles to both the velocity and the magnetic field so the particle will follow a circular path; the centripetal force is provided by the magnetic force.
  Radius $R = \frac{mv}{qB}$, time to make one complete revolution $T = \frac{2\pi m}{qB}$ (independent of the speed).
- If the positively charged particle loses energy as it moves along the circular path, the speed will decrease, and from the formula for the radius we see that the radius will also decrease. This means that the circular path will become a spiral.

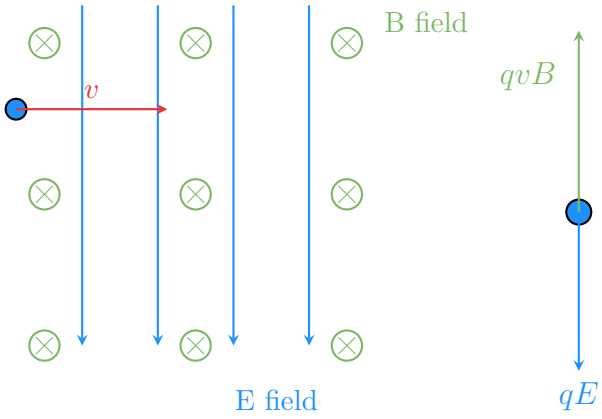


Work done and magnetic forces

Since the magnetic force is always normal to the velocity of the charge, it follows that it cannot do any work. This means the kinetic energy of the particle cannot change while in the region of magnetic field.

Motion in crossed electric and magnetic fields

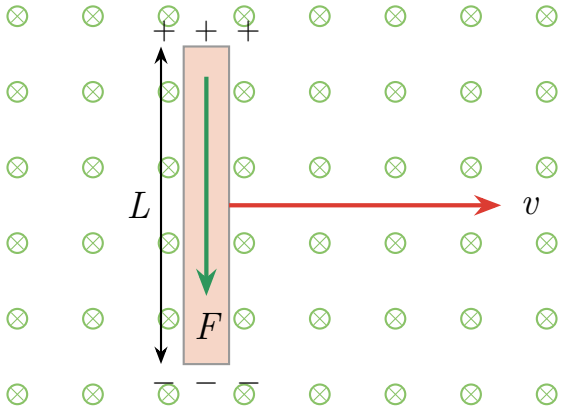
- If two forces are equal, the particle will continue moving on a straight line undeflected from its original path, $E = vB$.
- If the particle enters with speed greater than $v = \frac{E}{B}$, the particle will move upwards (larger F_m).
- If the speed is less than v, the particle will move downwards.



20 Electromagnetic induction

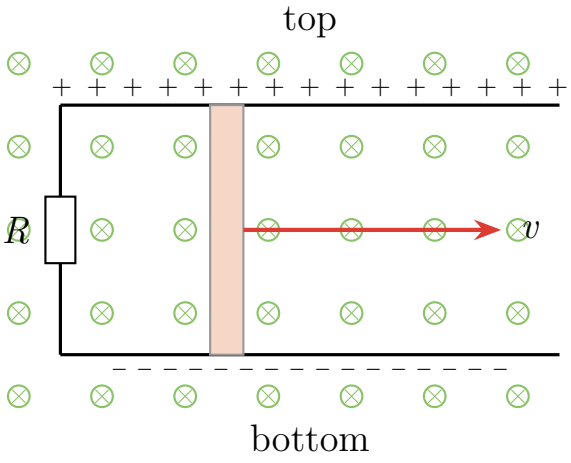
20.1 Electromagnetic induction

Motional *e.m.f*



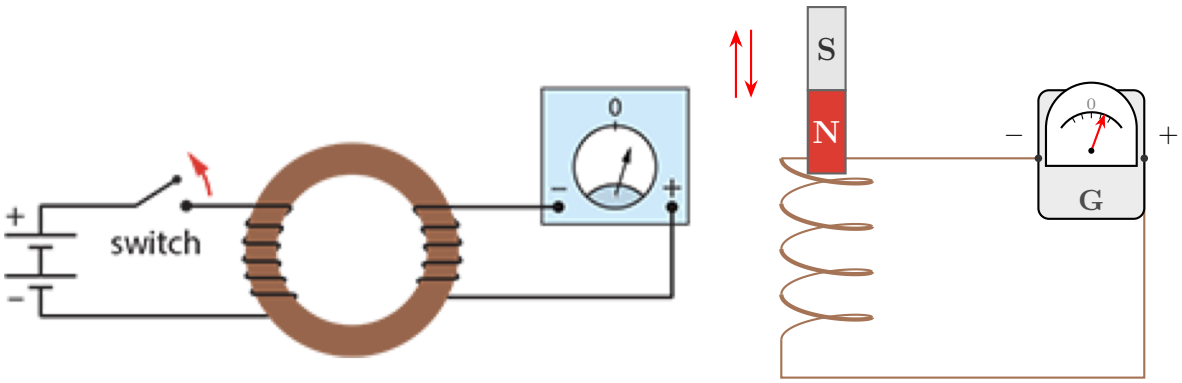
The rod is conducting. As it moves, the electrons within it also move from left to right. The magnetic field will exert a force on these moving electrons. The force on the electrons is directed downward. This means that the lower end of the rod has a net negative charge and the top end has an equal net positive charge. The flow of electrons towards the bottom end of the rod will stop when the electrons already there are numerous enough to push any new electrons back by electrostatic repulsion. There is, in other words, an electric field established in the rod whose direction is from top to bottom.

$E = \frac{\varepsilon}{L}$, where ε is the induced *e.m.f* and L is the length of the rod. The induced *e.m.f* is given by $\varepsilon = BLv$. It is important to note that except for a very short interval of time initially, no current exists in the rod.



Now there exists a current of $I = \dfrac{BvL}{R}$ in the circuit, and the moving rod behaves as a battery. In this diagram the current is going counterclockwise.
Notice also that now that we have a current, the rod needs to be pushed if it is to continue to move at constant speed. Now it's experiencing a magnetic force of $F = \dfrac{vB^2L^2}{R}$ directed to the left.
For speed to remain constant a force of equal magnitude needs to act on the rod, directed to the right. The power P generated by this force $= \dfrac{(BvL)^2}{R}$, consistent with the power dissipated in the circuit as thermal energy.

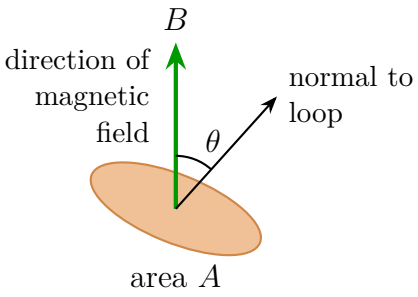
Magnetic flux and magnetic linkage



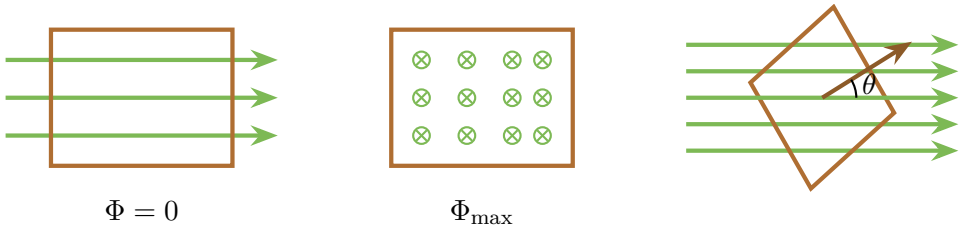
The current is created as a result of the motion of the magnet relative to the coil. The current registered by the galvanometer increases when:

- the relative speed of the magnet and the coil increases
- the strength of the magnet increases
- the number of turns increases
- the area of the loop increases
- the magnet moves at right angles to the plane of the loop.

The magnetic flux through the loop of wire is given as $\Phi = BA\cos\theta$, where B is the magnetic flux density, A is the area of the loop and θ is the angle between the normal to the plane of the loop and the direction of the magnetic field.



The magnetic flux multiplied by the number of turns in a loop is called the flux linkage, $\Phi = NBA\cos\theta$.
The intuitive picture of magnetic flux is the number of magnetic field lines that pierce the loop area.



$\Phi = 0$ Φ_{max}

Faraday's Law

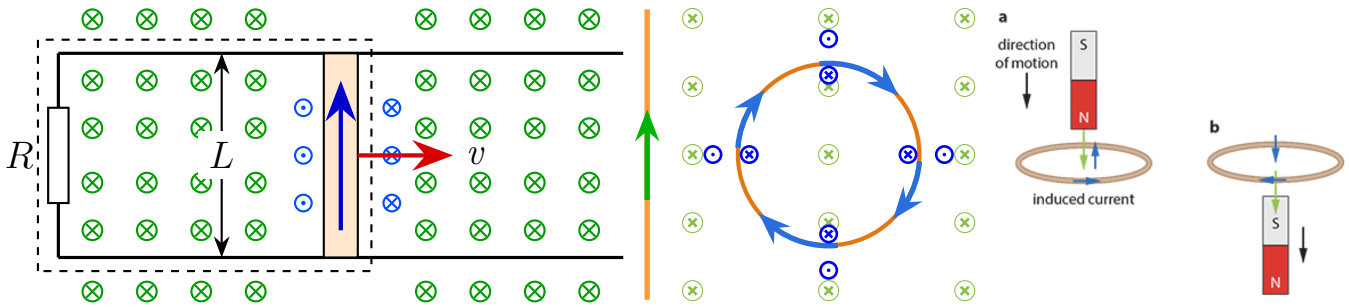
The current generated by a magnetic field is correlated with the rate of change of magnetic flux linkage through the loop. Changing flux creates an induced $e.m.f$, not necessarily a current. There will only be a current if the loop is conducting; that is, if the resistance of the circuit is not infinite. The induced $e.m.f$ is equal to the (negative) rate of change of magnetic flux linkage:

$$\varepsilon = -N\frac{\Delta\Phi}{\Delta t}\left(= -N\frac{\mathrm{d}\Phi}{\mathrm{d}t}\right)$$

Lenz's Law

It states that the direction of the induced current is such that it opposes the change in magnetic flux that produced it. This is a consequence of the conservation of energy and is reflected in the negative sign in Faraday's Law.

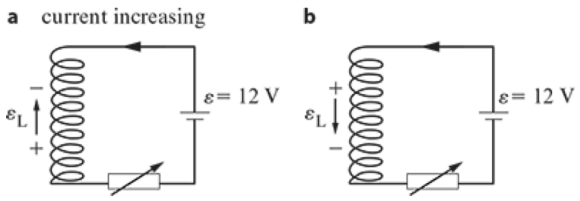
- If the flux increases, the induced $e.m.f$ will create a current whose magnetic field is opposite to the external field.
- If the flux decreases, the induced $e.m.f$ will create a current whose magnetic field is parallel to the external field.



- The rod is made to move to the right. The magnetic flux through the loop is increasing and a current will be established in the rod. By Lenz's law, the current must produce a magnetic field opposite to the existing magnetic field, i.e. out of the page. (Opposing the increase of the magnetic flux ⇒ decrease the magnetic flux, and one way is to set up a magnetic field inside the loop of the opposite direction) So the current must be counterclockwise.
- A loop of wire near a straight wire in which the current is decreasing. The current in the straight wire creates a magnetic field into the page at the position of the loop. The induced current in the loop produces a magnetic field in the same direction as to oppose the change in flux. From the right-hand rule, the current must be clockwise.
- The flux in the loop is increasing because the magnetic field at the loop is getting larger as the magnet approaches. The induced current must then oppose the increase in the flux. This can be done if the induced current produces a magnetic field in the opposite direction to that of the bar magnet. Thus, the current will flow in a counterclockwise direction when looked at from above.

Self induction

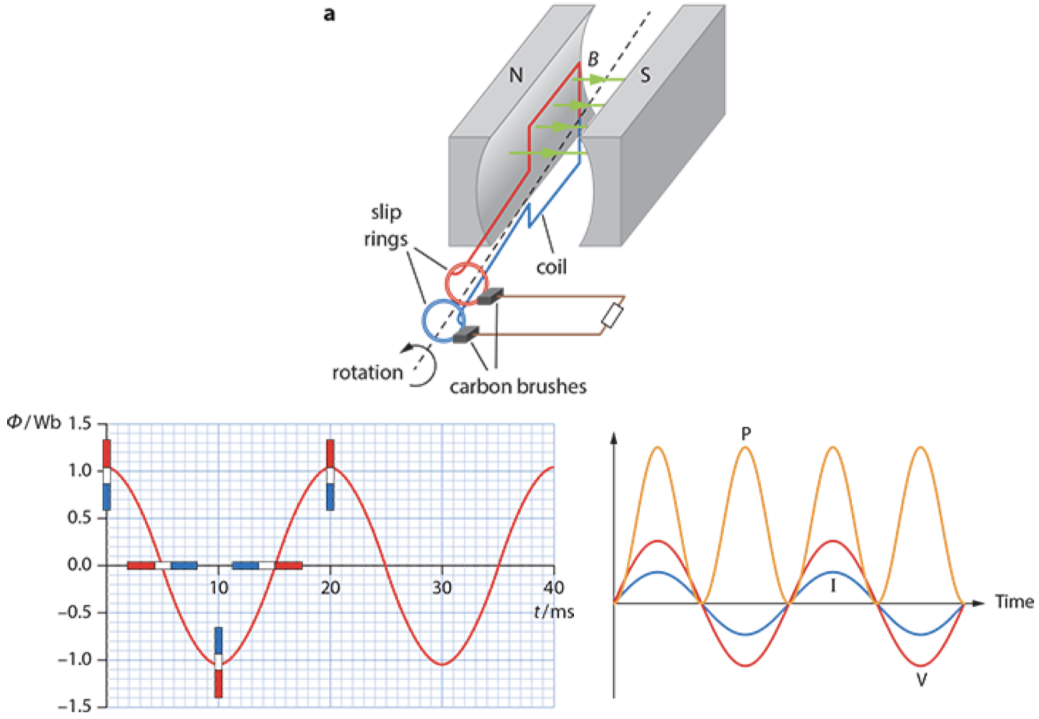
By changing the resistance of R the current in the circuit will be changing; the magnetic flux linkage in the coil will therefore also be changing and an $e.m.f$ ε_L will be induced in the coil. By Lenz's law, the induced $e.m.f$ will oppose the change in the current. The polarity of the induced $e.m.f$ is such as to drive current in the opposite direction to that of the current from the cell. In this way the induced $e.m.f$ opposes the increase in the current by decreasing the current in the circuit.



20.2 Generators and alternating current

AC Generator

A coil is made to rotate in a region of magnetic field. The ends of the coil are firmly attached to two slip rings that rotate along with the coil. The slip rings touch carbon brushes that transfer the current into an external circuit.



The flux linkage in the coil changes as time goes on according to a cosine function. The red, white and blue bar that is superposed on the graph indicates the position of the coil as we look at it along the axis of rotation. θ is the angle between the magnetic field and the normal to the coil. The coil rotates with angular speed of rotation, $\therefore \theta = \omega t = 2\pi f t \Rightarrow \Phi = NBA\cos(2\pi f t)$. ($f$ = revolutions per second). The $e.m.f$ induced is thus $V = -\dfrac{d\Phi}{dt} = 2\pi f NBA\cos(2\pi f t)$. $V_{\max} = 2\pi f NBA$ when magnetic flux=0, $V_{\min} = 0$ when the magnetic flux is at its maximum / minimum.

For alternating voltage and alternating current, the electrons oscillate back and forth with the same frequency as that of the voltage. The flux and the $e.m.f$ are out of phase by $90°/\dfrac{\pi}{2}$. The current is in phase with the $e.m.f$, which can be given by $I = I_{\max}\sin(2\pi f t) = \dfrac{V_{\max}}{R}\sin(2\pi f t)$, where the resistance R is given in the circuit.

Power in AC circuits

$$P = V_{\max} I_{\max}\sin^2(2\pi f t)$$
$$P_{\max} = V_{\max} I_{\max}$$

The average power dissipated is half the peak value. Key conclusion: $V \propto f, I \propto f, P \propto f^2$.

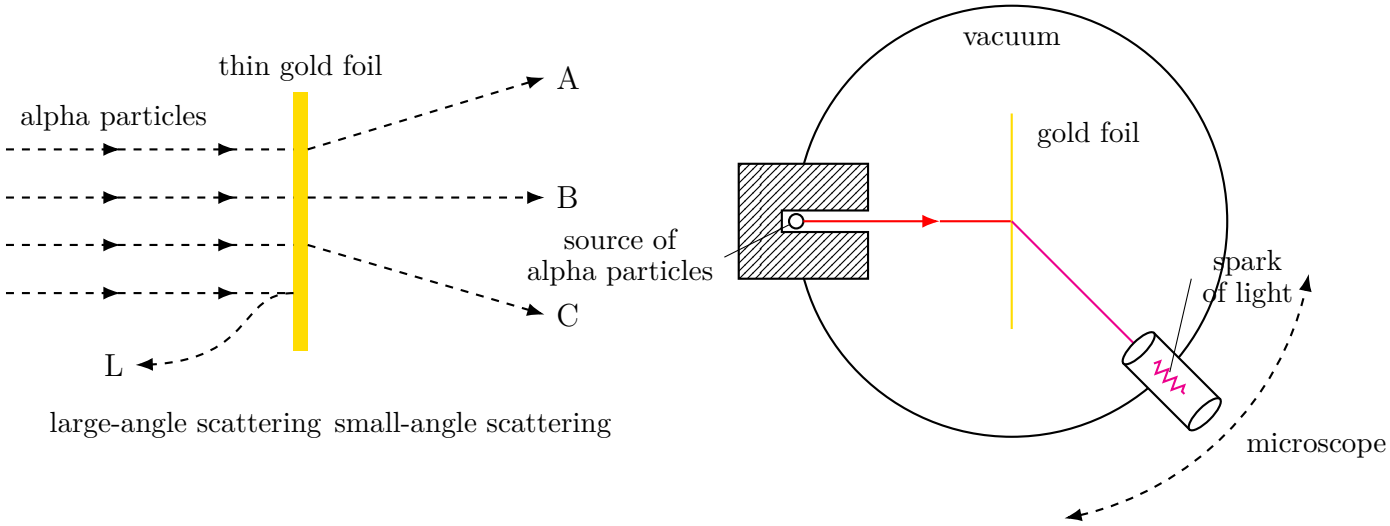
Unit E Nuclear and quantum physics

21 Atomic physics

21.1 The structure of the atom

α particles were directed at a thin gold foil in an evacuated chamber. 你Rutherford哥哥又限时返场了。They found that:

- the great majority of the alpha particles went straight through the foil with no or little deflection; they were detected at very small scattering angles: they were due to the electric force of repulsion between the positive charge of the gold atoms and the positive charge of the alpha particles. The effect of the electrons of the gold atoms on the path of the alpha particles is negligible.
- very occasionally, alpha particles were detected at very large scattering angles. **The repulsion force must be very large and thus the separation must be very small. This could only be explained if the positive charge on the atom is within a sphere that is much smaller than the sphere that Thomson had imagined.** This and subsequent experiments confirmed the existence of the nucleus inside the atom: a small, massive object carrying the positive charge of the atom and most of its mass.



结论: The Rutherford experiment established the Rutherford model of the atom: the atom consists of a very small, massive, positively charged nucleus, and the electrons orbit the nucleus like planets around the sun.

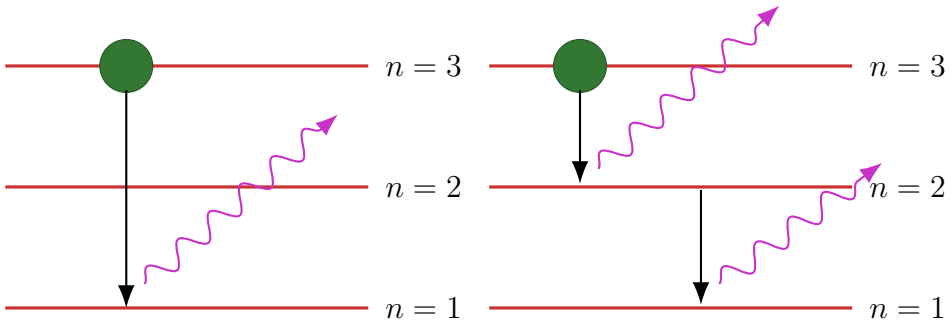
Discrete energy

If you expose a container of gas at low pressure to a strong electric field, light will be emitted from the gas. You can show that the emitted light contains many different wavelengths by passing it through a prism or diffraction grating. The result is a series of bands of light at different wavelengths. The set of possible wavelengths that can be emitted is called the emission spectrum.

Bohr argued that an atom has discrete energy, i.e. it can have one out of a specific set of values. He represented the possible energies (refers to $E_T = E_K + E_P$) with an energy level diagram. For example, the hydrogen atom in the n^{th} state has an energy of $E = -\dfrac{13.6}{n^2}$ eV, $n \in \mathbb{Z}^+$. The negative sign in the energy tells us that the electron is trapped within the atom. To have the electron leave the atom, energy must be provided.
Bohr suggested that an atom can make a transition from a state of higher energy to a state of lower energy by emitting a photon. The energy of the emitted photon is the difference in energy of the two levels.

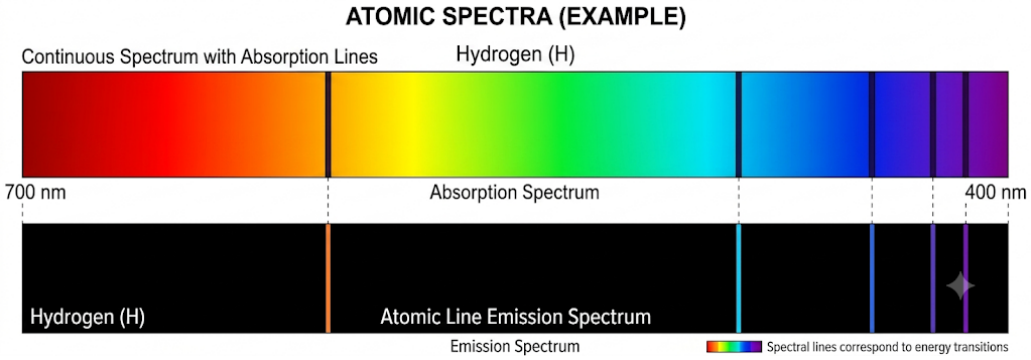
The energy of a photon is given by $E = hf = h\frac{c}{\lambda}$, where $h = 6.63 \times 10^{-34}$ J s is Planck's constant.

In a transition from a high to lower energy state, the wavelength of the emitted photon is given by $\lambda = \frac{hc}{\Delta E}$.
When undisturbed, the electron in the atom will occupy the lowest energy state, called the ground state. When energy is supplied to the atom, the electron can be excited to a higher energy state, called an excited state. From the excited state, the electron can return to the ground state by emitting photons, either going to ground state straightaway or passing through multiple transition energy levels, and the decision between the two chances is an issue of chance.



Absorption Spectra

Now imagine sending a beam of white light through a gas. The majority of the atoms in the gas are in their ground state. Electrons in the atoms may absorb photons in the beam and move to an excited state. This will happen only if the photon that is to be absorbed has exactly the right energy. This means that the light that is transmitted through the gas will be missing the photons that have been absorbed. This gives rise to absorption spectra.
Note: Why are the photons missing? This is because the photons are emitted in all directions and not necessarily along the direction the observer is looking.



Experiments show that no two elements have the same wavelengths in their emission and absorption spectra.

21.2 Quantisation of angular momentum

Bohr assumed that the electron could only exist in certain special orbits satisfying the condition

$$mvr = n\frac{h}{2\pi}, \ n \in \mathbb{Z}^+$$

where $L = mvr$, angular momentum of the electron, is quantised. The electron can only occupy orbits for which the angular momentum is an integer multiple of $\frac{h}{2\pi}$.

The total energy E_T of the orbiting electron is

$$E_T = \underbrace{\frac{1}{2}mv^2}_{E_K} + \underbrace{\left(-k\frac{e^2}{r}\right)}_{E_P}$$

The electron is acted upon the electric force, therefore

$$\begin{aligned}\frac{ke^2}{r^2} &= \frac{mv^2}{r}\\ \frac{ke^2}{r} &= mv^2\end{aligned}$$

$$\begin{aligned}E_T &= \frac{1}{2}\frac{ke^2}{r} - \frac{ke^2}{r}\\ &= -\frac{1}{2}\frac{ke^2}{r}\\ &= -\frac{1}{2}\frac{ke^2}{a_0} \times \frac{1}{n^2}\end{aligned}$$

Squaring the quantisation of angular momentum equation gives

$$\begin{aligned}m^2v^2r^2 &= n^2\frac{h^2}{4\pi^2}\\ mv^2 &= n^2\frac{h^2}{4\pi^2r^2}\\ \frac{ke^2}{r} &= n^2\frac{h^2}{4\pi^2r^2}\\ r &= a_0n^2\\ &= \frac{n^2h^2}{4\pi^2ke^2m},\ n \in \mathbb{Z}^+\end{aligned}$$

Therefore the orbital radius can't be arbitrary, but must be one of the discrete values given by the above equation. The Bohr radius is the smallest possible orbital radius, given by

$$a_0 = \frac{h^2}{4\pi^2ke^2m} \approx 5.29 \times 10^{-11}\ \text{m}$$

22 Quantum physics

22.1 Photons and the photoelectric effect

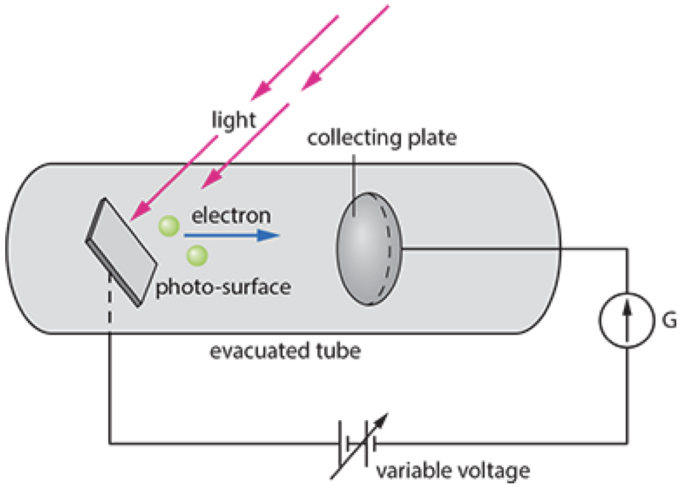
Photons

Einstein suggested that light should be thought of as a collection of quanta, or bundles of energy. Each quantum or bundle of light has energy $E = hf$, where f is the frequency of light. A beam of light is now to be thought of as a very large number of these quanta moving at the speed of light. The total energy of the beam is then Nhf, where N is the number of quanta in the beam. The energy of the beam is therefore an integral multiple of the basic unit hf. The photon is a particle with zero mass and zero electric charge.

In Einstein's theory of special relativity the total energy E, the momentum p and the mass m of a particle are related according to $E^2 = (pc)^2 + (mc^2)^2 = p^2c^2 + m^2c^4$. Therefore the photon has a momentum of $p = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda}$.

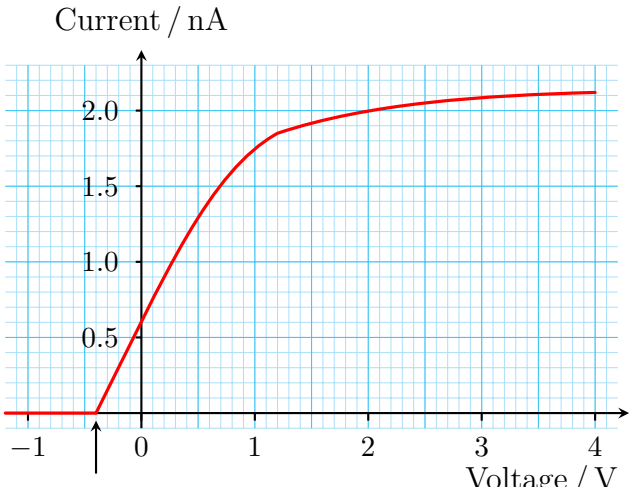
Photoelectric effect

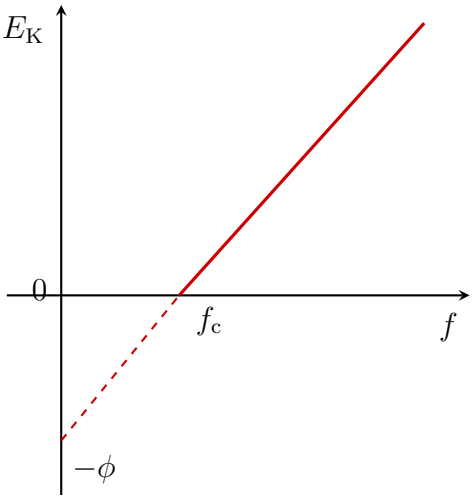
The photoelectric effect is the phenomenon in which light (or other forms of electromagnetic radiation) incident on a metallic surface causes electrons to be emitted from the surface.



The emitted electrons are called photoelectrons. Those electrons that make it to the collecting plate complete the circuit and so we have an electric current that is recorded by the sensitive galvanometer.

- Note that the terminal of the variable power supply connected to the collecting plate is the negative terminal. This means that the collecting plate actually repels the emitted electrons. Only the very energetic electrons will make it to the plate. As the magnitude of the voltage is increased (i.e. made more negative) fewer and fewer electrons make it to the plate; eventually no electron will arrive there and at that point the current becomes zero. The voltage at which the current becomes zero is called the stopping voltage V_s, which constitutes the formula for maximum kinetic energy for emitted photons, $E_{\max} = eV_s$.
- Now the positive terminal is connected to the collecting plate. The electrons are now attracted to the collecting plate and the current increases. As the voltage is increased even more the current saturates, i.e. it approaches a constant value. This is because the collecting plate is so positive that it attracts every single emitted electron.





Several conclusions were made:

1. Changing the intensity of light does not affect the stopping voltage, they release electrons of the same kinetic energy.

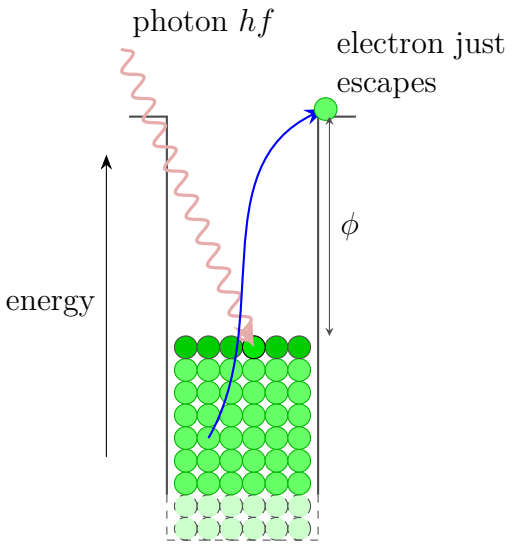
2. As f of light increases, $E_{\max}$ and V_s increases as well.

3. If we plot the kinetic energy of the electrons (which equals eV_s) versus frequency, we find a straight line. There exists a critical / threshold frequency f_c such that no electrons at all are emitted if the frequency of the light source is less than f_c. The frequency varies with different photo-surfaces.

4. Electrons are emitted with no time delay.

Einstein's explanation

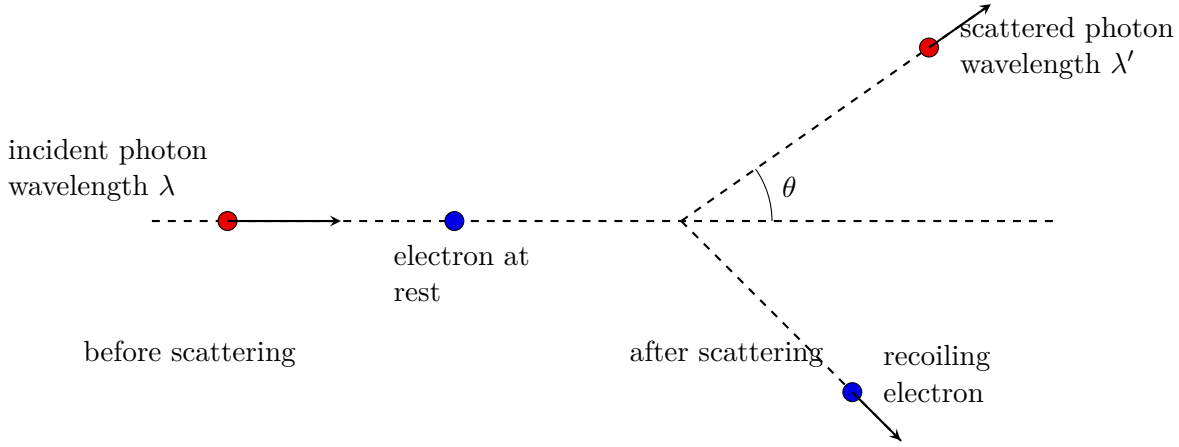
A single photon of frequency f is absorbed by a single electron in the photo-surface, and so the electron's energy increases by hf. The electron will have to spend a certain amount of energy, ϕ (the work function, given by $\phi = hf_c = hf - eV_s$), to free itself from the pull of the nuclei of the atoms of the photo-surface. The electron will be emitted (become free) if hf is bigger than ϕ. The difference $E_K = hf - \phi$ will be the kinetic energy of the (now) free electron.



We are thus able to derive this formula for stopping voltage: $V_s = \frac{h}{e}f - \frac{\phi}{e}$.

Compton scattering

This is the scattering of light off an electron. The scattered light bounces off the electron with a longer wavelength. We can understand this in terms of photons easily: the light gives some energy to the electron which recoils with kinetic energy. The energy of the light has therefore been reduced.



有人注意到这个图其实之前出现过么?

Applying the law of conservation of momentum:

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)$$

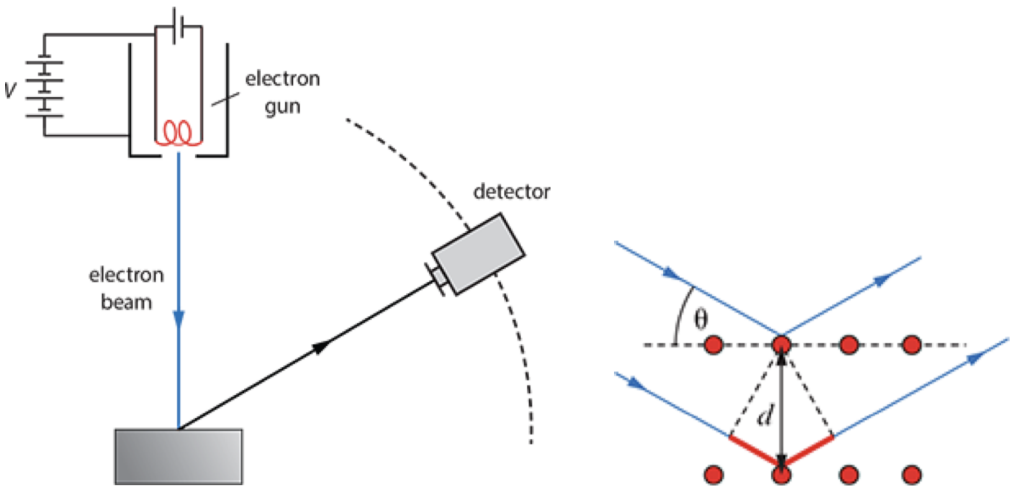
The largest wavelength shift occurs when $\theta = \pi$: $\lambda' - \lambda = 2\frac{2}{m_e c}$, and the energy transferred to the electron is the largest possible, $\frac{hc}{\lambda} - \frac{hc}{\lambda'}$. The Compton wavelength $\frac{h}{m_e c} \approx 2.4 \times 10^{-12}$ m.

22.2 Matter waves

In 1923, Louis de Broglie suggested that to any particle of momentum p there corresponds a wave of wavelength given by the formula $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqV}}$. Because wave-like properties were assigned to particles, this characteristic is called the **duality of matter**. Having wave-like properties means exhibiting the two basic phenomena of waves: diffraction and interference.

Schrödinger and Born gave the meaning for matter waves in the research about quantum mechanics: the interpretation given was that the matter wave is a wave of probability. The square of the amplitude of the wave is proportional to the probability of finding the particle near a particular point in space at a particular time.

Electrons emitted from the hot filament of the electron gun are accelerated through a known potential difference V and are then allowed to fall on a crystal. The positions of the scattered electrons are recorded by a detector. The wave scatters off atoms in two layers. The path difference between the two rays is indicated in thick lines, equal to $2d\sin\theta$. When this path difference is an integer number of the wavelength, $2d\sin\theta = n\lambda, n \in \mathbb{Z}^+$, constructive interference takes place.



23 Nuclear physics

23.1 Mass defect and binding energy

Nuclear notation

Very *professional vocabs*:

- Nucleon: denotes a proton / neutron
- Proton number: number of protons, denoted as Z

- Nucleon number: number of protons + neutrons
- Number of neutrons: $N = A - Z$
- Electric charge of the nucleus: eZ, where $e = 1.6 \times 10^{-19}$ C is the charge of a proton
- ${}^{A}_{Z}X$ is the nucleus of element X.
- Nuclide: A nucleus with a specific number of protons and neutrons

A table of common particles and a summary of them:

Particle	**Symbol**
Proton, Antiproton	${}^{1}_{1}p,\ {}^{1}_{-1}\overline{p}$
Neutron, Antineutron	${}^{1}_{0}n,\ {}^{1}_{0}\overline{n}$
Electron, Positron	${}^{0}_{-1}e,\ {}^{0}_{+1}\overline{e}$
Photon	${}^{0}_{0}\gamma/\gamma$
Neutrino, Antineutrino	${}^{0}_{0}\nu/\nu,\ {}^{0}_{0}\overline{\nu}/\overline{\nu}$
Alpha particle	${}^{4}_{2}\mathrm{He}/{}^{4}_{2}\alpha$

(Four) Fundamental forces

1. The electromagnetic interaction: it acts on any particle that has electric charge. The force is given by Coulomb's law. It has infinite range.
2. The weak nuclear interaction: it acts on protons, neutrons, electrons and neutrinos in order to bring about beta decay. It has very short range.
3. The strong nuclear interaction: it acts on protons and neutrons to keep them bound to each other inside nuclei. It has short range
4. The gravitational interaction: it acts on any particle with mass. The small mass of atomic particles makes this force irrelevant for atomic and nuclear physics. It has infinite range.

Isotopes

Nuclei of the same element have the same number of protons but may have different number of neutrons; these are called isotopes of each other. Since isotopes have the same number of protons, their atoms have the same number of electrons as well. This means that isotopes have identical chemical properties but different physical properties. The existence of isotopes is evidence for the existence of neutrons inside atomic nuclei.

The atomic mass unit

Defined to be $\dfrac{1}{12}$ of the mass of carbon-12, ${}^{12}_{6}\mathrm{C}$. The symbol for atomic mass unit is $u = 1.6605402 \times 10^{-27}$ kg.

The mass defect and binding energy

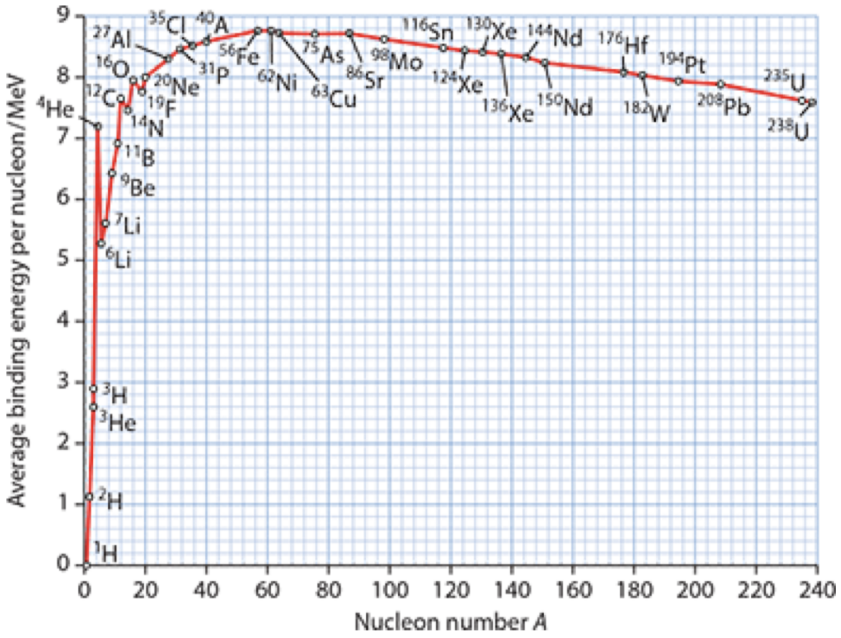
Protons and neutrons are very tightly bound to each other in a nucleus through the action of the strong nuclear force. To separate them, energy must be supplied to the nucleus. Conversely, energy is released when a nucleus is assembled from its constituent nucleons. In Einstein's theory of relativity, $E = mc^2$. Therefore, for a mass of 1u, the energy is 931.5 MeV. Since we must supply energy to a nucleus to separate the nucleons

and since energy is equivalent to mass, it follows that the total mass of the separated nucleons is greater than the mass of the nucleus. The mass defect, μ, is the total mass of the nucleons minus the mass of the nucleus.

$$\begin{aligned}\mu &= Zm_p + Nm_n - M_{\text{nucleus}} \\ &= Zm_p + Nm_n - (M_{\text{atom}} - Zm_e) \\ &= Z(m_p + m_e) + Nm_n - M_{\text{atom}} \\ &= Zm_{\text{Hydrogen}} + Nm_n - M_{\text{atom}}\end{aligned}$$

The energy equivalent to the mass defect is called binding energy, $E_b = \mu c^2$.
The energy required to remove one nucleon from the nucleus is roughly the binding energy divided by the total number of nucleons. Thus, the binding energy per nucleon, b, is a measure of how tightly bound the nucleus is.

The binding energy curve



- The binding energy per nucleon for hydrogen is 0 because there is only one particle in the nucleus and there is nothing to separate.
- The curve rises sharply for low values of A.
- The curve has a maximum for A = 62 corresponding to nickel, which makes this nucleus particularly stable.
- There are peaks at the position of the nuclei $^{4}_{2}$He,$^{12}_{6}$C,$^{16}_{8}$O, which makes these nuclei unusually stable compared to their immediate neighbors.
- The curve drops gently from the peak at A = 62 and onwards.
- Most nuclei have a binding energy per nucleon between 7 and 9 MeV.

The energy released in nuclear reactions

There's a conservation of electric charge and nucleon number for nuclear reaction (fusion or fission).
To decide whether energy is released in any nuclear reaction, we have to calculate the mass difference $\Delta m = m_{\text{reactants}} - m_{\text{products}}$.

- If $\Delta m > 0$, then energy will be released and the decay will occur.
- If $\Delta m < 0$, the reactants will not react and the reaction can take place only if energy is supplied.

The quantity of energy Q released is the energy equivalent of the mass difference. This energy is released in the form of kinetic energy.

23.2 Radioactivity

Radioactive decays

An unstable nucleus is one that randomly and spontaneously emits particles that carry energy away from the nucleus. The emission of particles and energy from a nucleus is called radioactivity. Radioactive decay is random and spontaneous.

Properties of alpha, beta and gamma radiations

Characteristic	**Alpha**	**Beta**	**Gamma**
Nature	Helium nucleus	Fast electron	Photon
Charge	+2e	-e	0
Mass	6.64×10^{-27} kg	9.1×10^{-31} kg	0
Penetrating power	A few cm of air	A few mm of metal	Many cm of lead
Ions per mm of air for 2 MeV particles	10000	100	1
Description	An alpha particle is emitted from the parent nucleus creating the daughter nucleus.	β^- : a neutron turns into a proton, emitting an electron and an anti-neutrino. β^+: the nucleus emits positron and neutrino.	In gamma decay a nucleus emits a gamma ray, a photon of high-f EM radiation. λ of the photon emitted is $\lambda = \frac{hc}{E}$.
Detection	Affects photographic film Is affected by electric and magnetic fields	Affects photographic film Is affected by electric and magnetic fields	Affects photographic film Is not affected by electric and magnetic fields

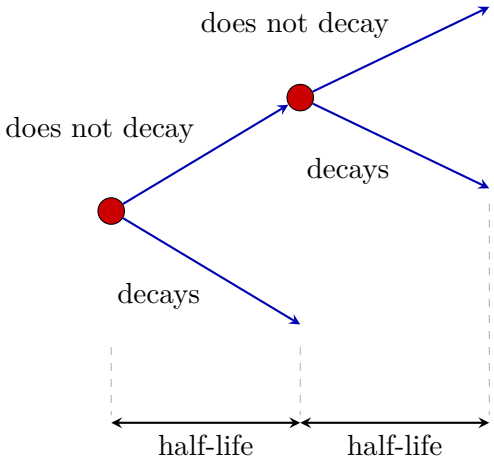
The law of radioactive decay

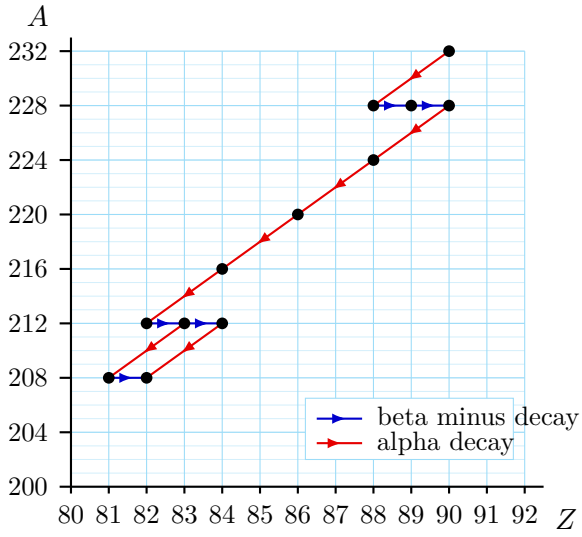
The law of radioactive decay states that the rate of decay is proportional to the number of nuclei that have not yet decayed: $\frac{\Delta N}{\Delta t} \propto N$. A consequence of this law is that the number of radioactive nuclei decreases exponentially.

Half-life: the time after which the number of the radioactive nuclei is reduced by a factor of 2, or the interval of time after which the activity (A, number of decays, unit Bq= 1 decay / s) of a radioactive sample is reduced by a factor of 2.
Background radiation sources: cosmic rays from the sun, radioactive material in rocks and the ground, radiation from nuclear weapons testing grounds etc. that provides background radiation.

Half-life and probability

The meaning of a half-life can also be understood in terms of probability. Any given nucleus has a 50% chance of decaying within a time interval equal to the half-life. If a half-life goes by and the nucleus has not decayed, the chance that it will decay in the next half-life is still 50%.



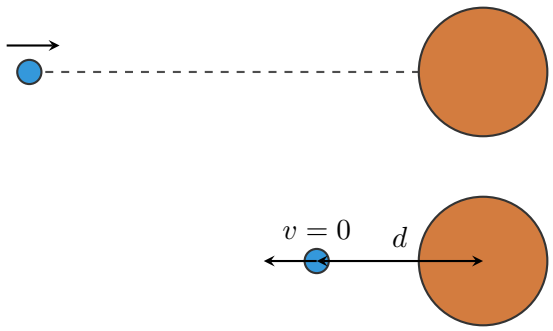


23.3 Nuclear properties and the radioactive decay law

Nuclear radii

When an alpha particle of kinetic energy E_K is directed at a nucleus of proton number Z, the distance of closest approach, d, is found by equating the kinetic energy far from the nucleus to the electric potential energy at the point of closest approach.

$$E_K = k\frac{(2e)(Ze)}{d} \Rightarrow d = \frac{2kZe^2}{E_K}$$



As the energy of the incoming alpha particles increases, the distance of closest approach decreases and will eventually approach the nuclear surface.

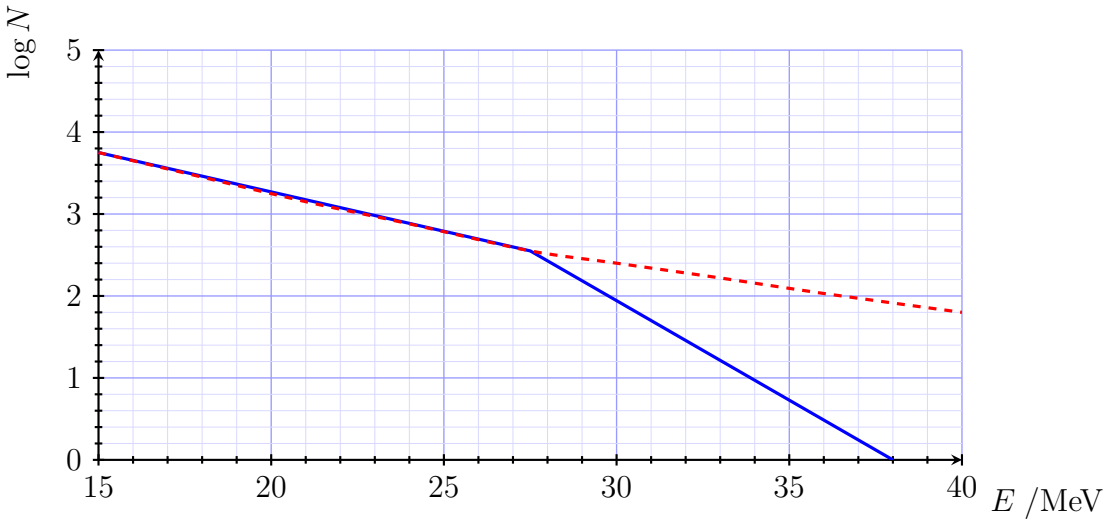
When the alpha particle is near the nuclear surface deviations from Rutherford scattering will be observed because the nuclear force will start acting on the alpha particle. The number of scattered alpha particles will not be what is expected based on Rutherford scattering. Thus, the distance of closest approach measured in this case will essentially be the nuclear radius.

The radius of a nucleus is given by $R = R_0 A^{1/3}$, where $R_0 \approx 1.2 \times 10^{-15}$ m and A is the nucleon number. (This formula also implies that the volume of a nucleus is proportional to the number of nucleons in it, and that the density of nuclei is constant, $\rho = 2 \times 10^{17}$ kg m^{-3}.)

Deviations from Rutherford scattering — evidence for the strong nuclear force

The Rutherford formula states that as the scattering angle increases, the number of alpha particles scattered at that angle decreases very sharply. It states that the number N of alpha particles scattering at an angle θ is proportional to $\frac{1}{\sin^4(\theta/2)}$. (If this was the case, $N \sin^4\left(\frac{\theta}{2}\right)$ would be a constant for all angles of θ.)

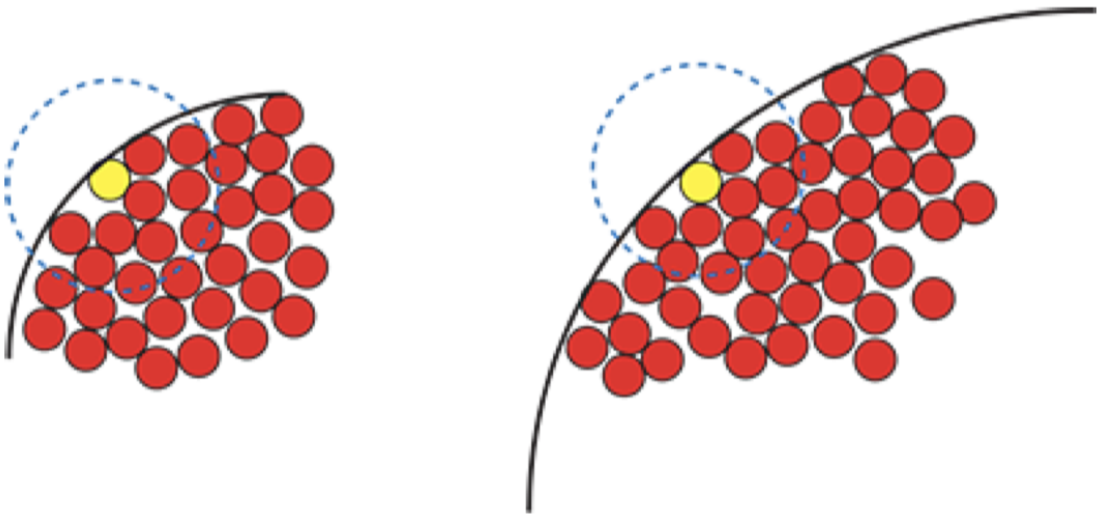
The derivation of the Rutherford formula is based on a number of assumptions. The most important is that the only force in play during the scattering process is the electric force. As the energy of the alpha particles increases, the alpha particles can get closer to the nucleus. When the distance of closest approach gets to be about 10^{-15} m or less, deviations from the Rutherford formula are observed. This is due to the fact that the alpha particles are so close to the nucleus that the strong nuclear force begins to act on the alpha particles.



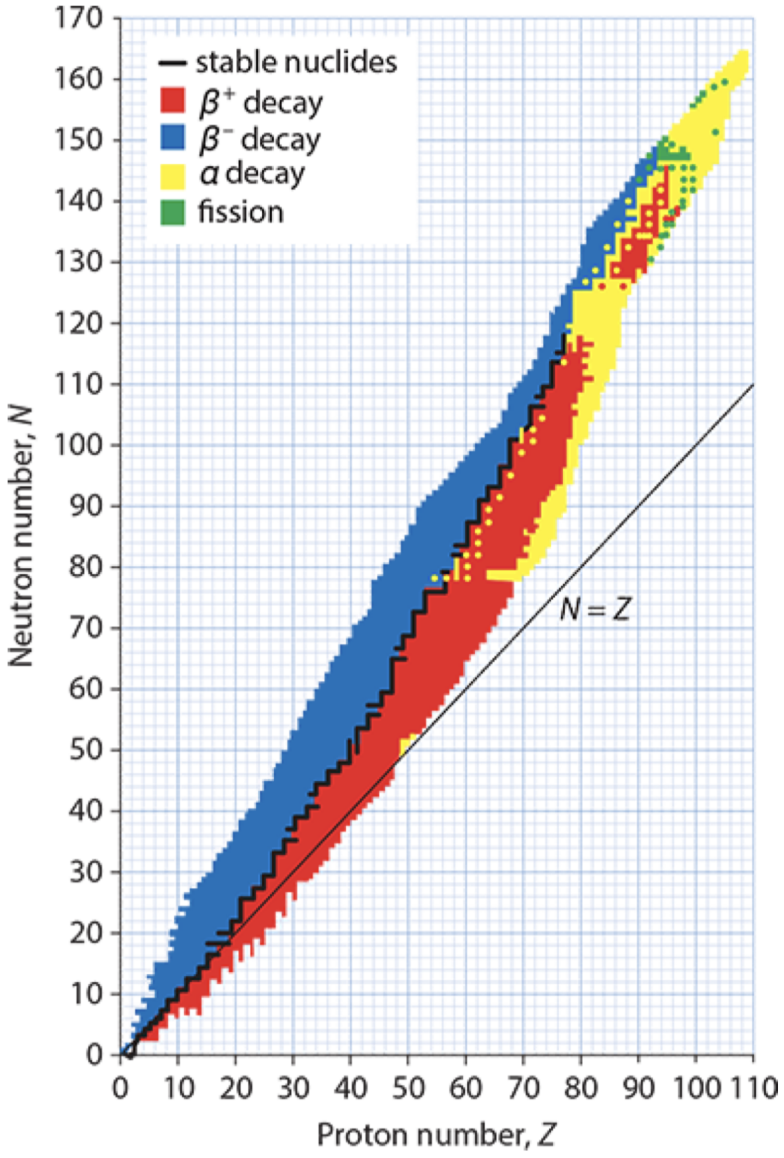
The dotted curve is based on Rutherford scattering. The blue curve is the observed curve. We see deviations when the energy exceeds about 28 MeV. The energy at which deviations start may be used to estimate the nuclear radius.

Approximate constancy of the binding energy per nucleon

The short range of the force implies that a given nucleon can interact with its immediate neighbours only and not with all of the nucleons in the nucleus. So, for large nuclei (roughly A > 20) any one nucleon is surrounded by the same number of immediate neighbours, and so the energy needed to remove that nucleon from the nucleus is the same.



Nuclear stability - the role of neutrons



Notice that stable nuclei with small values of Z have equal numbers of neutrons and protons. As Z increases, stable nuclei have more neutrons than protons, both because the extra neutrons make the distance between protons larger (EM force acts on protons, and it is repulsive, acts on infinite range, increasing the tendency for the nucleus to break apart. The strong nuclear force does not help because for large enough nuclei the number of nearest neighbours of a given proton is the same and because of the short range of the force only these act on a given proton.) and because the neutrons contribute to binding.

Nuclear energy levels

The nucleus, like the atom, exists in discrete energy levels. The main evidence for the existence of nuclear energy levels comes from the fact that the energies of the alpha particles and gamma ray photons that are emitted by nuclei in alpha and gamma decays are discrete. This is to be contrasted with beta decays, in which the electron has a continuous range of energies.

The neutrino

The IG / 1930s version of the beta minus decay can be given by

$$ {}^{1}_{0}n \rightarrow {}^{1}_{1}p + {}^{0}_{-1}e $$

It has a mass difference $\Delta m = 0.00084$ u and energy of $Q = 0.783$ MeV. If only the electron and the proton are produced, then the electron, being the lighter of the two, will carry most of this energy away as kinetic energy.
To see this, assume that the neutron is at rest when it decays. Then the total momentum before the decay is zero. After the decay the electron and the proton will have equal and opposite momenta, each of magnitude p. Equating the total E_K after the decay to the energy released, Q:

$$\begin{aligned} \frac{p^2}{2m_e} + \frac{p^2}{2m_p} &= Q \\ p^2\left(\frac{1}{2m_e} + \frac{1}{2m_p}\right) &= Q \\ p^2 &= \frac{2Qm_em_p}{m_e + m_p} \\ E_K = \frac{p^2}{2m_e} &= \frac{Qm_p}{m_e + m_p} \approx Q \end{aligned}$$

Thus, we should observe electrons with kinetic energies of about 0.783 MeV. In experiments, however, the electron has a **range** of energies 0 to 0.783 MeV. It means that a third very light particle must also be produced so that it carries the remainder of the available energy, and it is named "neutrino", ν. Some properties:

- Electrically neutral
- Negligible mass, gravitational interaction is negligible
- No strong nuclear interaction

The radioactive decay law

The rate of decay is proportional to the number of nuclei present that have not yet decayed,

$$\frac{dN}{dt} = -\lambda N$$

The decay constant is $\lambda = \dfrac{\text{probability of decay per unit time}}{dt}$, the constant of proportionality / probability of decay per unit time. The solution to this differential equation is

$$N(t) = N_0 e^{-\lambda t}$$

where N_0 is the initial number of nuclei.
The (negative) rate of decay (i.e. the number of decays per second) is called activity, A:

$$A = -\frac{dN}{dt} = \lambda N_0 e^{-\lambda t} = \lambda N;\ A_0 = \lambda N_0$$

A graph of A against N gives a straight line whose slope is the decay constant.
The relationship of half-life and decay constant is given by

$$\lambda T_{1/2} = \ln 2$$

24 Nuclear fission

24.1 Nuclear fission

Nuclear fission is the process in which a heavy nucleus splits up into two lighter nuclei. The production of neutrons is a key feature of fission reactions. The energy released appears as kinetic energy of the products, mostly for the produced neutrons.

1. This can happen when the nucleus is bombarded by a neutron, and in this case we speak of **induced fission**.
2. In **spontaneous fission**, a nucleus may split into two smaller nuclei without neutron absorption.

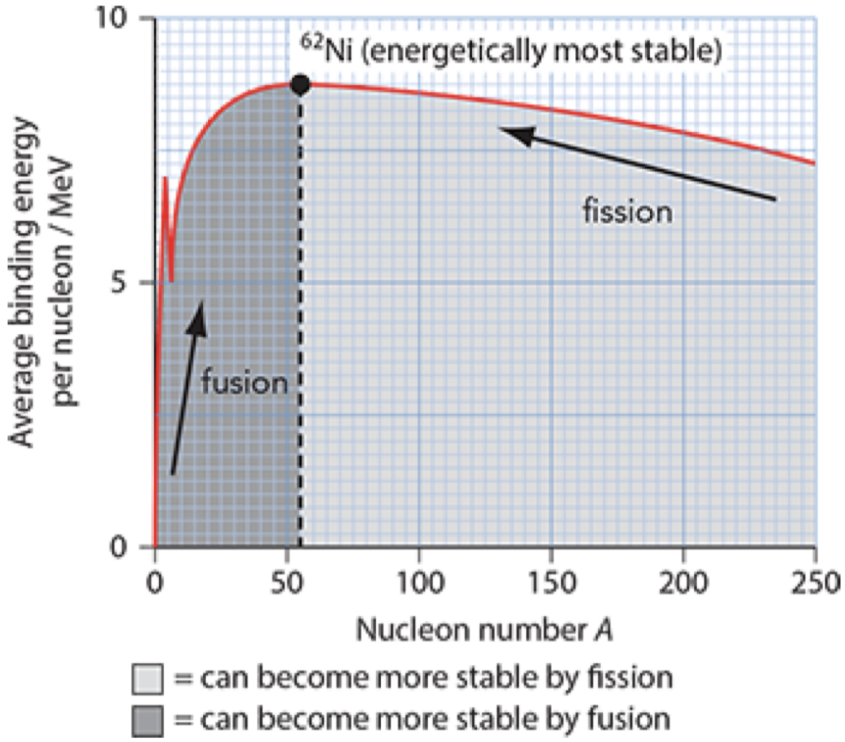
Fission and the binding energy curve

The energy released, other than calculating mass difference, can also be calculated by subtracting from the total binding energy on the right of the reaction the total binding energy on the left of the reaction.

The energy released in terms of binding energy per nucleon, b, is expressed as

$$Q = \sum N(b_{\text{products}} - b_{\text{reactants}})$$

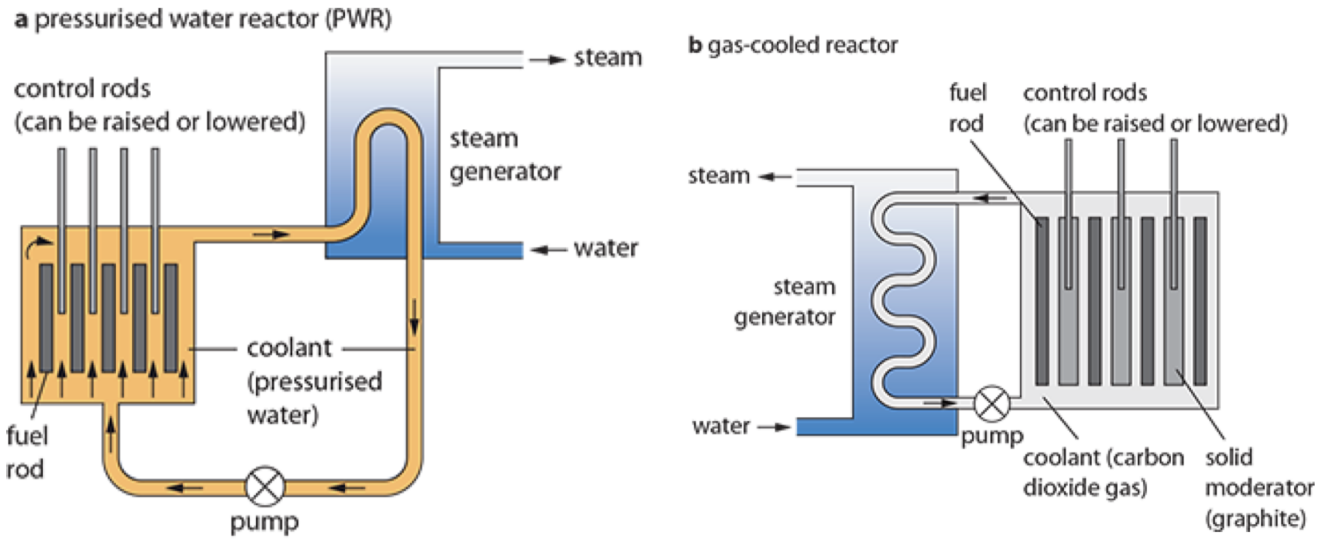
where N is the number of nucleons in the nucleus.
For fission, the products will be at the left of the reactants on the energy binding curve, and thus $b_{\text{products}} > b_{\text{reactants}}$, so $Q > 0$, indicating energy release. Fission means heavy nuclei split into lighter nuclei with higher binding energy per nucleon. Fusion means light nuclei join to form heavier nuclei with higher binding energy per nucleon.



Nuclear fission reactors

A nuclear reactor is a machine in which nuclear fission reactions take place, producing energy. The fuel of a nuclear reactor is typically uranium-235. Neutrons initiate the reaction. One possible fission reaction is:

$$^{1}_{0}n + ^{235}_{92}\text{U} \rightarrow ^{140}_{54}\text{Xe} + ^{94}_{38}\text{Sr} + 2^{1}_{0}n$$



The neutrons produced can be used to collide with other nuclei of uranium-235 in the reactor, producing more fission, more energy and more neutrons. The reaction is thus self-sustaining, called a chain reaction. For the chain reaction to get going, a certain minimum mass, called the critical mass (about 15kg for pure uranium-235), of uranium-235 must be present, otherwise the neutrons escape without causing further reactions.

- **A moderator** must be used so that it can slow down the neutrons produced in the fission reaction. This is because the neutrons produced are fast neutrons, and they are less likely to cause further fission reactions (not captured by uranium-235). The moderator is typically water or graphite.
- **Control rods** are used to absorb excess neutrons and thus control the rate of the reaction. The control rods are made of materials that absorb neutrons, such as boron or cadmium. The control rods can be raised or lowered into the reactor core to control the rate of fission.
- **A heat exchanger** is needed to decrease the temperature of the moderator (due to the energy increment caused by neutrons colliding with the moderator). This can be done using cold water that circulates in pipes throughout the moderator.

The water is turned into steam at high temperature and pressure. The steam is then used to turn the turbines of a generator, finally producing electricity.

The rate of the reactions is determined by the number of neutrons available to be captured by uranium-235.

- Too few neutrons would result in the reactions stopping.
- Too many neutrons would lead to an uncontrolled chain reaction, which is dangerous.

Risks with nuclear power

The fast neutrons produced may be used to bombard uranium-238 and produce plutonium-239.

$$
\begin{aligned}
{}^{1}_{0}n + {}^{238}_{92}\mathrm{U} &\rightarrow {}^{239}_{92}\mathrm{U} \\
{}^{239}_{92}\mathrm{U} &\xrightarrow{\beta^-} {}^{239}_{93}\mathrm{Np} \\
{}^{239}_{93}\mathrm{Np} &\xrightarrow{\beta^-} {}^{239}_{94}\mathrm{Pu}
\end{aligned}
$$

- The plutonium-239 produced can then be used as the nuclear fuel in other reactors. It can also be used in the production of nuclear weapons, which therefore raises serious concerns.

- The spent fuel in a nuclear reactor and the products of the reactions are all highly radioactive with long half-lives. Radioactive waste is broadly classified as higher level and low level waste.
  - Higher level waste is mainly the left-over uranium fuel in a reactor's fuel rods and the lighter elements produced in the fission of the uranium fuel. These elements are highly radioactive. The radiation produced when they decay is very penetrating and delivers fatal doses even during short exposures, decades after they have been removed from the reactor.
  - Low level waste includes material that has become radioactive due to neutron absorption or has been contaminated with radioactive material.
- There is always the possibility of an accident due to uncontrolled heating of the moderator, which might start a fire or explosion. In the event of an explosion, radioactive material would leak from the sealed core of a reactor, dispersing radioactive material into the environment.

Advantages	**Disadvantages**
High power output	Radioactive waste products difficult to dispose of
Large reserves of nuclear fuels	Major public health hazard should 'something go wrong'
No greenhouse gas emissions	Problems associated with uranium mining
No acid rain	Possibility of producing materials for nuclear weapons

25 Nuclear fusion and stars

25.1 Nuclear fusion

Nuclear fusion is the joining of two light nuclei into a heavier one with the associated production of energy.

Conditions for fusion to take place

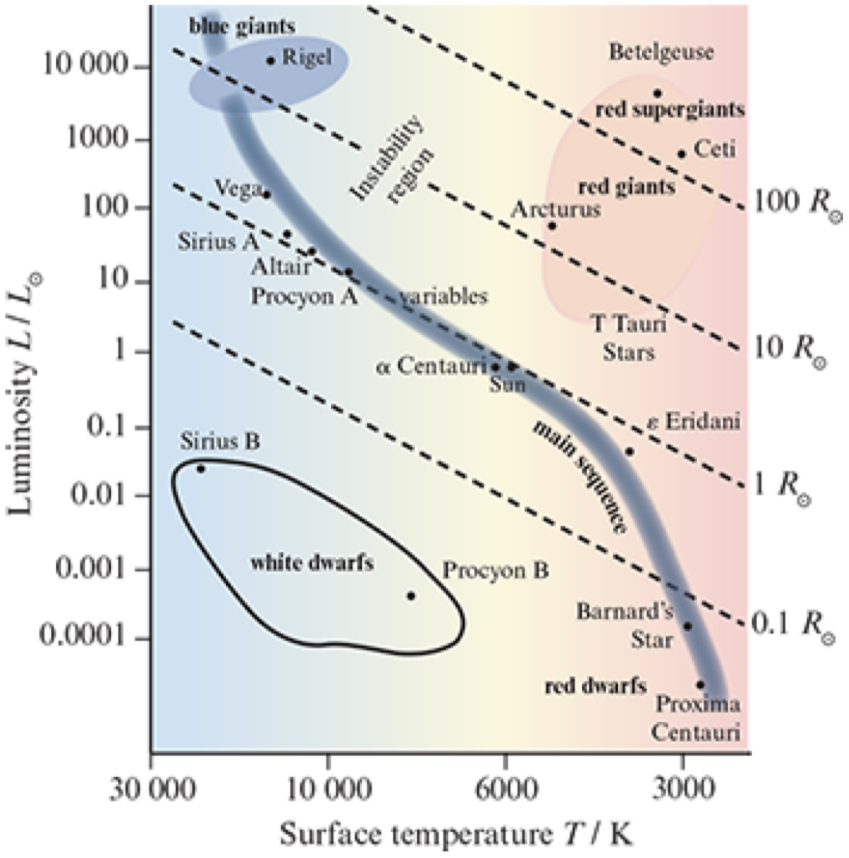
- Temperature
  - For fusion to occur, the reacting nuclei must join.
  - But nuclei are positively charged so they repel each other.
  - At high temperatures (namely, millions of degrees, at plasma state already (electrons left atom)) the nuclei would be moving fast enough to overcome the Coulomb repulsion and crash into each other.
- Density
  - The number density of the nuclei (number of nuclei per unit volume) must be high so that there are lots of them around for fusion to occur.
- Confinement time (not easy to be satisfied)
  - The nuclei must be maintained at high temperature and density for a time long enough for many fusion reactions to take place.

Nuclear fusion provides the energy that is radiated by stars into space, the pressure to keep a star from collapsing under its own weight and the nuclear fusion reactions and related processes that are responsible for nucleosynthesis, the creation of the chemical elements.

25.2 Stellar properties and the Hertzsprung-Russell diagram

The Hertzsprung-Russell diagram

In this diagram, luminosity (given in terms of the sun's luminosity, $L_\odot = 3.9 \times 10^{26}$W)was plotted on the vertical axis and surface temperature (in K) on the horizontal axis. The slanted dotted lines represent stars with the same radius. $R_\odot$ stands for the sun's radius.



The instability region contains stars that have a variable luminosity.

- **Main sequence stars**
  - Most stars fall on a strip extending diagonally across the diagram from top left to bottom right.
  - As we move along the main sequence toward hotter stars, the mass of the stars increases. Thus, the right end of the main sequence is occupied by red dwarfs and the left by blue giants.
  - Main sequence stars produce energy in their core from nuclear fusion of hydrogen into helium.
  - The outflow of energy from the core creates an outward radiation pressure that exactly counterbalances the tendency of the star to collapse under its own weight.
- **Red giants / super red giants**
  - Some large, cool stars, reddish in colour, occupy the top right. Super red giants are even larger and brighter than red giants.
  - The luminosity of red giants is considerably greater than the luminosity of main sequence stars of the same temperature.
  - The mass of a red giant can be as much as 1000 times the mass of our sun, but their huge size also implies small densities.
  - A red giant will have a central hot core surrounded by an enormous envelope of extremely tenuous gas.

- **White dwarfs**
  - The bottom left is a region of small stars (hot).
  - These are common stars, but their faintness makes them hard to detect.

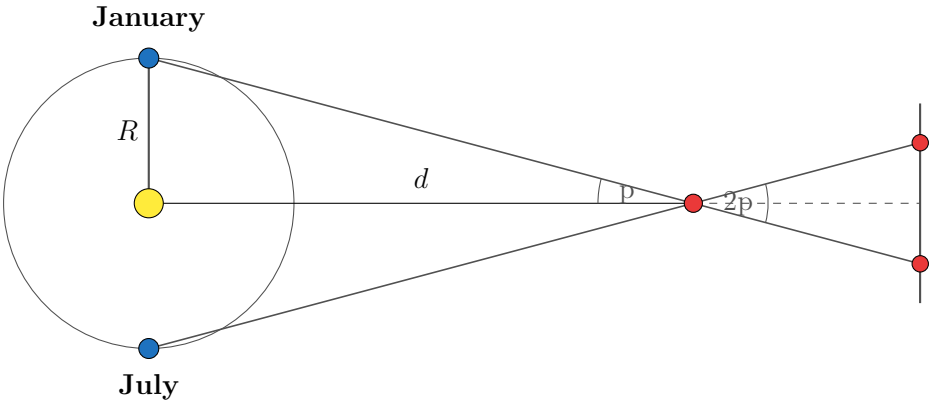
Type of star	**Main characteristic**
Main sequence star	Fuses Hydrogen into Helium
Red giant	Bright, large, cool, reddish, tenuous
Super red giant	Even larger and brighter than red giants
White dwarf	Dim, small, hot, whitish, dense
Neutron star	Dimmer and smaller than a white dwarf with a density similar to that of nuclei. Too dim to show on an HR diagram
Black hole	The end point of the very massive stars

The stellar radii can be given by $R = \sqrt{\dfrac{L}{4\pi\sigma T^4}}$.

To measure the radius of a star through its temperature and luminosity assumes we know the temperature and luminosity! Temperature can be measured using Wien's law, while luminosity can be found by apparent brightness and distance.

Parallax methods

It can be used to measure the distance. It is based on the fact that when you look at an object from two different positions, the object appears to shift relative to the very distant background. The method is restricted to nearby stars. Stars far away have parallax angles that are too small to measure accurately. The parallax angle is the angle at which the star subtends a distance equal to the earth orbit radius R.



Parallax angles are very small. It is convenient to express them in arc seconds: $1^\circ = 60$ arcmin $= 3600$ arcsec. One parsec is the distance at which the parallax is 1 arcsec, $d(/\text{pc}) = \dfrac{1}{p(/\text{arcsec})}$, 1 pc $= \dfrac{R}{\tan p} = \dfrac{1.5 \times 10^{11}}{\tan(1/3600)^\circ} \approx 3.09 \times 10^{16}$ m $= 3.27$ ly.

The proton-proton chain

The main series of nuclear fusion reactions taking place in the cores of low mass main sequence stars is the proton-proton chain:

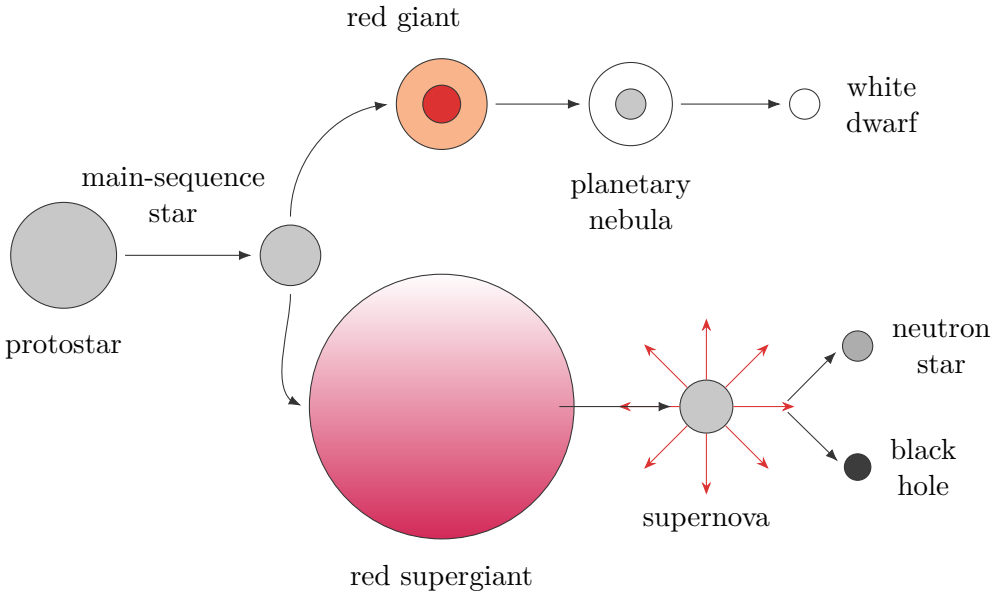
$$
\begin{aligned}
{}^1_1\mathrm{H} + {}^1_1\mathrm{H} &\rightarrow {}^2_1\mathrm{H} + {}^0_1 e^+ + \nu \\
{}^1_1\mathrm{H} + {}^2_1\mathrm{H} &\rightarrow {}^3_2\mathrm{He} + \gamma \\
{}^3_2\mathrm{He} + {}^3_2\mathrm{He} &\rightarrow {}^4_2\mathrm{He} + 2{}^1_1\mathrm{H}
\end{aligned}
$$

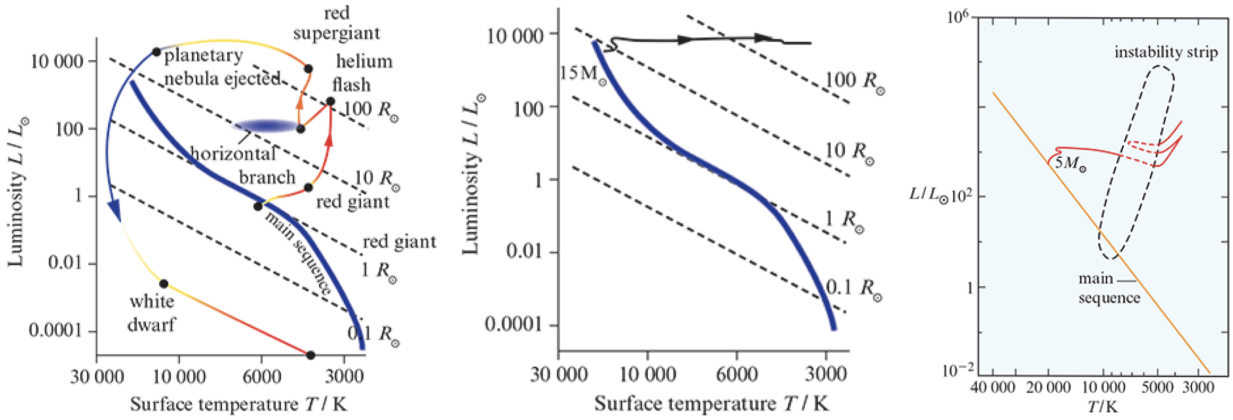
In this cycle, the net effect is to turn four hydrogen nuclei into one nucleus of helium.

$$4{}^1_1\mathrm{H} \rightarrow {}^4_2\mathrm{He} + 2{}^0_1 e^+ + 2\nu + 2\gamma$$

Stellar evolution

When about 12% of the hydrogen in the star is used up in nuclear fusion, a series of instabilities develop in the star. These have the result that the delicate balance between the radiation and gravitational pressures is upset. The star will then begin to move away from the main sequence. Other types of nuclear fusion reactions will take place and the star changes in size, surface temperature, and color.





A star more massive than the sun may go back and forth on the HR diagram several times. The delicate balance between radiation and gravitational pressures has been disturbed; the star tries to find a new equilibrium state but instead the star pulsates, i.e. it grows and then shrinks in size. So the star cools and heats up during these pulsations and that affects the rate and type of nuclear fusion reactions taking place. This makes its luminosity variable, and this is a general characteristic of all stars in the instability region.

The Chandrasekhar limit

It is the largest mass a white dwarf star can have, $1.4M_{\odot}$.
前情提要:

- Low mass stars after ejecting mass into space in the planetary nebula explosion leave behind a core.
- The conditions in the core mean that the electrons behave as a gas, and the pressure they generate is what keeps the core from collapsing further under its weight. (driven by quantum effect)
- The core has now become a white dwarf star. Now exposed, and with no further energy source, the star is doomed to cool down and will then become what is called a black dwarf.
- This pressure is able to stop the further collapse of the core only if the mass of the core $<$ the Chandrasekhar limit.

The Oppenheimer-Volkoff limit

It is the largest mass a neutron star can have and equals about $3M_{\odot}$.
前情提要:

- If the Chandasekhar limit is exceeded, electrons will be forced into protons, turning them into neutrons.
- The star is now made almost entirely of neutrons, and it is still contracting rapidly.
- The neutrons get too close to each other, and a pressure develops to prevent them from getting any closer.
- But they would still become too close together, and the entire core will now rebound to a larger size.
- The rebounding of the core is catastrophic for the star, creating an enormous shockwave travelling outward that tears apart the outer layers of the star. The violent explosion is called the supernova.
- The core that is left behind is called a neutron star. The core is kept stable by quantum effects similar to those in the white dwarf if the mass of the core is not more than the Oppenheimer-Volkoff limit.

25.3 Stellar evolution extension

The creation of elements

The hydrogen in the universe and most of the helium (and some lithium) were produced at the very earliest times in the life of the universe. The rest of the elements were produced in a process known as nucleosynthesis in stars, in the process of stellar evolution.

For stars more massive than about 1.5 solar masses there is a second way to fuse hydrogen into helium. This is the CNO (carbon, nitrogen, oxygen) cycle:

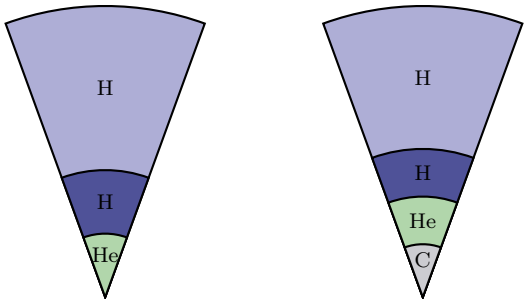
$$
\begin{aligned}
{}^{1}_{1}\mathrm{H} + {}^{12}_{6}\mathrm{C} &\rightarrow {}^{13}_{7}\mathrm{N} + \gamma \\
{}^{13}_{7}\mathrm{N} &\rightarrow {}^{13}_{6}\mathrm{C} + {}^{0}_{1}e^{+} + \nu \\
{}^{1}_{1}\mathrm{H} + {}^{13}_{6}\mathrm{C} &\rightarrow {}^{14}_{7}\mathrm{N} + \gamma \\
{}^{1}_{1}\mathrm{H} + {}^{14}_{7}\mathrm{N} &\rightarrow {}^{15}_{8}\mathrm{O} + \gamma \\
{}^{15}_{8}\mathrm{O} &\rightarrow {}^{15}_{7}\mathrm{N} + {}^{0}_{1}e^{+} + \nu \\
{}^{1}_{1}\mathrm{H} + {}^{15}_{7}\mathrm{N} &\rightarrow {}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He}
\end{aligned}
$$

- For low mass stars, the helium produced in the p-p chain is collecting in the core of the star surrounded by a thin shell of hydrogen and a bigger hydrogen envelope.

- In more massive stars, carbon will replace helium in the core once carbon starts getting produced in the triple alpha process:

$$^4_2\text{He} +^4_2 \text{He} \rightarrow^8_4 \text{Be} + \gamma$$
$$^4_2\text{He} +^8_4 \text{Be} \rightarrow^{12}_6 \text{C} + \gamma$$

  - Only hydrogen in the thin shell is undergoing nuclear fusion.
  - The temperature and pressure of helium build up and eventually helium begins to fuse through the triple alpha process, called the "Helium flash".
  - Carbon collects in the core, with helium now fusing in a thin shell around carbon.
  - This release of energy blows away the outer layers of the star in an explosion called a planetary nebula; mass is thrown into space leaving behind a helium or carbon core.



For stars with different sizes of mass:

- $M < 0.25M_\odot$: white dwarf with Helium core:

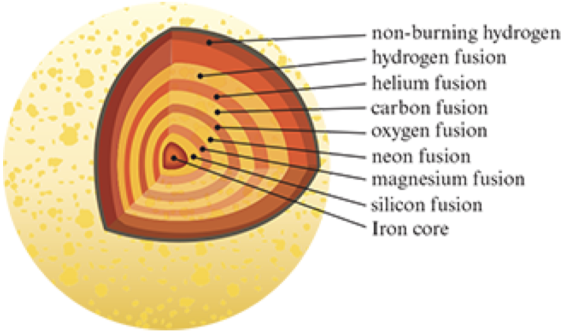
$$4^1_1\text{H} \rightarrow^4_2 \text{He} + 2^0_1 e^+ + 2\nu + 2\gamma$$

- $0.25M_\odot < M < 8M_\odot$: white dwarf with Carbon core (triple alpha process just mentioned)
- $8M_\odot < M < 12M_\odot$: white dwarf with Oxygen / Neon / Magnesium core:

$$^4_2\text{He} +^{12}_6 \text{C} \rightarrow^{16}_8 \text{O} + \gamma$$
$$^{12}_6\text{C} +^{12}_6 \text{C} \rightarrow^{20}_{10} \text{Ne} +^4_2 \text{He}$$
$$^{12}_6\text{C} +^{12}_6 \text{C} \rightarrow^{23}_{11} \text{Na} +^2_1 \text{H}$$
$$^{12}_6\text{C} +^{12}_6 \text{C} \rightarrow^{23}_{12} \text{Mg} +^1_0 n$$
$$^{16}_8\text{O} +^{16}_8 \text{O} \rightarrow^{28}_{14} \text{Si} +^4_2 \text{He}$$

- $12M_\odot < M < 40M_\odot$: Neutron star
- $M > 40M_\odot$: Black hole

Fusion cannot produce elements heavier than iron, since the binding energy per nucleon peaks near iron and further fusion is not energetically possible. Elements heavier than iron are produced during supernova explosions through neutron absorption.



Appendices

A versatile method for doning phyGUESS method:

- G: given information
- U: unknown information
- E: equation to connect the given and unknown info
- S: substitute
- S: solve problem

A Common constants

Symbol	Value	Name & significance / usage
c	$3.00 \times 10^{8}\ \mathrm{m\,s^{-1}}$	Speed of light in vacuum; maximum speed limit, central to relativity and $E = mc^2$.
h	$6.63 \times 10^{-34}\ \mathrm{J\,s}$	Planck constant; relates photon energy to frequency ($E = hf$) and appears in de Broglie wavelength.
e	$1.60 \times 10^{-19}\ \mathrm{C}$	Elementary charge; basic quantum unit of charge (proton charge, electron charge magnitude).
G	$6.67 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$	Gravitational constant; proportionality constant in Newton's law of gravitation.
g	$9.8\ \mathrm{m\,s^{-2}}$	Acceleration of free fall at Earth's surface (directed downward).
k	$8.99 \times 10^{9}\ \mathrm{N\,m^2\,C^{-2}}$	Coulomb constant; $k = \dfrac{1}{4\pi\varepsilon_0}$, proportionality constant in Coulomb's law.
ε_0	$8.85 \times 10^{-12}\ \mathrm{C^2\,N^{-1}\,m^{-2}}$	Permittivity of free space; relates electric field to charge in vacuum.
μ_0	$4\pi \times 10^{-7}\ \mathrm{T\,m\,A^{-1}}$	Magnetic permeability of free space; relates magnetic field to current.
k_B	$1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$	Boltzmann constant; relates average molecular kinetic energy to temperature ($k_B = R/N_A$).
N_A	$6.02 \times 10^{23}\ \mathrm{mol^{-1}}$	Avogadro constant; number of particles per mole of substance.
R	$8.31\ \mathrm{J\,mol^{-1}\,K^{-1}}$	Molar gas constant; universal constant in the ideal gas law $PV = nRT$.
σ	$5.67 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$	Stefan–Boltzmann constant; proportionality constant in $P = \sigma A T^4$.

Continued on next page…

Table 1 – continued

Symbol	Value	Name & significance / usage
b	2.90×10^{-3} K m	Wien's displacement constant; links peak wavelength of black-body spectrum to temperature.
u	1.66×10^{-27} kg	Atomic mass unit; $\frac{1}{12}$ mass of carbon-12, standard nuclear mass unit.
uc^2	931.5 MeV	Energy equivalent of 1 atomic mass unit (uc^2); used in nuclear mass–energy calculations.
m_e	9.11×10^{-31} kg	Electron rest mass; mass of the electron (beta particle).
m_p	1.67×10^{-27} kg	Proton rest mass; mass of a proton (hydrogen nucleus).
m_n	1.675×10^{-27} kg	Neutron rest mass; mass of a free neutron.
a_0	5.29×10^{-11} m	Bohr radius; smallest allowed orbital radius of the hydrogen electron.
R_0	1.2×10^{-15} m	Nuclear radius constant; characteristic length scale in $R = R_0 A^{1/3}$.
$h/(m_e c)$	2.43×10^{-12} m	Compton wavelength of the electron; characteristic wavelength-shift scale in Compton scattering.
γ_{mono}	5/3	Adiabatic index for a monatomic ideal gas; ratio of specific heats.
Astronomical & planetary data		
S	≈ 1400 W m^{-2}	Solar constant; solar intensity at the top of Earth's atmosphere.
$L_\odot$	3.9×10^{26} W	Sun's luminosity; total radiated power of the Sun.
$R_\odot$	7.0×10^{8} m	Sun's radius; used in stellar surface intensity calculations.
d_{au}	1.5×10^{11} m	Astronomical unit (Earth–Sun distance); used in solar constant and parallax.
α_{Earth}	≈ 0.30	Earth's average albedo; fraction of incident solar radiation reflected.
1 atm	1.01×10^{5} Pa	Standard atmospheric pressure at sea level.
1 pc	3.09×10^{16} m = 3.27 ly	Parsec; distance at which 1 AU subtends 1 arcsecond.

B Formula list

B.1 Unit A — Space, time and motion

Topic	Formula	Significance / usage
Kinematics	$v = \dfrac{\Delta s}{\Delta t}$	Average speed (total distance over total time).
Kinematics	$s = s_i + vt$	Position for uniform motion with initial position.
Kinematics	$v = u + at$	Velocity as a function of time (constant acceleration).
Kinematics	$s = ut + \dfrac{1}{2}at^2$	Displacement as a function of time.
Kinematics	$v^2 = u^2 + 2as$	Velocity–displacement relationship (no time).
Kinematics	$s = \dfrac{u+v}{2}\,t$	Displacement from average velocity.
Kinematics	$v = \dfrac{ds}{dt}, \quad a = \dfrac{dv}{dt} = \dfrac{d^2s}{dt^2}$	Calculus definitions of velocity and acceleration.
Projectile	$V_x = u\cos\theta, \quad V_y = u\sin\theta - gt$	Horizontal and vertical velocity components.
Projectile	$x = ut\cos\theta, \quad y = ut\sin\theta - \frac{1}{2}gt^2$	Horizontal and vertical position vs. time.
Projectile	$y_{\max} = \dfrac{u^2\sin^2\theta}{2g}$	Maximum height of a projectile.
Projectile	$R = \dfrac{u^2\sin 2\theta}{g}$	Range (maximum horizontal displacement).
Projectile	$t_{\text{flight}} = \dfrac{2u\sin\theta}{g}$	Total time of flight (level ground).
Drag	$F = kv$ (low v), $\quad F = kv^2$ (high v)	Fluid resistance force.
Drag	$v_T = \dfrac{mg}{k}$	Terminal speed (net force = 0).
Forces	$W = mg$	Weight of a body.
Forces	$F = kx$	Hooke's law for a spring (extension x).
Forces	$F_D = 6\pi\eta r v$	Stokes' law drag on a sphere.
Forces	$F_B = \rho g V_{\text{imm}}$	Buoyant force (upthrust).
Forces	$f_d = \mu_d R, \quad f_s = \mu_s R$	Dynamic and maximum static friction.
Newton's 2nd law	$F = ma$	Net force equals mass times acceleration.
Newton's 2nd law	$F = \dfrac{\Delta p}{\Delta t}$	Newton's second law in momentum form.

Continued on next page…

Table 2 – continued

Topic	Formula	Significance / usage
Inclined plane	$R = mg\cos\theta, \quad f = mg\sin\theta$	Normal force and friction on an incline.
Inclined plane	$\mu_s = \tan\theta_{\max}$	Coefficient of static friction from equilibrium angle.
Circular motion	$v = r\omega = \dfrac{2\pi r}{T}$	Linear speed in circular motion.
Circular motion	$\omega = \dfrac{2\pi}{T} = 2\pi f$	Angular speed for a full revolution.
Circular motion	$a_c = \dfrac{v^2}{r} = \omega^2 r$	Centripetal acceleration (toward centre).
Energy	$W = Fs\cos\theta$	Work done by a constant force.
Energy	$E_k = \dfrac{1}{2}mv^2$	Kinetic energy.
Energy	$E_p = mg\Delta h$	Gravitational potential energy.
Energy	$E_H = \dfrac{1}{2}kx^2$	Elastic potential energy of a spring.
Energy	$W_{\text{net}} = \Delta E_k$	Work–kinetic energy theorem.
Energy	$E_T = E_k + E_p$	Conservation of total mechanical energy.
Power	$P = \dfrac{W}{t} = Fv$	Power (rate of work; force × speed).
Power	$\eta = \dfrac{E_{\text{out}}}{E_{\text{in}}}$	Efficiency.
Momentum	$p = mv$	Linear momentum.
Momentum	$J = \Delta p = F\,\Delta t$	Impulse (area under F–t graph).
Momentum	$\sum p_{\text{before}} = \sum p_{\text{after}}$	Conservation of momentum.
Momentum	$E_k = \dfrac{p^2}{2m}$	Kinetic energy in terms of momentum.
Momentum	$M\dfrac{dv}{dt} = u\dfrac{dm}{dt}$	Rocket equation (u = exhaust speed).
Rotation	$\omega = \omega_1 + \alpha t$	Rotational kinematics (velocity–time).
Rotation	$\theta = \omega_1 t + \frac{1}{2}\alpha t^2$	Rotational kinematics (displacement–time).
Rotation	$\omega^2 = \omega_1^2 + 2\alpha\theta$	Rotational kinematics (velocity–displacement).
Rotation	$\tau = Fr\sin\theta$	Torque (force × perpendicular distance).
Rotation	$I = \sum m_i r_i^2$	Moment of inertia definition.
Rotation	$\tau_{\text{net}} = I\alpha$	Newton's second law for rotation.
Rotation	$E_k = \dfrac{1}{2}I\omega^2$	Rotational kinetic energy.

Continued on next page…

Table 2 – continued

Topic	**Formula**	**Significance / usage**
Rotation	$L = I\omega$	Angular momentum.
Rotation	$P = \tau\omega$	Rotational power.
Rotation	$u = \omega R$	Rolling-without-slipping condition.
Rotation	$E_k = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$	Total KE of a rolling body.
Rotation	$a = \frac{g\sin\theta}{1 + I/(MR^2)}$	Linear acceleration of a body rolling down an incline.
Mom. of inertia	$I = MR^2$ (ring)	Moment of inertia of a ring.
Mom. of inertia	$I = \frac{1}{2}MR^2$	Moment of inertia of a disk / cylinder.
Mom. of inertia	$I = \frac{2}{5}MR^2$	Moment of inertia of a sphere.
Mom. of inertia	$I = \frac{1}{12}ML^2$	Moment of inertia of a rod.
Relativity	$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$	Lorentz factor.
Relativity	$x' = \gamma(x - vt), \quad t' = \gamma\left(t - \frac{v}{c^2}x\right)$	Lorentz position and time transformations.
Relativity	$\Delta t = \gamma\,\Delta t'$	Time dilation ($\Delta t'$ = proper time).
Relativity	$L = \frac{L_0}{\gamma}$	Length contraction (L_0 = proper length).
Relativity	$u = \frac{u' + v}{1 + u'v/c^2}$	Relativistic velocity addition.
Relativity	$(c\Delta t)^2 - (\Delta x)^2 = \text{invariant}$	Invariant spacetime interval (all observers agree).

B.2 Unit B — The particulate nature of matter

Topic	Formula	Significance / usage
Thermal	$\rho = \dfrac{m}{V}$	Density (mass per unit volume).
Thermal	$\overline{E}_k = \dfrac{3}{2}k_B T$	Average random kinetic energy of a molecule.
Thermal	$T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273$	Celsius to kelvin conversion.
Thermal	$Q = mc\Delta T$	Heat to change temperature (specific heat capacity c).
Thermal	$Q = mL$	Latent heat for phase change ($L = L_F$ fusion, L_V vaporisation).
Thermal	$\dfrac{\Delta T}{\Delta t} = k(T - T_s)$	Newton's law of cooling.
Thermal	$\dfrac{\Delta Q}{\Delta t} = kA\dfrac{\Delta T}{\Delta x}$	Rate of heat conduction.
Radiation	$P = \sigma A T^4$	Stefan–Boltzmann law (total black-body power).
Radiation	$\lambda_{\max} T = 2.90 \times 10^{-3}\ \mathrm{m\,K}$	Wien's displacement law.
Radiation	$P = e\sigma A T^4$	Modified Stefan–Boltzmann for real bodies (emissivity e).
Radiation	$I = \dfrac{P}{4\pi d^2}$	Intensity at distance d from an isotropic source.
Radiation	$e + \alpha = 1$	Emissivity–albedo relation.
Radiation	$(1 - \alpha)\dfrac{S}{4} = \sigma T^4$	Earth's radiative energy balance (no greenhouse).
Gas laws	$PV = nRT = k_B NT$	Ideal gas law (equation of state).
Gas laws	$\dfrac{P_1 V_1}{T_1} = \dfrac{P_2 V_2}{T_2}$	Combined gas law (fixed n).
Gas laws	$P = \dfrac{1}{3}\rho c^2$	Kinetic theory pressure (rms speed c).
Gas laws	$\overline{E}_k = \dfrac{1}{2}mc^2 = \dfrac{3}{2}k_B T$	Average molecular KE from rms speed / temperature.
Gas laws	$U = \dfrac{3}{2}Nk_B T = \dfrac{3}{2}nRT$	Internal energy of a monatomic ideal gas.
Thermo	$\Delta U = Q - W$	First law (W = work done *by* gas).
Thermo	$W = P\Delta V$	Work done during volume change at constant pressure.
Thermo	$\Delta S = \dfrac{Q}{T}$	Entropy change for reversible heat transfer.
Thermo	$S = k_B \ln \Omega$	Statistical definition of entropy (microstates Ω).

Continued on next page…

Table 3 – continued

Topic	**Formula**	**Significance / usage**
Thermo	$\eta = 1 - \frac{T_C}{T_H}$	Carnot (maximum) efficiency of a heat engine.
Thermo	$P_1 V_1^\gamma = P_2 V_2^\gamma$	Adiabatic process relation ($\gamma = 5/3$ for monatomic).
Circuits	$V = \frac{W}{q}$	Potential difference (work per unit charge).
Circuits	$I = \frac{\Delta q}{\Delta t}$	Electric current (rate of flow of charge).
Circuits	$R = \frac{V}{I}$	Ohm's law / definition of resistance.
Circuits	$R = \rho \frac{L}{A}$	Resistance from resistivity, length, area.
Circuits	$P = IV = I^2 R = \frac{V^2}{R}$	Electric power (multiple forms).
Circuits	$\varepsilon = \frac{W}{q} = \frac{P}{I}$	Electromotive force (e.m.f.).
Circuits	$R_T = R_1 + R_2 + \cdots$	Resistors in series.
Circuits	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$	Resistors in parallel.

B.3 Unit C — Wave behaviour

Topic	Formula	Significance / usage
SHM	$a = -\omega^2 x$	Defining condition of SHM (acceleration $\propto$ displacement).
SHM	$x = x_0 \sin(\omega t + \phi)$	Displacement as a function of time.
SHM	$\omega = \sqrt{\frac{k}{m}}, \quad T = 2\pi\sqrt{\frac{m}{k}}$	Angular frequency and period of a mass–spring system.
SHM	$T = 2\pi\sqrt{\frac{L}{g}}$	Period of a simple pendulum.
SHM	$v_{\max} = \omega x_0, \quad a_{\max} = \omega^2 x_0$	Maximum velocity and acceleration in SHM.
SHM	$E_T = \frac{1}{2} m\omega^2 x_0^2$	Total energy of an SHM system.
SHM	$E_k = \frac{1}{2} m\omega^2 (x_0^2 - x^2), \quad E_p = \frac{1}{2} m\omega^2 x^2$	KE and PE at displacement x.
SHM	$v = \pm\omega\sqrt{x_0^2 - x^2}$	Velocity as a function of displacement.
Waves	$v = f\lambda$	Wave speed (frequency $\times$ wavelength).
Waves	$\Delta\phi = 2\pi\frac{\Delta x}{\lambda}$	Phase difference between two points on a wave.
Waves	$I \propto A^2$	Intensity proportional to amplitude squared.
Refraction	$n_1 \sin\theta_1 = n_2 \sin\theta_2$	Snell's law of refraction.
Refraction	$n = \frac{c_{\text{vacuum}}}{c_{\text{medium}}}$	Definition of refractive index.
Refraction	$\sin\theta_c = \frac{1}{n}$	Critical angle for total internal reflection.
Interference	$\lvert S_1P - S_2P\rvert = n\lambda$	Path difference for constructive interference.
Interference	$\lvert S_1P - S_2P\rvert = (n + \frac{1}{2})\lambda$	Path difference for destructive interference.
Interference	$s = \frac{\lambda D}{d}$	Fringe spacing in Young's double-slit experiment.
Diffraction	$b\sin\theta = \lambda$	First minimum of single-slit diffraction (slit width b).
Diffraction	$d\sin\theta = n\lambda$	Diffraction-grating equation for the nth-order maximum.
Diffraction	$I_{\text{central}} = N^2 I_{\text{single}}$	Central-maximum intensity with N slits.
Standing waves	$\lambda_n = \frac{2L}{n}, \quad f_n = n\frac{v}{2L}$	Harmonics on a string fixed at both ends.

Continued on next page…

Table 4 – continued

Topic	**Formula**	**Significance / usage**
Standing waves	$\lambda_n = \dfrac{4L}{n}$ (n odd)	Allowed wavelengths in a pipe closed at one end.
Doppler	$f' = f\left(\dfrac{v}{v \pm u_s}\right)$	Observed frequency for a moving source.
Doppler	$f' = f\left(\dfrac{v \pm u_o}{v}\right)$	Observed frequency for a moving observer.
Doppler	$\dfrac{\Delta f}{f} \approx \dfrac{v}{c}$	Approximate Doppler shift for light ($v \ll c$).

B.4 Unit D — Fields

Topic	Formula	Significance / usage
Gravitation	$F = \frac{GMm}{r^2}$	Newton's law of gravitation (force between two masses).
Gravitation	$g = \frac{GM}{r^2}$	Gravitational field strength at distance r from mass M.
Gravitation	$V_g = -\frac{GM}{r}$	Gravitational potential (zero at infinity).
Gravitation	$E_p = mV_g = -\frac{GMm}{r}$	Gravitational potential energy of two masses.
Gravitation	$g = -\frac{dV_g}{dr}$	Field strength as the negative potential gradient.
Gravitation	$v_{\text{orb}} = \sqrt{\frac{GM}{r}}$	Orbital speed of a satellite in circular orbit.
Gravitation	$T^2 = \frac{4\pi^2 r^3}{GM}$	Kepler's third law (period squared vs. radius cubed).
Gravitation	$v_{\text{esc}} = \sqrt{\frac{2GM}{r}}$	Escape speed from distance r.
Gravitation	$E_T = -\frac{GMm}{2r}$	Total energy of a body in circular orbit.
Electric	$F = \frac{kQq}{r^2}$	Coulomb's law (force between two charges).
Electric	$E = \frac{F}{q} = \frac{kQ}{r^2}$	Electric field strength from a point charge.
Electric	$V_e = \frac{kQ}{r}$	Electric potential at distance r from a point charge.
Electric	$E_p = qV_e = \frac{kQq}{r}$	Electric potential energy of two charges.
Electric	$E = -\frac{dV_e}{dr}$	Electric field as the negative potential gradient.
Electric	$E = \frac{V}{d}$	Uniform field between parallel plates.
Magnetic	$F = qvB\sin\theta$	Magnetic force on a moving charge.
Magnetic	$F = BIL\sin\theta$	Magnetic force on a current-carrying wire.
Magnetic	$B = \frac{\mu_0 I}{2\pi r}$	Field at distance r from a long straight wire.
Magnetic	$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi r}$	Force per unit length between two parallel wires.

Continued on next page…

Table 5 – continued

Topic	**Formula**	**Significance / usage**
Motion in E	$a = \frac{qE}{m}, \quad y = \frac{qE}{2m}t^2$	Acceleration and deflection of a charge in a uniform field.
Motion in B	$R = \frac{mv}{qB}$	Radius of circular motion of a charge in a magnetic field.
Motion in B	$T = \frac{2\pi m}{qB}$	Period of circular motion in a magnetic field.
Velocity selector	$E = vB$	Condition for undeflected motion in crossed E and B fields.
Induction	$\varepsilon = BLv$	Motional e.m.f. in a conductor moving through a field.
Induction	$\Phi = BA\cos\theta$	Magnetic flux through a loop.
Induction	$\Phi_{\text{link}} = NBA\cos\theta$	Magnetic flux linkage for a coil of N turns.
Induction	$\varepsilon = -N\frac{d\Phi}{dt}$	Faraday's law of electromagnetic induction.
AC generator	$V = V_{\max}\sin(2\pi ft), \quad V_{\max} = 2\pi fNBA$	Induced e.m.f. in a rotating coil.

B.5 Unit E — Nuclear and quantum physics

Topic	Formula	Significance / usage
Atomic	$E_n = -\dfrac{13.6}{n^2}\ \text{eV}$	Energy levels of the hydrogen atom.
Atomic	$mvr = n\dfrac{h}{2\pi}$	Bohr quantisation of angular momentum.
Atomic	$r_n = a_0 n^2$	Allowed orbital radii of the electron.
Quantum	$E = hf = \dfrac{hc}{\lambda}$	Photon energy.
Quantum	$p = \dfrac{h}{\lambda}$	Photon momentum (and de Broglie relation).
Quantum	$E_k = hf - \phi$	Einstein's photoelectric equation (ϕ = work function).
Quantum	$eV_s = E_{k,\max}$	Maximum photoelectron KE from stopping voltage.
Quantum	$\lambda' - \lambda = \dfrac{h}{m_e c}(1 - \cos\theta)$	Compton scattering wavelength shift.
Quantum	$\lambda = \dfrac{h}{p} = \dfrac{h}{\sqrt{2mqV}}$	de Broglie wavelength of a particle.
Quantum	$2d\sin\theta = n\lambda$	Bragg / Davisson–Germer diffraction condition.
Nuclear	$N = A - Z$	Number of neutrons (A = mass no., Z = atomic no.).
Nuclear	$R = R_0 A^{1/3}$	Nuclear radius from mass number.
Nuclear	$\mu = Zm_p + Nm_n - M_{\text{nucleus}}$	Mass defect.
Nuclear	$E_b = \mu c^2$	Binding energy.
Nuclear	$E = mc^2$	Einstein mass–energy equivalence.
Nuclear	$b = \dfrac{E_b}{A}$	Binding energy per nucleon (stability measure).
Nuclear	$\dfrac{dN}{dt} = -\lambda N$	Differential radioactive decay law.
Nuclear	$N = N_0 e^{-\lambda t}$	Exponential decay of number of nuclei.
Nuclear	$A = \lambda N$	Activity (decays per second).
Nuclear	$\lambda T_{1/2} = \ln 2$	Relation between decay constant and half-life.
Nuclear	$Q = \sum n(b_{\text{products}} - b_{\text{reactants}})$	Energy released in a nuclear reaction.
Stars	$R = \sqrt{\dfrac{L}{4\pi\sigma T^4}}$	Stellar radius from luminosity and surface temperature.
Stars	$d(\text{pc}) = \dfrac{1}{p(\text{arcsec})}$	Parallax distance in parsecs.